De Montfort University

Beng (Hons) Mechanical Engineering – Year 3

Objective: Demonstrate Key Concepts for 3D Printing & Manufacturing for Three Different Scenarios, One Design Will Be Printed and & Tested to Meet Criteria

Coursework: Design For 3D Printing

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Course: Mechanical Engineering

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DE MONTFORT UNIVERSITY LEICESTER

Contents

Abstract

Introduction

Section 1 – Design Of Wrist Splint For 3D printing

i. Voronoi Pattern 3D Wrist Splint

Figure 1 Researched Design Idea: Velcro Strap Wrist Splint

[Advanced Black Wrist Support Splint | Actesso]

First design idea inspired by traditional Velcro strap splint. However, points of weakness would be the Velcro strap holding points on the splint and complexity would be increased. Decided a simpler flexing design would have less points of failure.

Figure 2 Example 3D Printed Wrist Splint on Google Images

Where a cutout through the splint would allow for bending to wear the splint easily and take it off too, without the splint losing its stabilising/protective purpose.

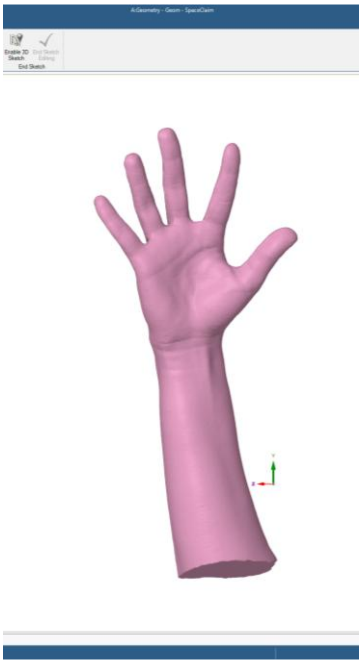


Figure 3 Provided Hand Scan

Opened hand scan stl. File into ANSYS Spaceclaim.

Figure 4 Initial Cut Out

Removed parts of the scan that are not needed such as fingers, thumb and excess wrist.

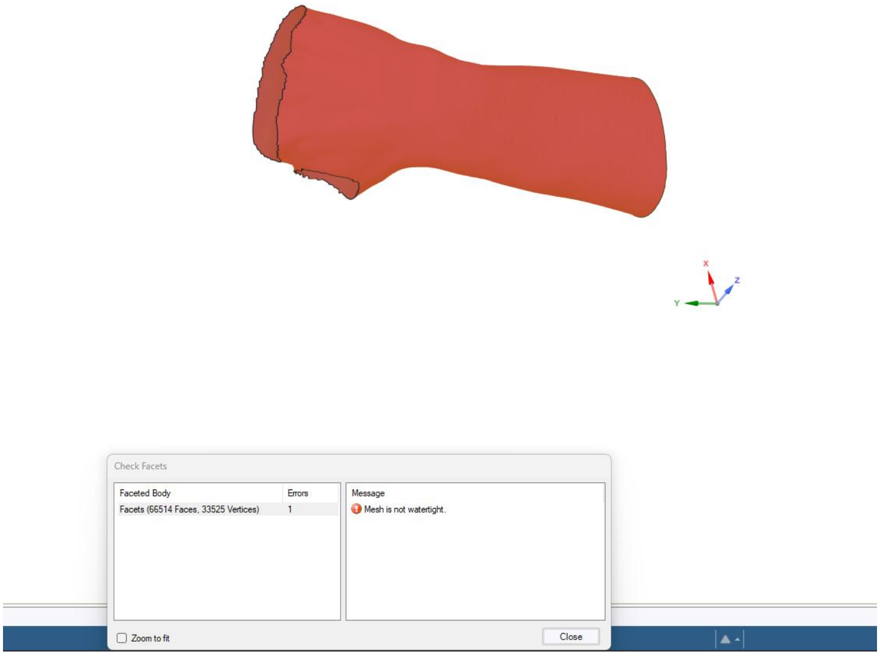


Figure 5 Check Facets

Scanned the facets with “check facets” to find any issues. The facets I removed left the model open, which means it cannot be used.

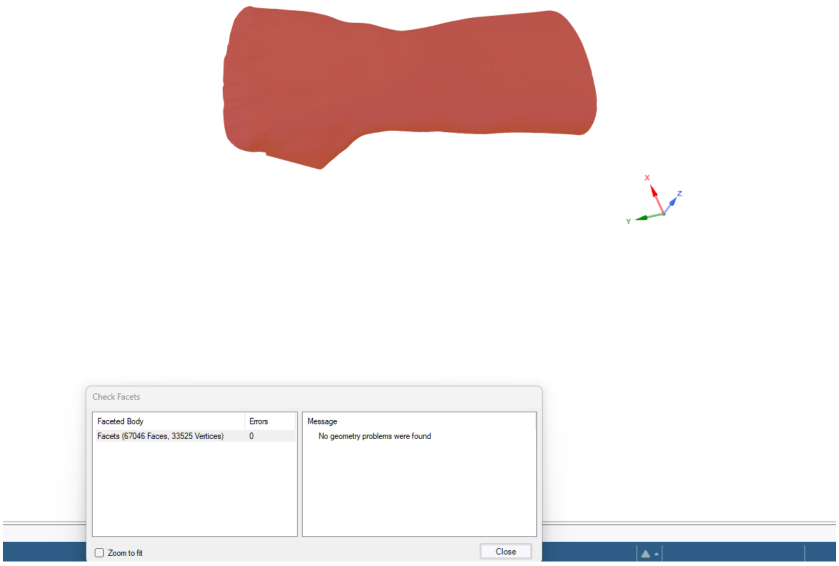


Figure 6 Auto-fix

Used “auto fix” tool to automatically repair and fill in empty spaces to provide a closed model.

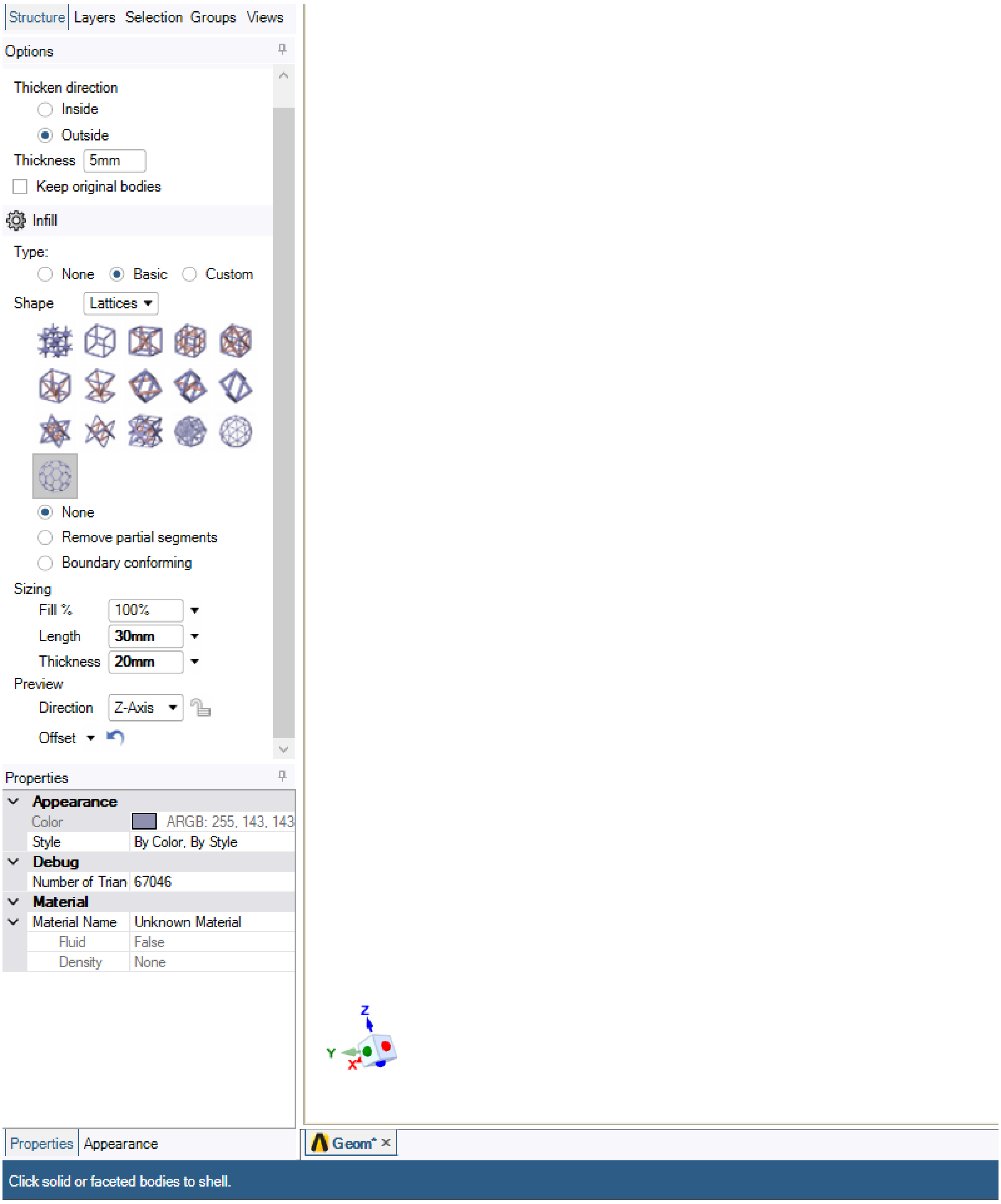


Figure 7 Shell Tool

Clicked on “shell”. Picked basic, lattice then hexagon surface faceted lattice. This will provide a Voronoi type lattice structure in Spaceclaim. Set outside thickness to 5mm, length to 30mm and thickness to 20mm.

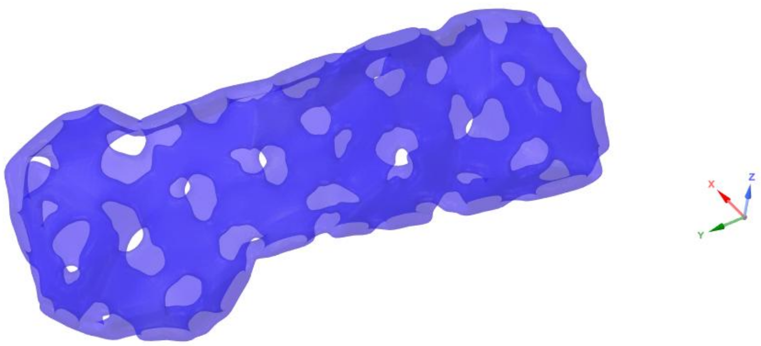


Figure 8 Initial Design

Was not happy with the amount of material in between Voronoi holes. Wanted it to be more efficient with material to speed up 3D printing, be less wasteful and more breathable for user.

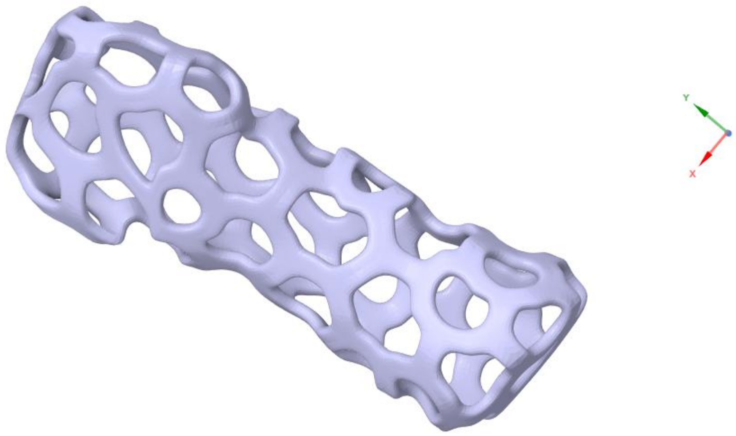


Figure 9 Second Shell Tool Iteration

Changed thickness to 10mm. Still not desired sizing, holes are large which is good for breathability in those areas, but thickness of infill means less breathability in those areas. Decided more holes, less thickness would be stronger, more breathable and just as material efficient.

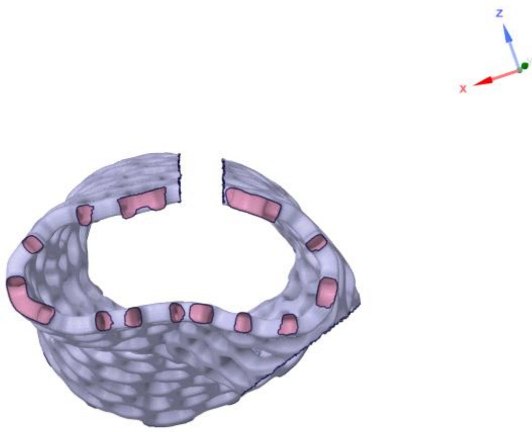


Figure 11 Final Shell Tool Sizing Iteration

Kept outside thickness to 5mm, changed length to 12mm and thickness to 4mm keeping the 3:1

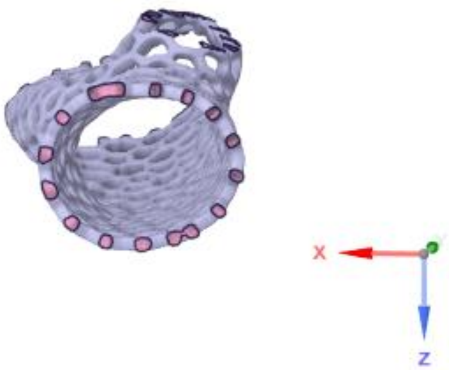


Figure 10 Excess Removed

ratio between length and thickness.Square & Lasso selection tool for removing material to make holes for wrist, fingers and thumb.

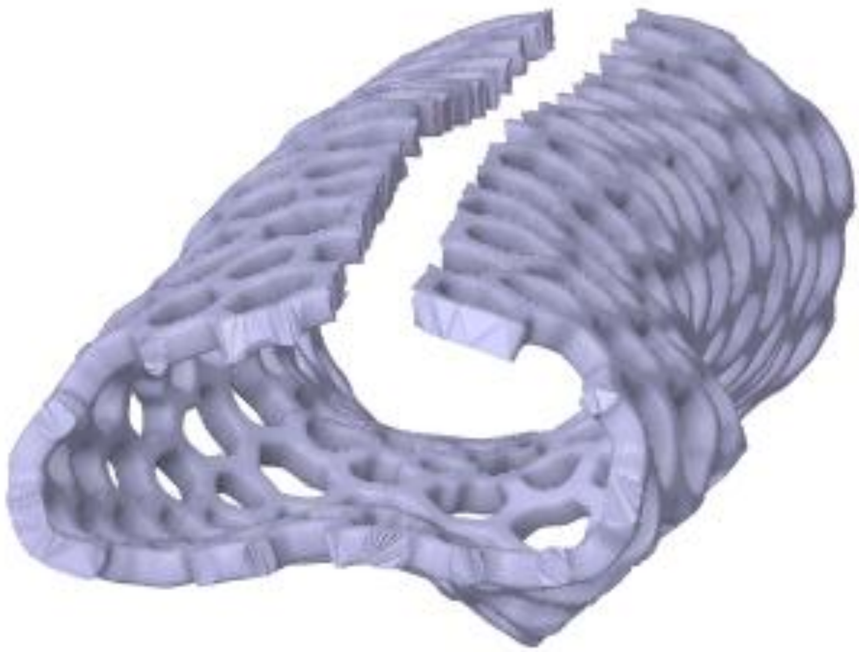


Figure 12 Flexible Design

Slit cut through whole of the splint so as to allow for flexibility of the shape when inserting patients hand. Autofix to fill in removed material edges.

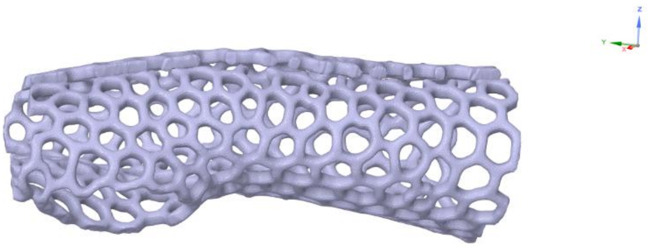


Figure 13 Finalisation/Post-Processing

Regularise 1mm 60 degrees. Then, smooth of 270 degrees, flatten peaks. To provide smooth round and safe surface: sharp edges can be uncomfortable and potentially cause harm to the user.

ii. Design Requirements

Stiff to support wrist and prevent excess movement causing injury.

Flexible enough for comfortability and being able to put it on and off without hurting wrist.

Fatigue resistant to handle repeated bending from user.

Lightweight.

Breathable.

Allows for use of fingers and thumb.

Nontoxic materials, safe for long exposure to skin (wrist splint worn for 12 weeks).

Recyclable material used (one person use and then unusable).

iii. Suitable material and additive manufacturing process

Selective Laser Sintering: bed of un-sintered powder provides support structure which is great for complex designs such as this Voronoi pattern wrist splint because printing support structures directly onto the design may cause difficulties removing the structure once the print is completed. Offers high precision. Uniform mechanical properties across all axes. Hardened steel nozzle at temperature of 240°C-270°C. heated enclosure at 50°C. filament drying oven (70°C for 6-12 hours) to prevent steam-voids. PA12 filament costs £45-£90 per kg. machine such as Prusa MK4 with enclosure costs ~£1000. These requirements are completely viable in a hospital; splints are required frequently enough for repeated printing but not so frequently that it would be impossible to provide enough splints using this process.

Nylon 12 (PA12): ultimate tensile strength of 48 MPA. Capable of lasting 10^6 cycles at around stress levels around 22 MPa. This is more than enough for the average 12 weeks. Tensile modulus of 1,400 MPa provides enough immobilisation for protection of injury under movement, and enough flexibility for patient to use comfortably. And elongation at break of up to 200% which is great for daily stresses without breaking the splint, this also means the patient can bend it enough to put it on without breaking it.

Section 2 – Design of a Laptop Holder Using Lattice Structures for 3D Printing

- Material: PLA
- Use lattice structure
- Prusa i3 MK3 or MK4 build volume 25x21x21cm
- Provided dimensional restrictions
- No print required only the design
- 70% must be lattice infill
- Any infill but must be manufacturable using the FDM process and follows FDM guidelines

Table 1 PLA Filament Material Properties

PLA Filament Material Properties	
Young’s (elastic) Modulus	3500 MPa
Poisson’s ratio	0.35
Ultimate Strength	65 MPa
Density	1.24 gr/cm^3

i. Design & design process documentation, CAD, FEA, optimisation & calculations

Heaviest laptop 15-inch laptop: MSI Titan GT77 3.5 kg round it to 5 kg. average key actuation force of a laptop is around 50 g. A person may accidentally lean or apply pressure on the keyboard without thinking; approximate 10 kg of extra random force.

Total downwards force to handle ~ 15 kg

Desired safety factor: 2.5

Figure 14 Google Image of Commercial Laptop Stand for Design Ideas

[Durable Premium Aluminium Foldable Laptop Stand, Adjustable Tilt, Silver | Paperstone]

This laptop stand used as inspiration for its slim design. Specifically design idea of two main pieces: one flat on the table, the second suspended and holding up the laptop at an angle by a supporting arm. Design has potential for sleek aesthetic and efficient use of material. Can be placed on its side for easy 3D printing with minimal supports used.

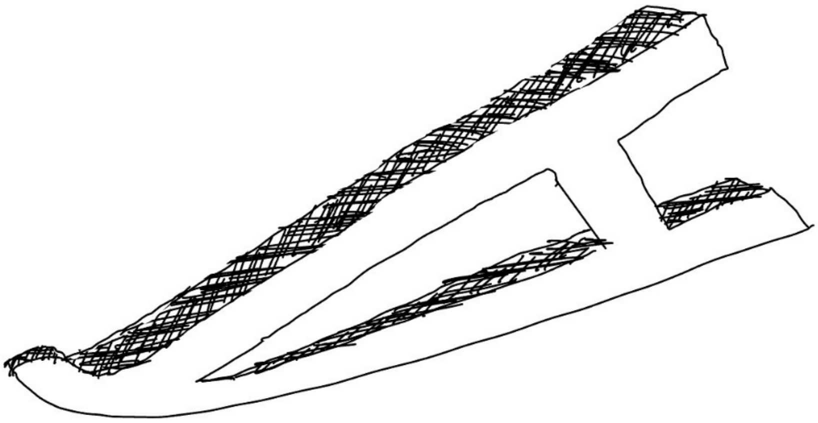


Figure 15 Initial Design Idea Hand Drawing

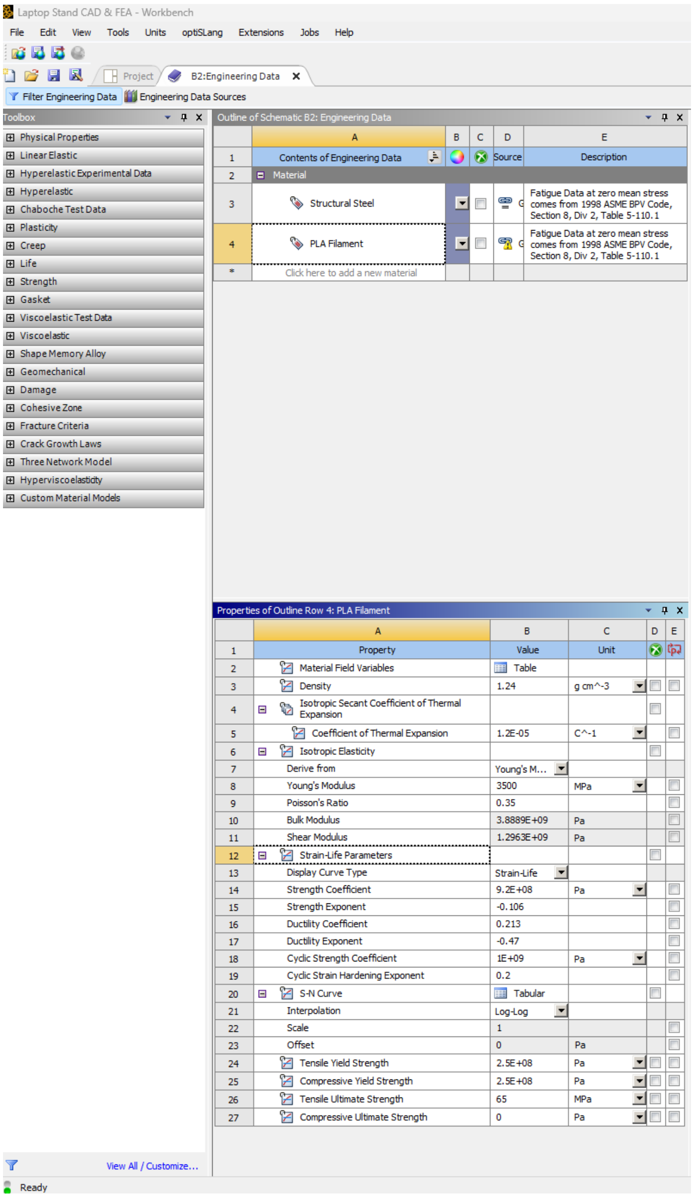


Figure 16 PLA Filament Material and Properties Added

Added new material selection with relevant PLA filament material properties applied.

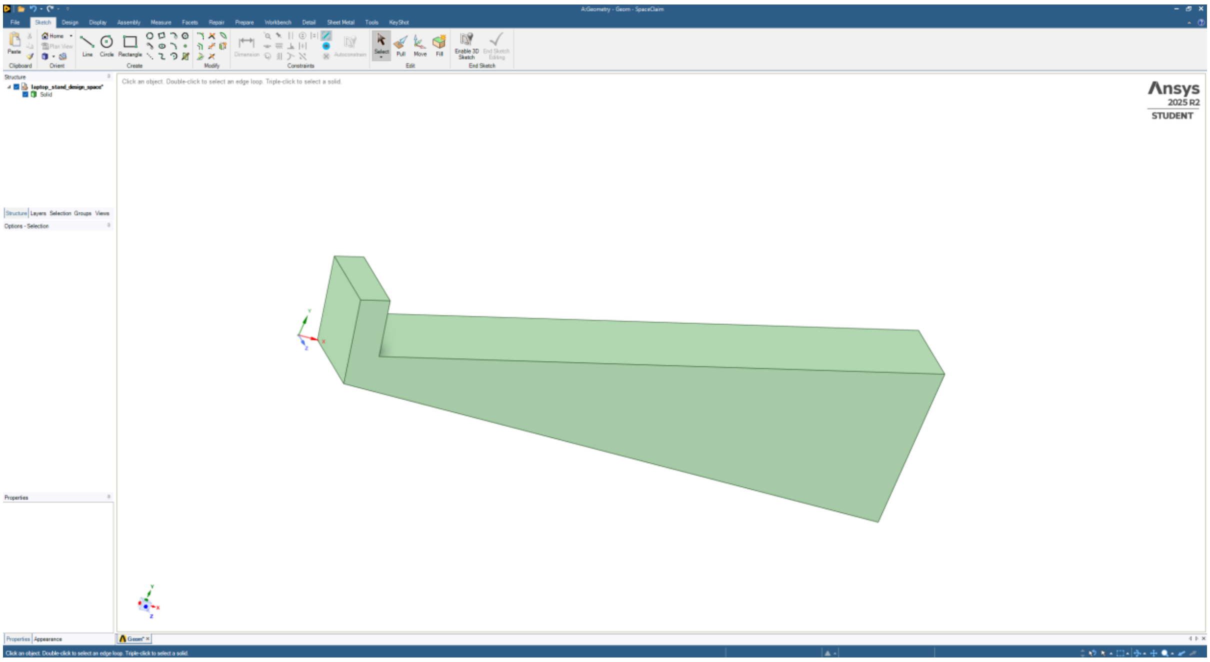


Figure 17 Provided 3D STL Geometry of Dimensional Constraints

Design constraints stl file opened in Spaceclaim for editing and converted to solid merge faces.

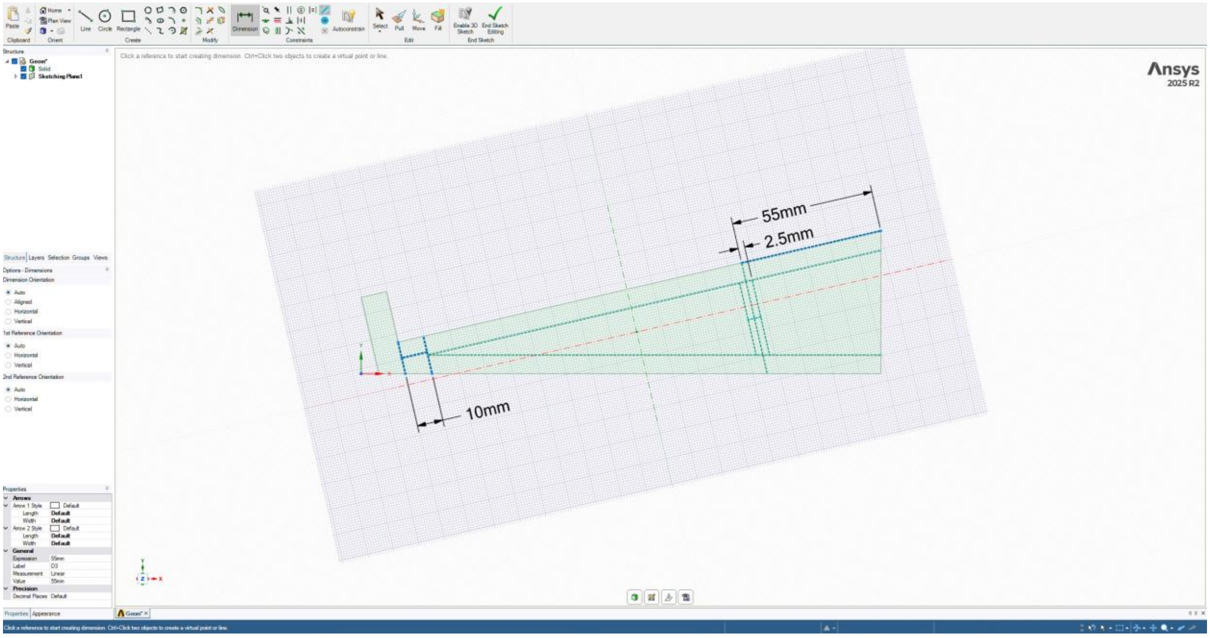


Figure 18 Initial Design

Sketched construction lines for proper alignment of sketch lines, all angles are 90 degrees, from middle or end of lines. Pillar following angle of top platform to better handle the force applied to angled platform. 10 mm given away from corner to allow for more material to support potentially high stress point.

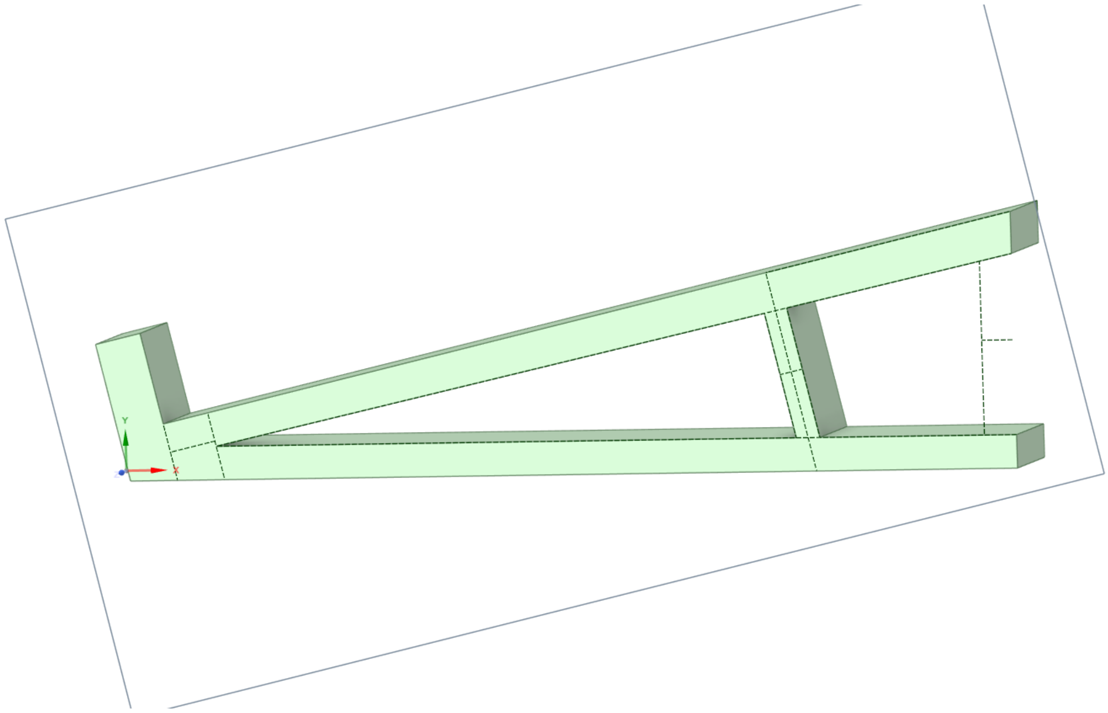


Figure 19 Pull Tool

Pull tool to remove material. Pillar seems a bit thin, may not support well enough. Ended up not using the 7 mm from the right edge as aesthetically seems better.

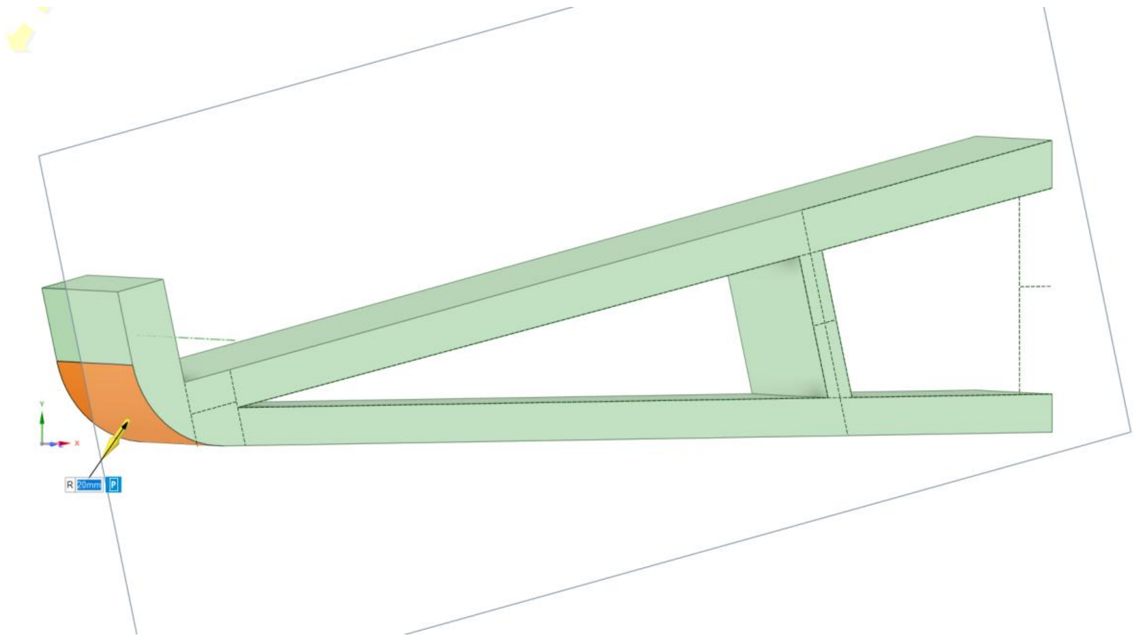


Figure 20 Rounding 1

20 mm rounding for aesthetics

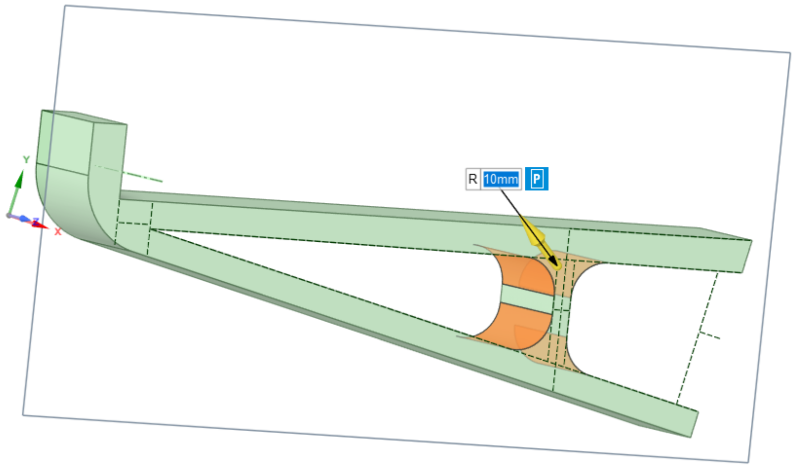


Figure 21 Rounding 2

10 mm rounding for aesthetic and to spread out stress through a round shape and added material. Spreading stress through a round shape and adding materials allows for more smooth change in stress direction through a material allowing the material to handle more stress.

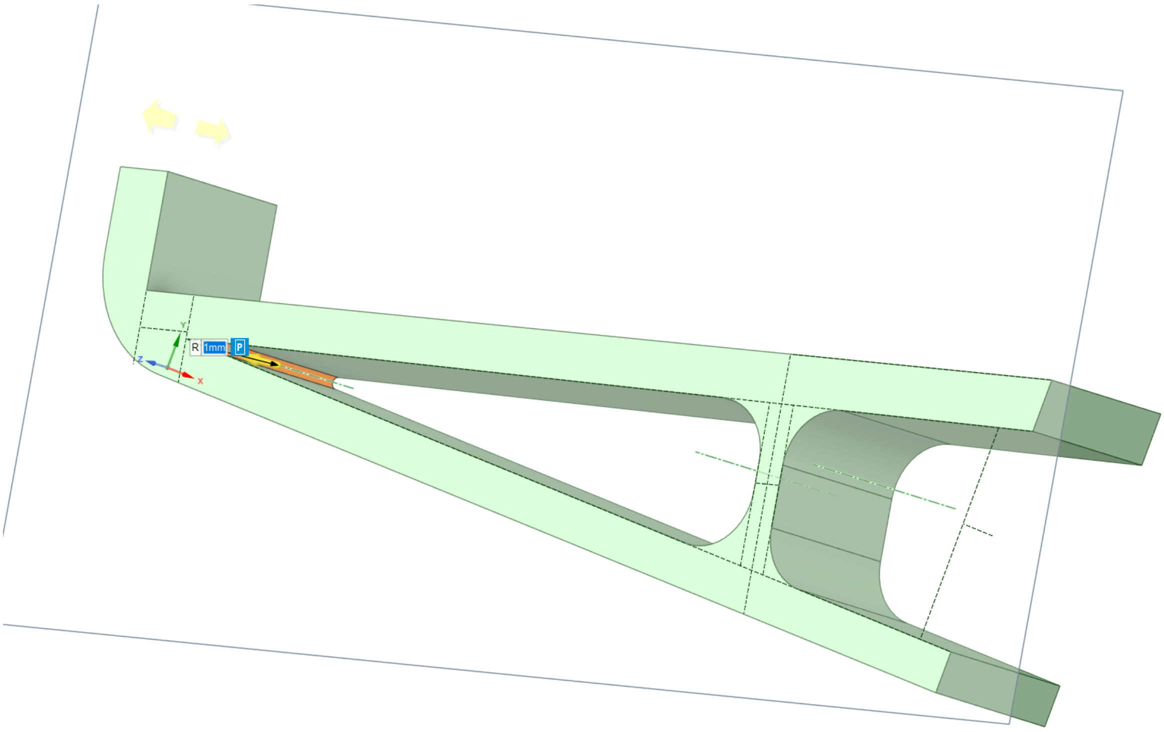


Figure 22 Rounding 3

1 mm rounding for aesthetics

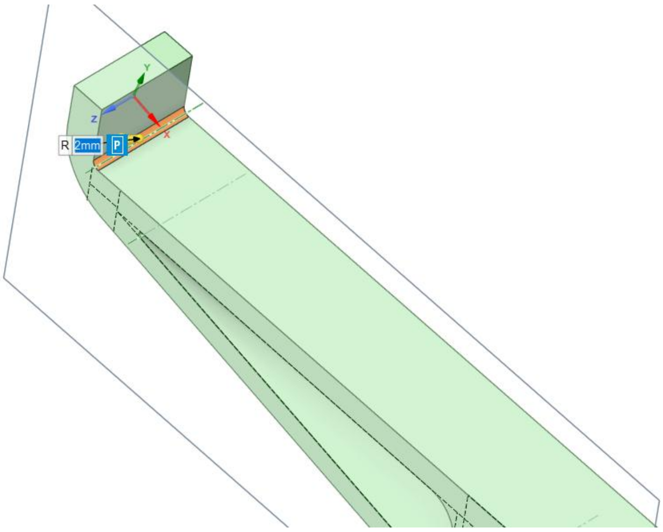


Figure 23 Rounding 4

2 mm rounding for aesthetics

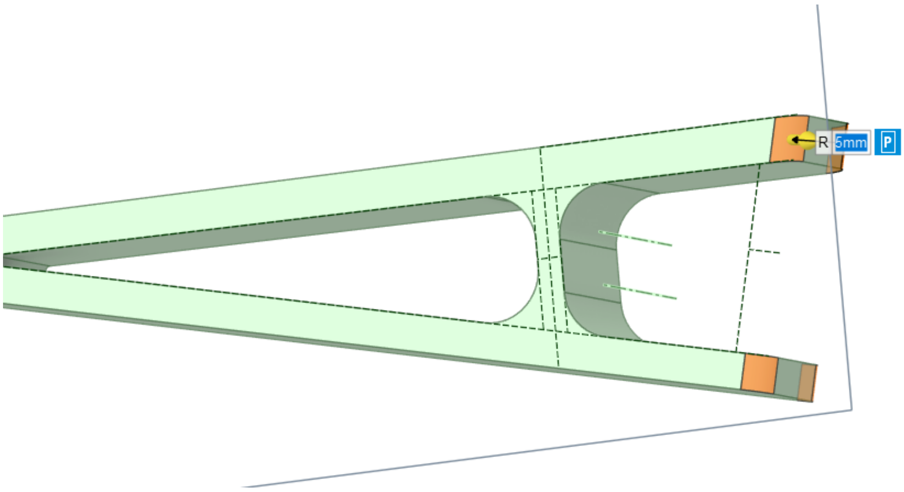


Figure 24 Rounding 5

5 mm rounding on edges for safety: consumer can hurt themselves on sharp edges.

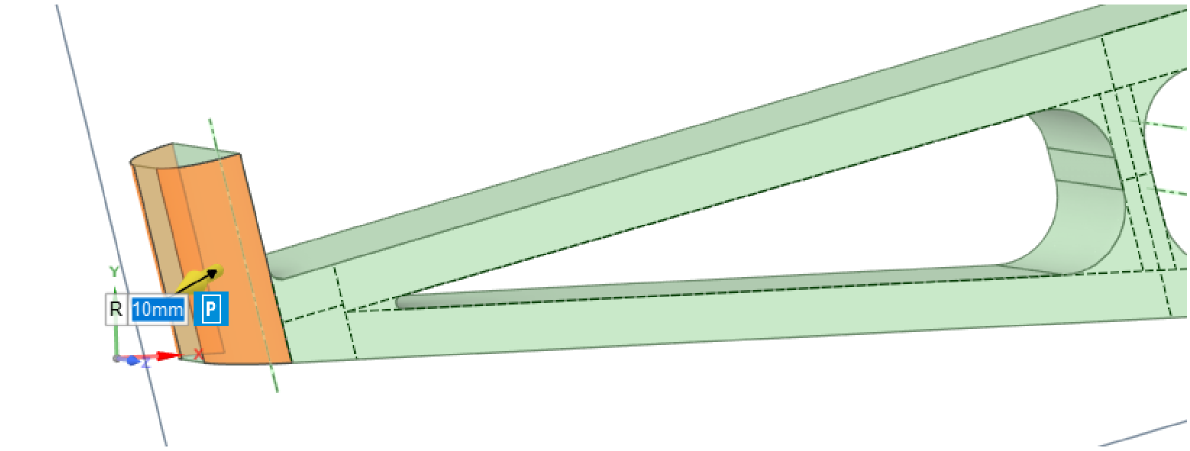


Figure 25 Rounding 6

Removed past bottom edge rounding for cleaner aesthetic of sides rounded 10 mm

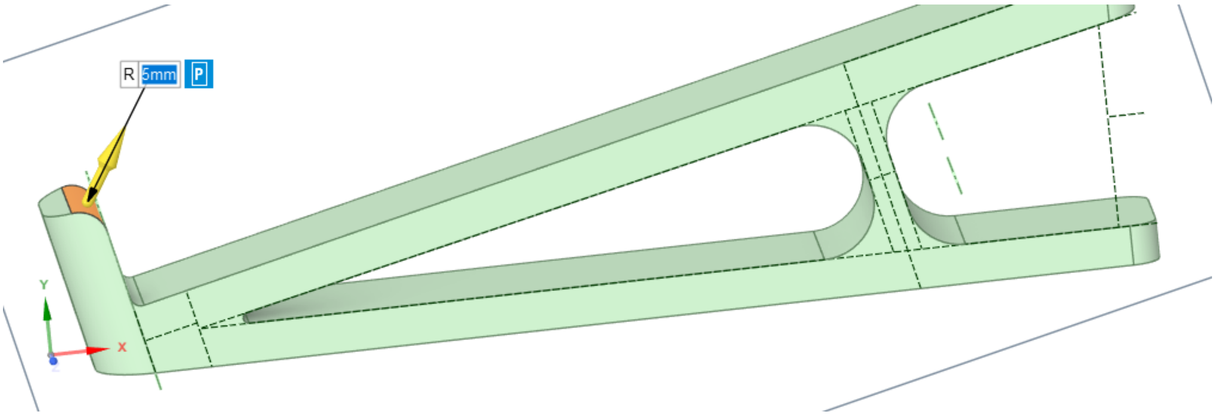


Figure 26 Rounding 7

5 mm rounded top edge for safety: sharp edge.

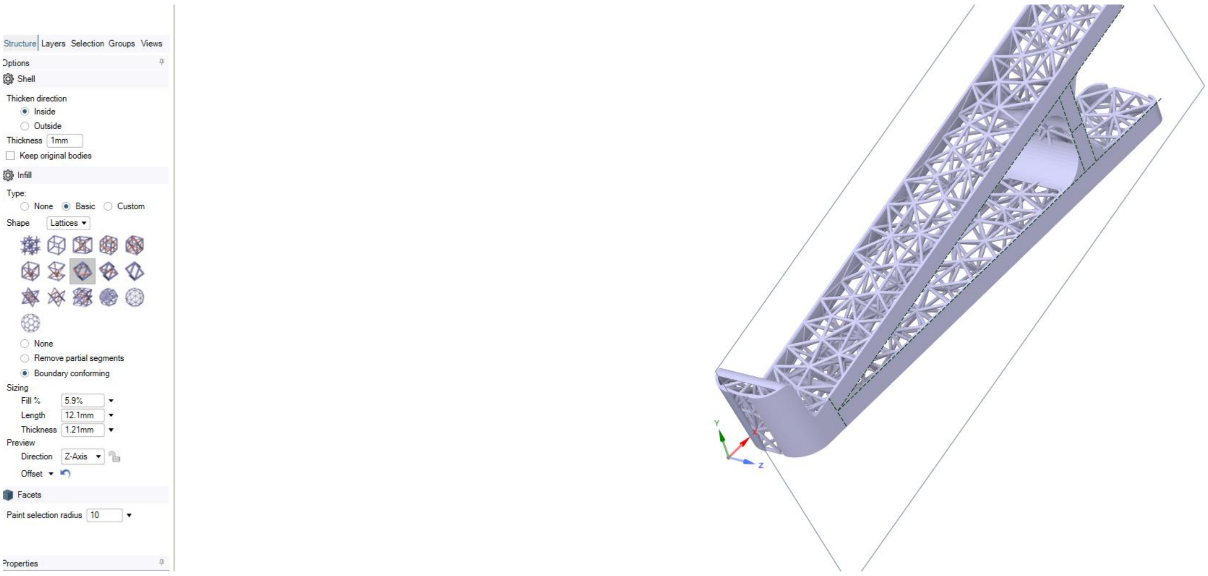


Figure 27 First Shell Idea

Used shell before converting into facets as solid merged faces are easier to select. Keeping side faces for aesthetics. Platform faces empty allows for better air flow, better cooling. 1 mm inner thickness, double pyramid basic lattice, 12.1 mm length, 1.2 mm thickness, 5.9% fill. Seems too weak and aesthetically bad.

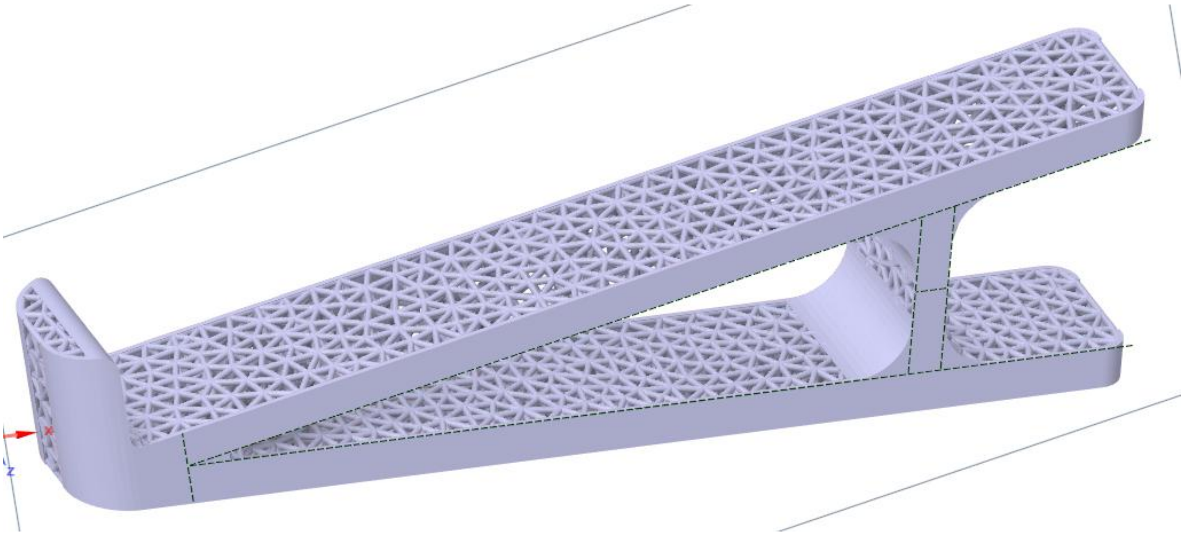


Figure 28 Sizing Modified for Aesthetics & Strength

Decreased length to 5 mm which brought the infill parts closer together, making the aesthetic better and increasing strength.

Warnings such as “foldovers” show up with this geometry with all infill types; when curvature is removed warnings stop.

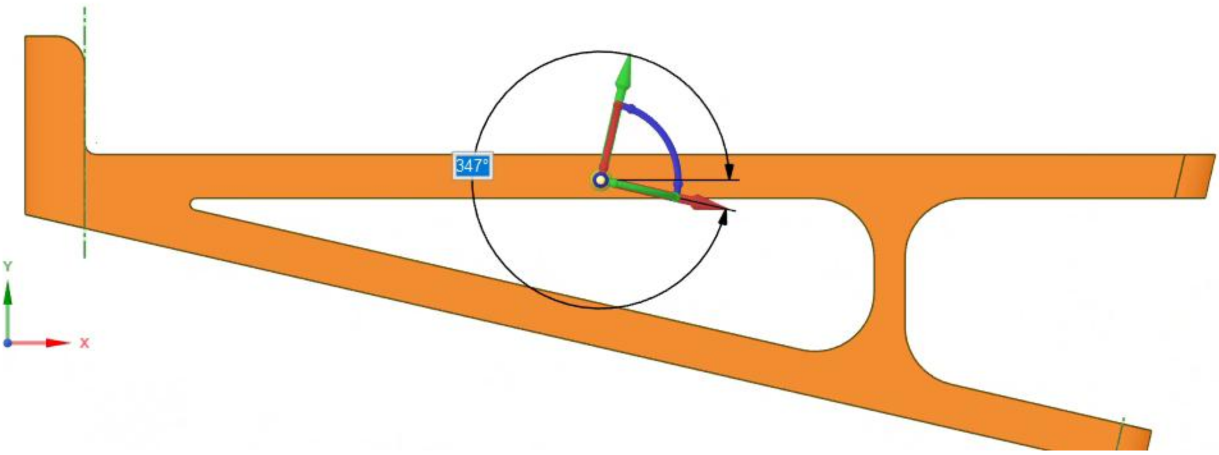


Figure 29 Rotated Using Move Tool

Went back and rotated so that lattice structures such as square, will fall in line with top platform for better performance under load.

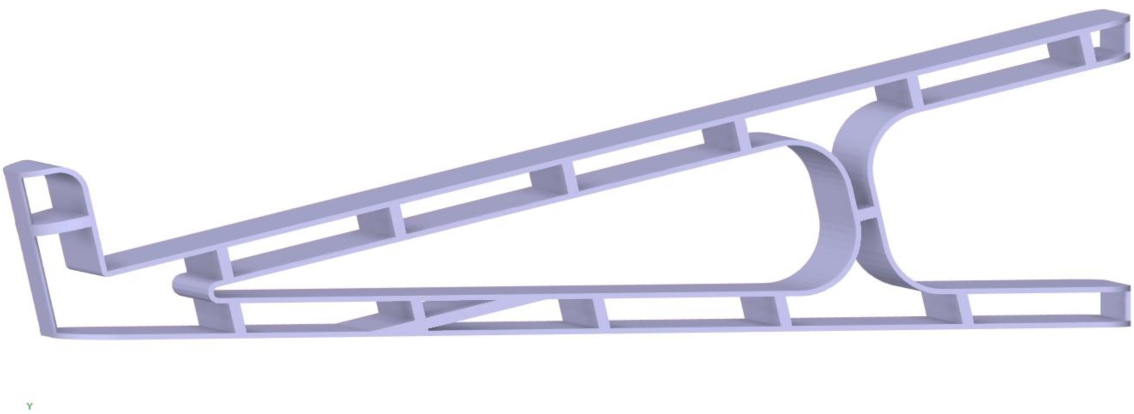


Figure 30 Second Shell Idea

Shell thickness remains 1mm. lattice changed to squares for computational and weight efficiency. lattice length changed to 30mm and thickness to 2mm for weight savings. Also changed the open faces to the flat sides and the curved sides for better aesthetics of a flat surface platform; cooling will not be affected greatly by open surface that touches the laptop since there the fan grills are generally towards the centre and designed for constricted airflow: this laptop stand design will keep the laptop in the air maintaining completely unrestricted airflow.

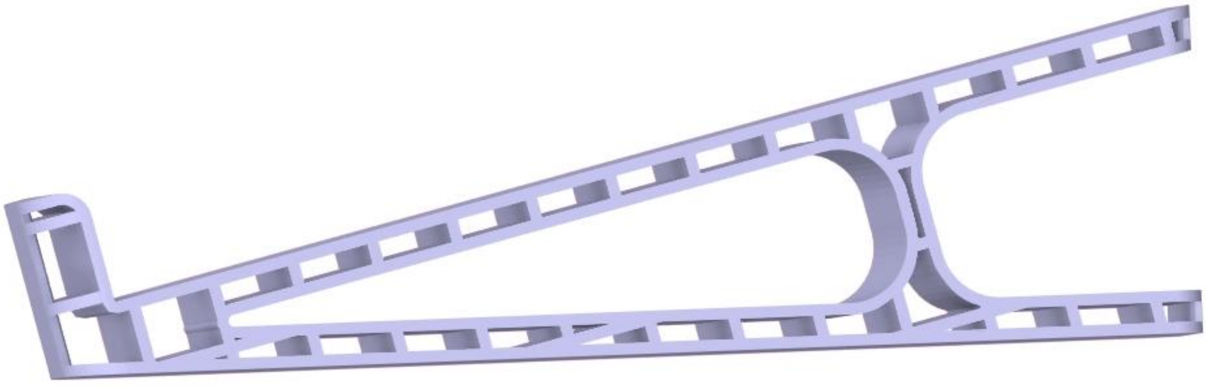


Figure 31 Sizing Changed for Strength & Aesthetics

Shell thickness and Infill length to 12mm for structural strength. Infill percentage at 26.2%. This design should eliminate the need for any support structures when being printed since it can be printed on its side as an extrusion. This design is very simple and should be computationally easy to simulate.

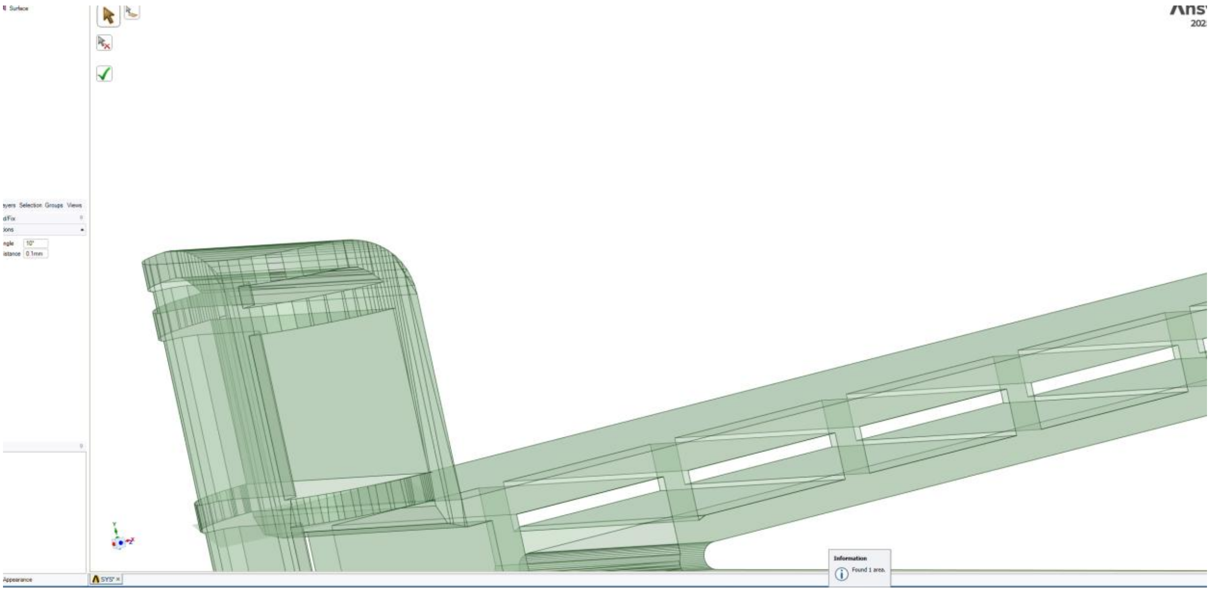


Figure 32 Facets to Solid Conversion Failure

Failed to convert to solid and remained surface. One missing face found.

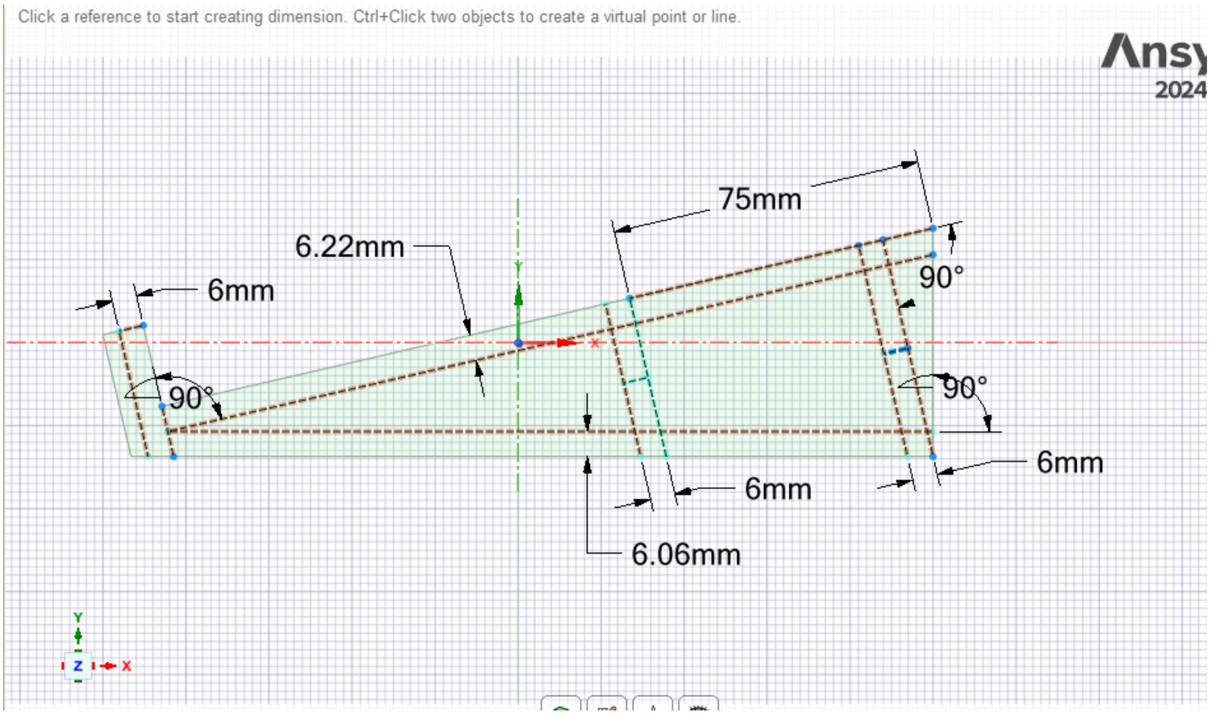


Figure 33 Second Geometry Design Plan

New design iteration based on past design using more consistent geometry. All sides are around 6mm thick. Including the front lip that will hold the laptop from falling down the slope. The prioritising of aesthetics by further removal of geometry volume means an extra support should be used to maintain strength. Since the moment is larger further from a support the added support is applied towards the rear of the geometry, and the centre support is moved further to the left at 75mm. all angles are kept at 90 degrees to at least one surface for consistent aesthetics.

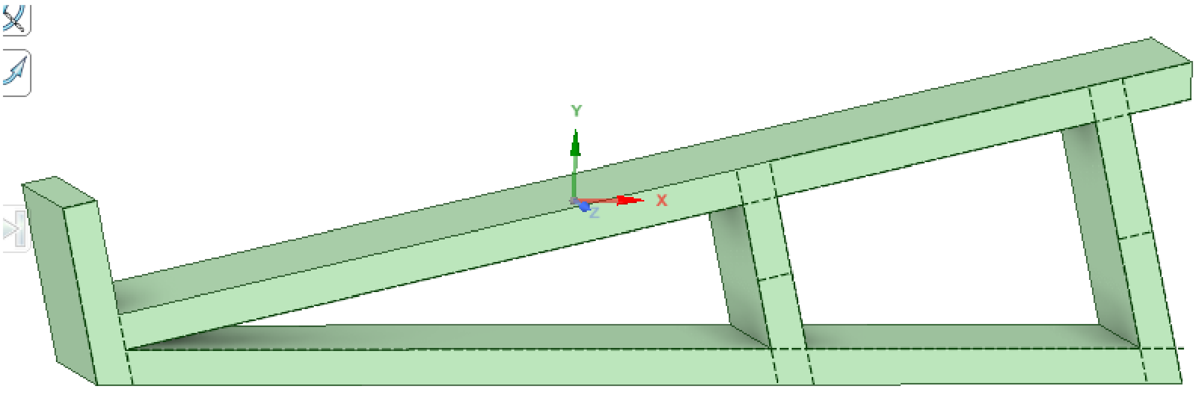


Figure 34 Pull Tool

Used pull tool to remove geometry. Leaving a sleek aesthetic.

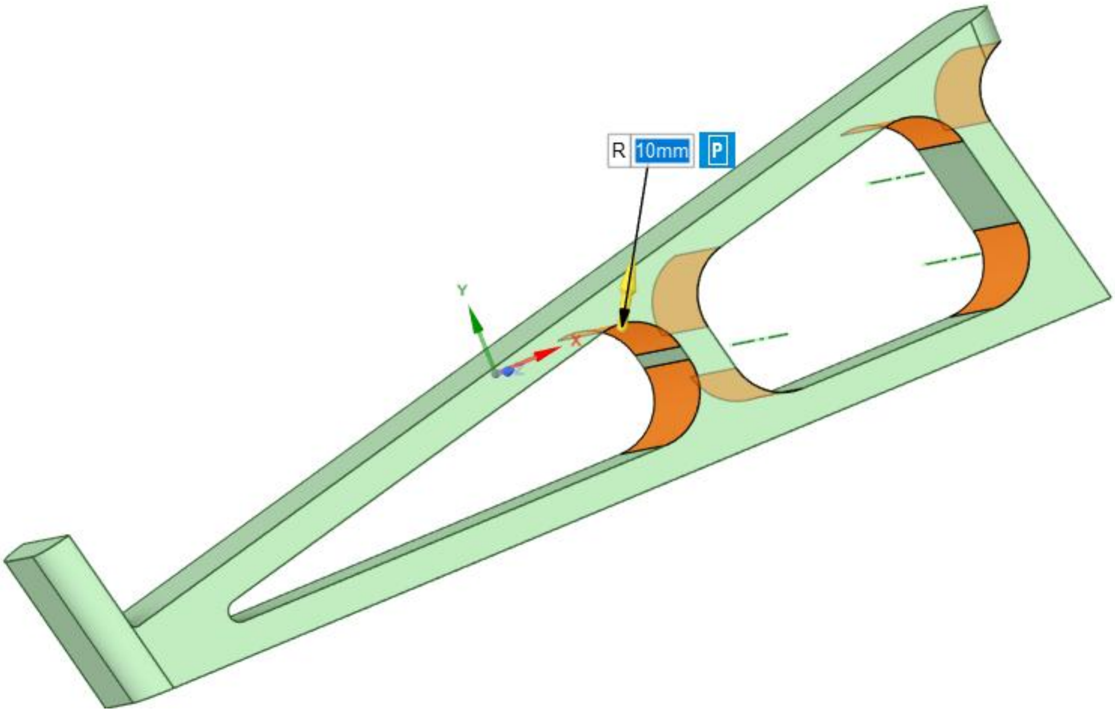


Figure 35 Rounding 1

Round using pull tool 10mm. Rounding is essential for distributing stress to avoid concentrated stress regions caused by sharp geometry. It also provides safety by avoiding sharp points that can injure the user.

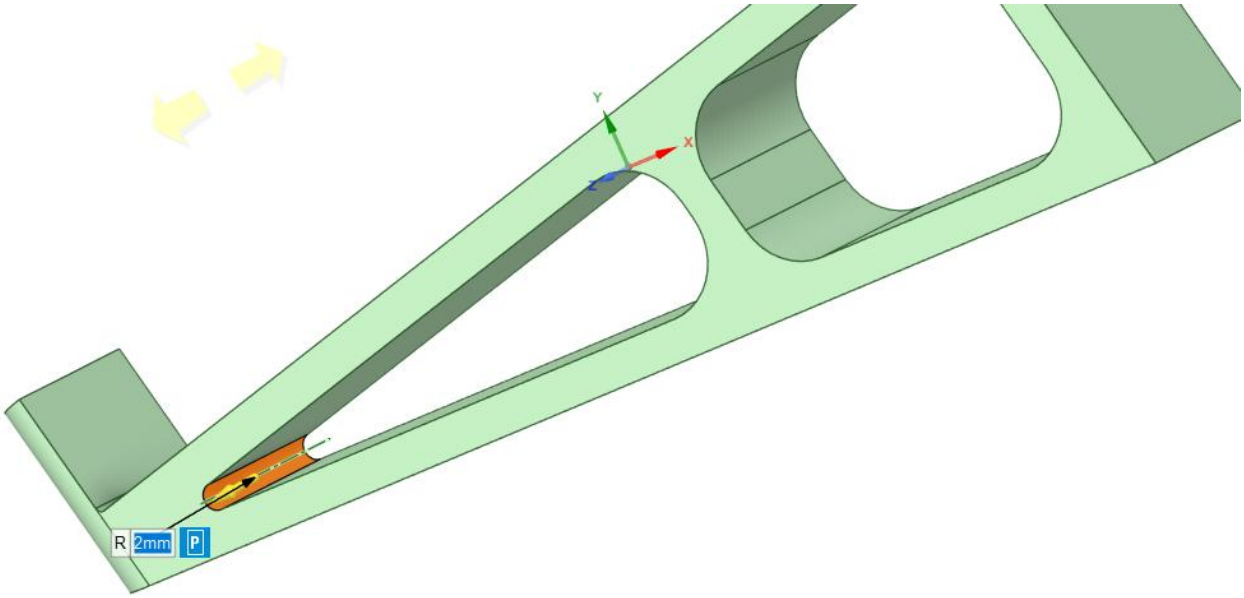


Figure 36 Rounding 2

2mm round.

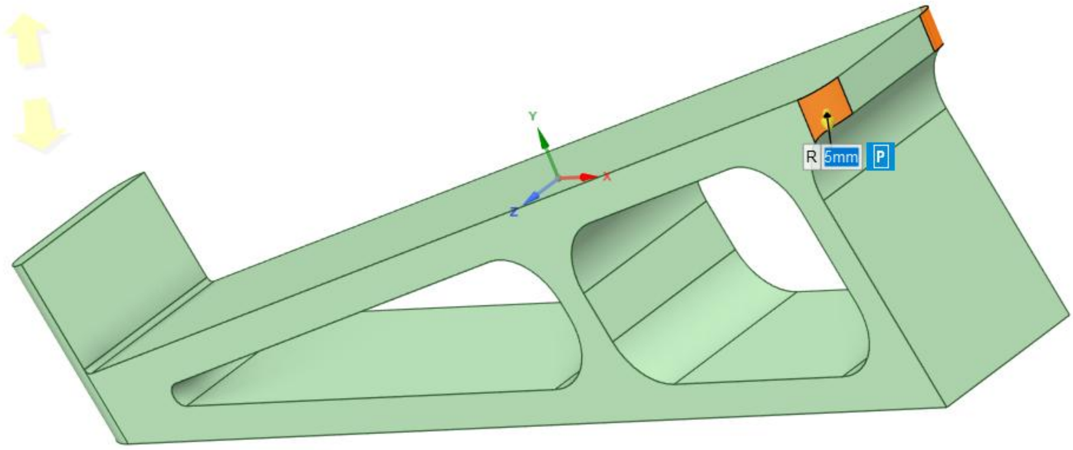


Figure 37 Rounding 3

5mm round

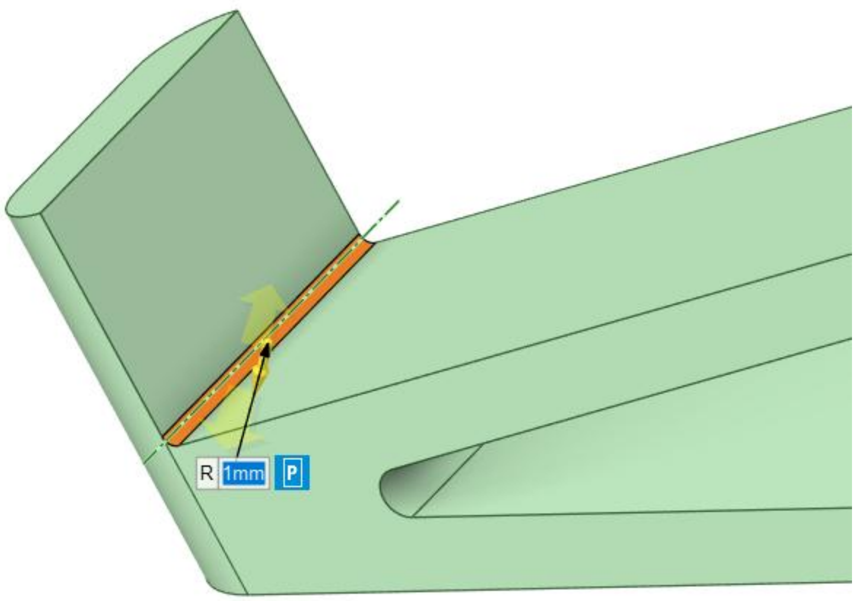


Figure 38 Rounding 4

1mm rounding. Small rounding to minimise disturbing movement from laptop running up the lip in use.

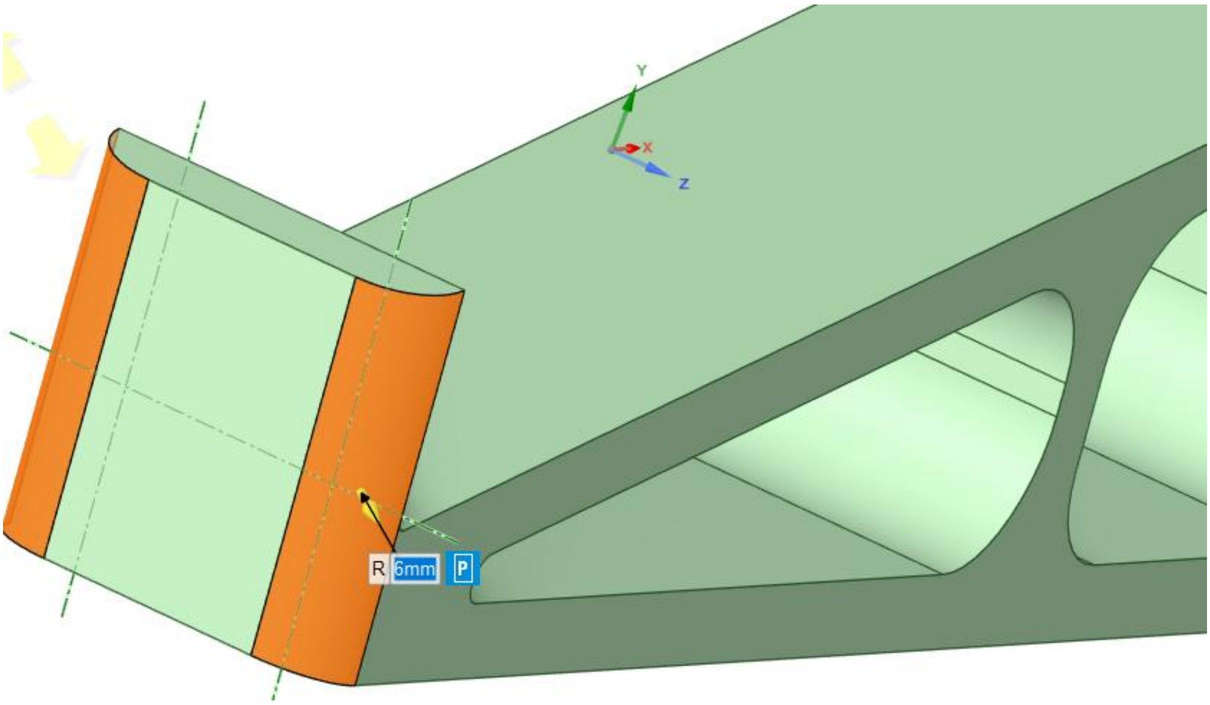


Figure 39 Rounding 5

6mm rounding for safety and aesthetics.

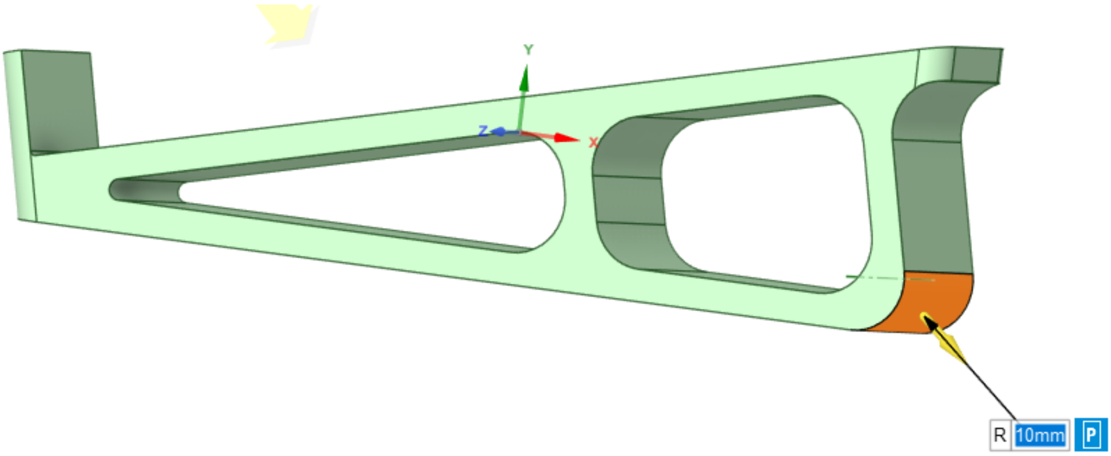


Figure 40 Rounding 5

10mm mainly for aesthetics but also for the off chance of injury.

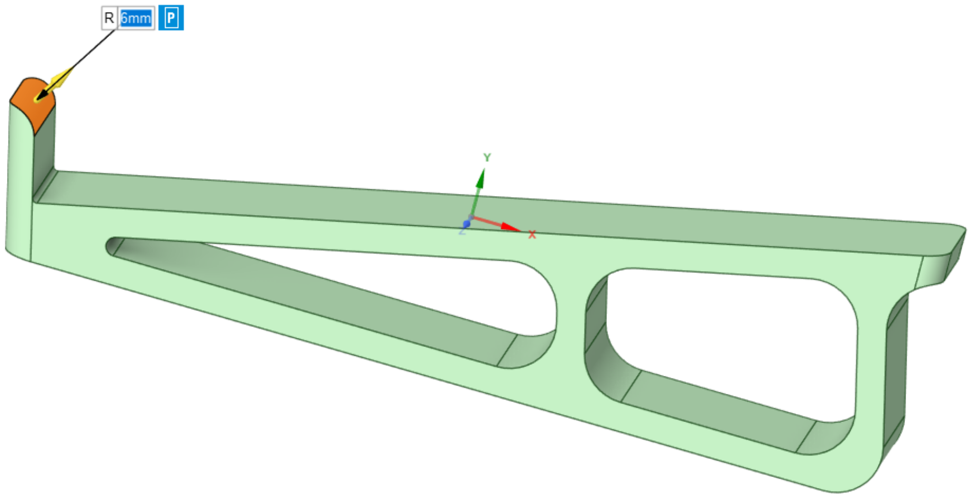


Figure 41 Rounding 6

6mm rounding for aesthetics and safety.

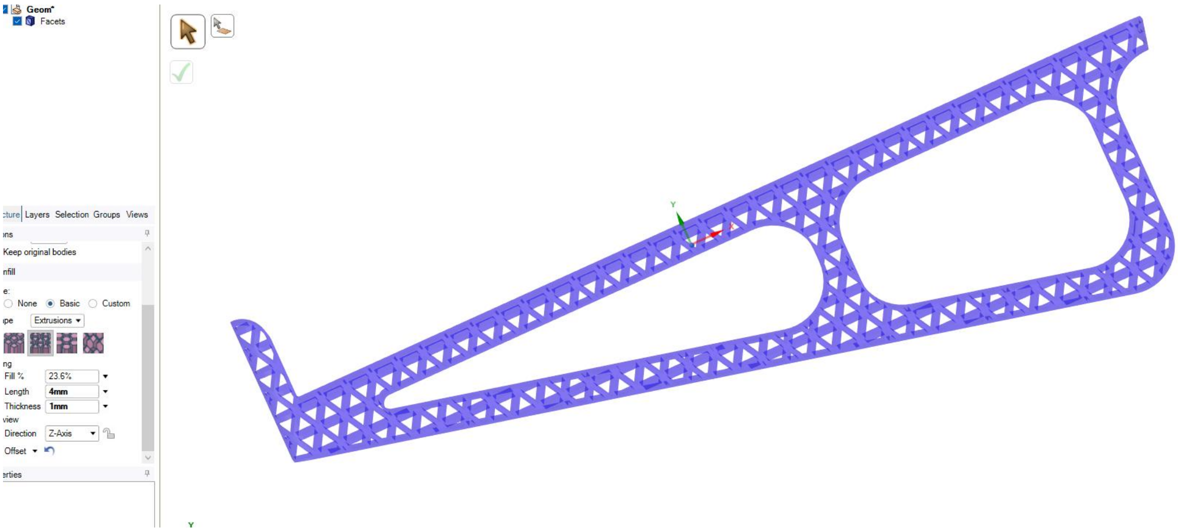


Figure 42 Triangle Extrusion Infill

Triangle extrusion infill selected. 0.5mm shell thickness. 4mm infill length and 1mm thickness. Side faces selected with rounded corners too. Geometry rotated using move tool so that top platform is in line with infill for better strength direction against force direction.

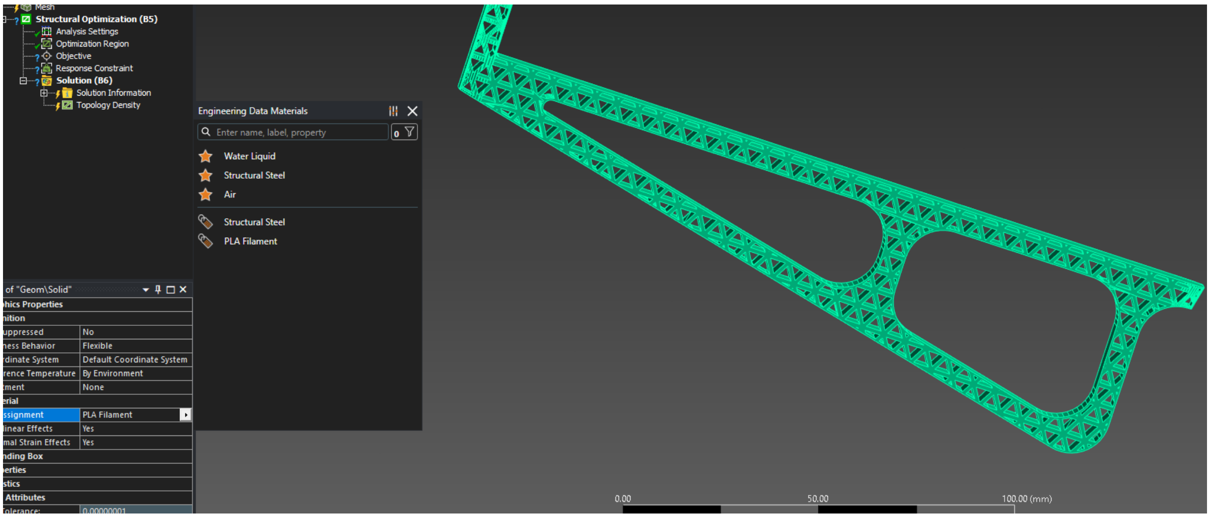


Figure 43 FEA Material Selection, Meshing Failure

Materials selection for PLA filament. Meshing failed due to complexity of geometry causing issues in Spaceclaim. Cannot continue with this geometry unless repaired, this may take too long.

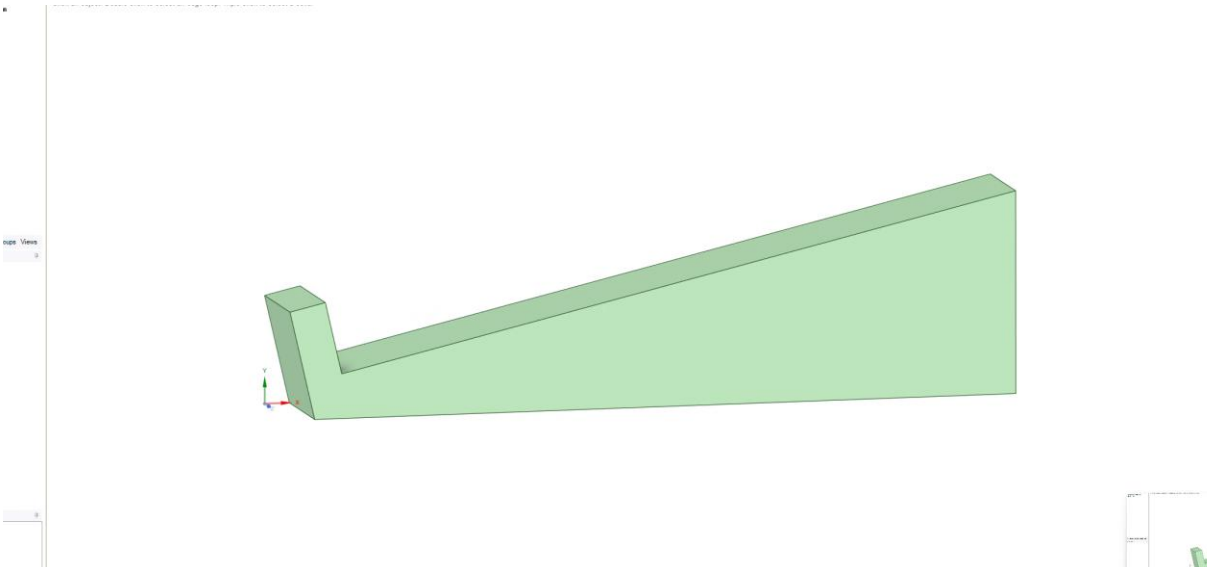


Figure 44 Restart

Fresh start from provided stand geometry.

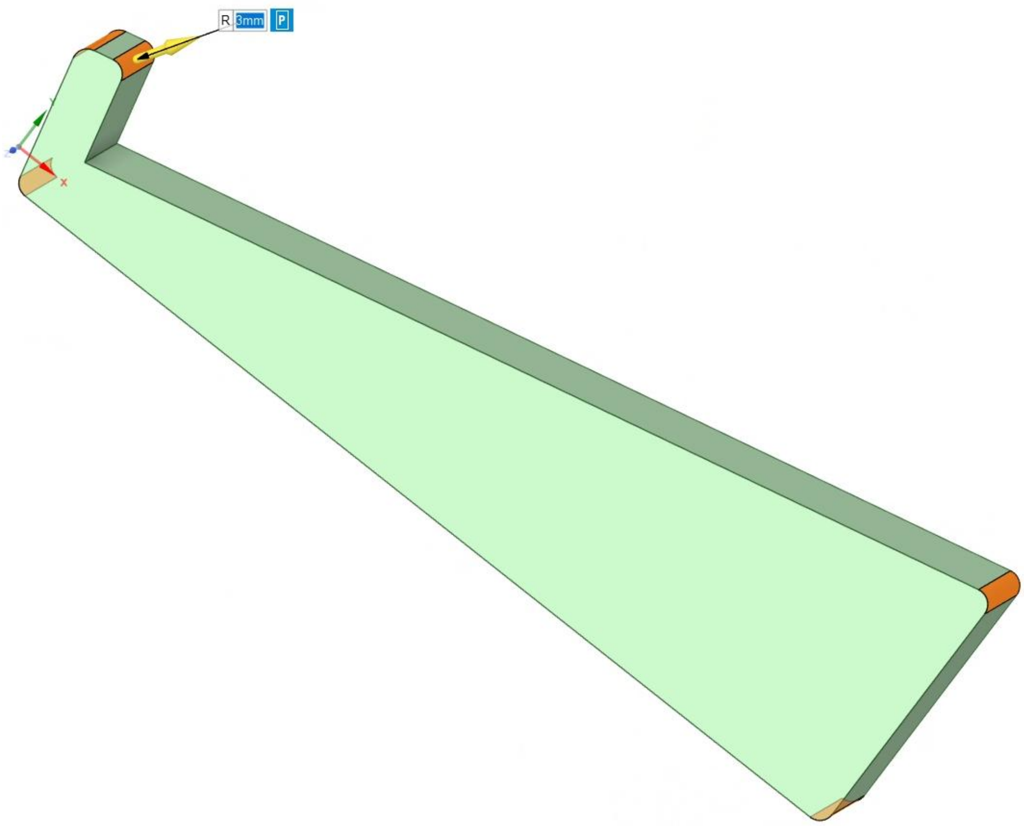


Figure 45 Rounding, All Corners Same This Time for Simplicity

Rounded 3mm to remove sharp corners; sharp corners can poke/cut user. Edges are not dangerous so they will be left to reduce complexity when applying infill.

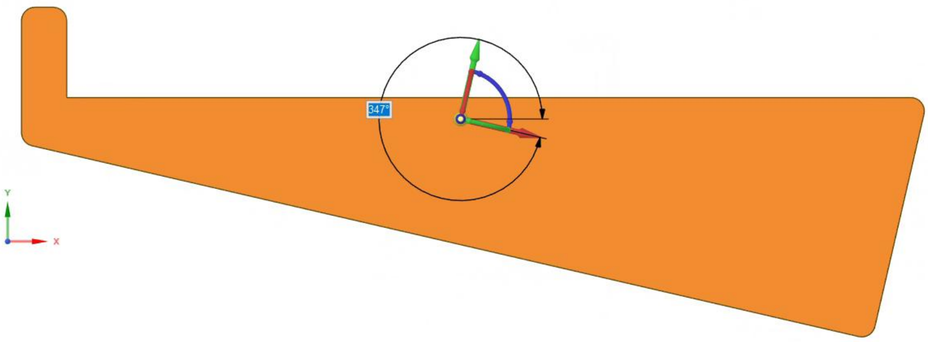


Figure 46 Rotation Using Move Tool

Rotated so that top platform is in line with x axis so that infill is at optimum angle to support weight placed on top platform.

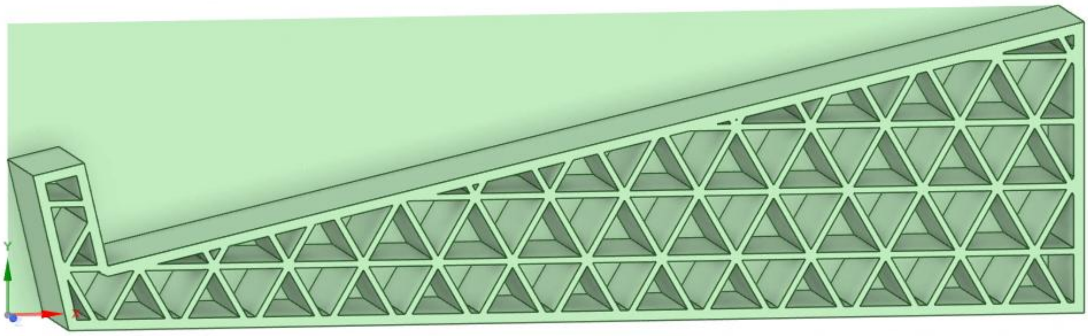


Figure 47 Facets to Solid Conversion Failure

Errors occurring yet geometry is simple.

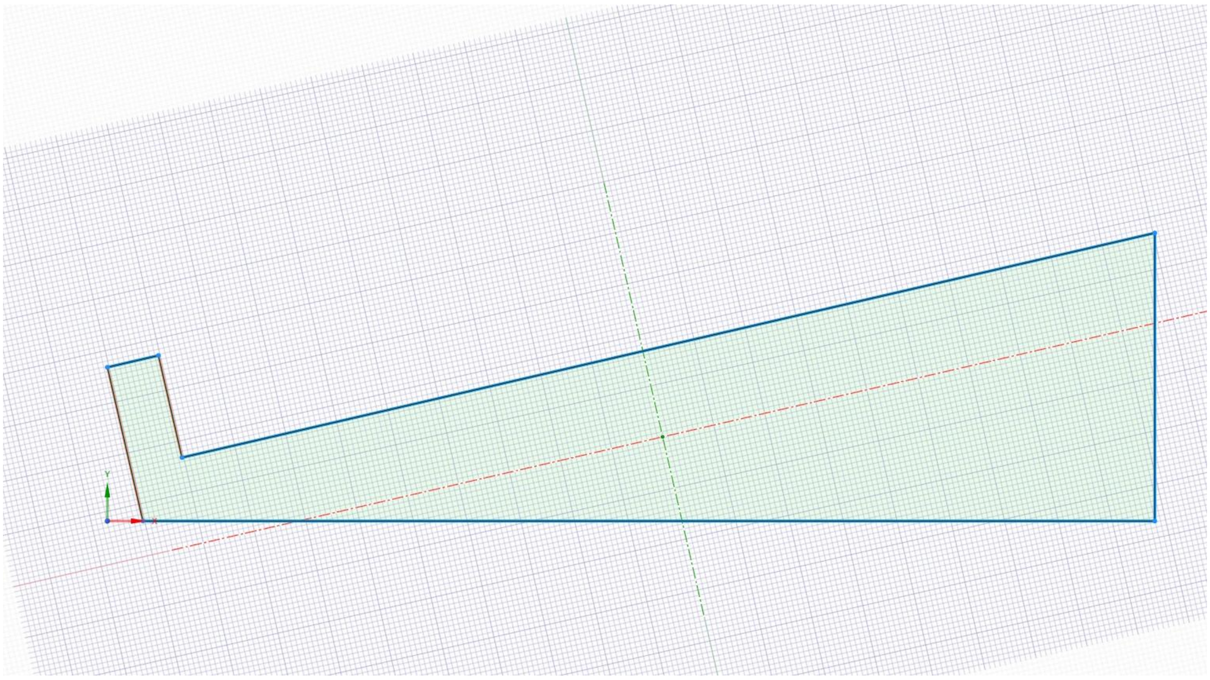


Figure 48 Designing Dimensional Limits From Scratch to Rule Out Issue with Provided File

Sketched on top of provided laptop stand file since errors were occurring

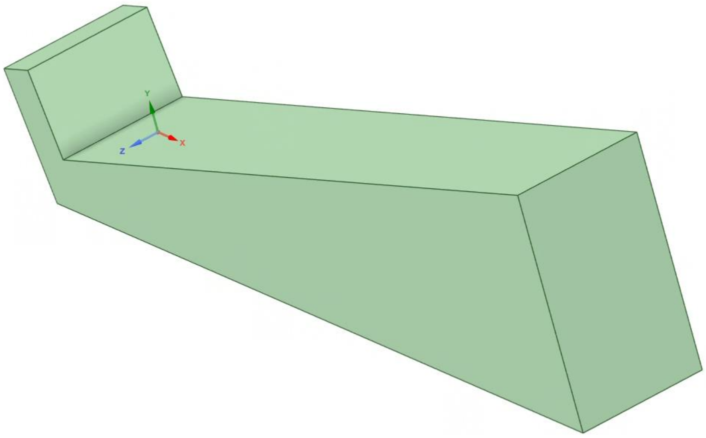


Figure 49 Pull Tool to Finalise

Pull tool used by 30 mm to follow specifications.

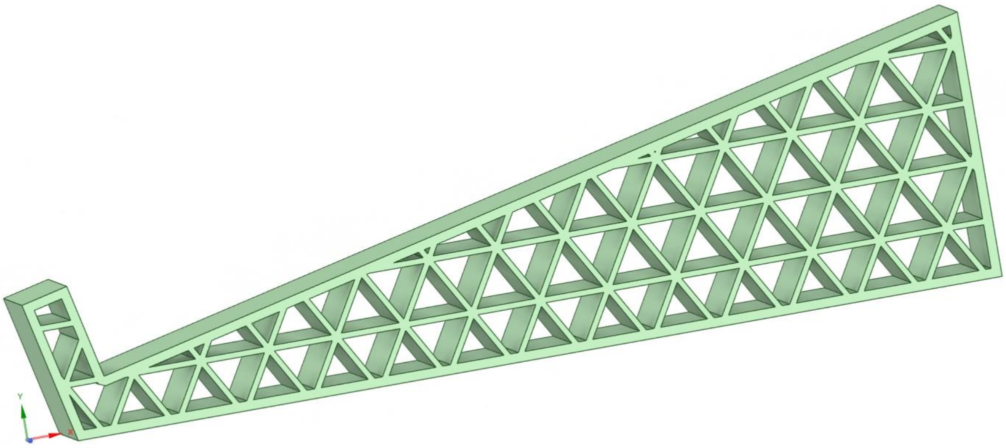


Figure 50 Last Iteration Method Repeated, Successfully

Errors stopped, problem fixed. Issue was imported geometry causing math errors when applying infill and converting to solid. Design going in new direction; more simple design with larger spaces for infill, to reduce chance of errors and speed up process.

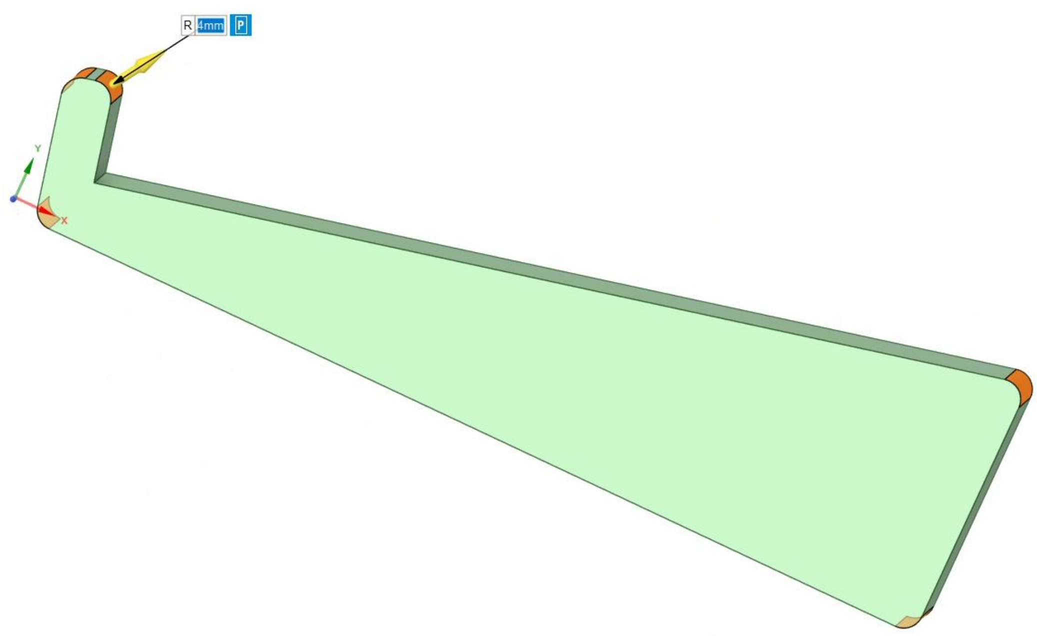


Figure 51 Went Back for Rounding

4mm rounding using pull tool for protection of user from sharp corners.

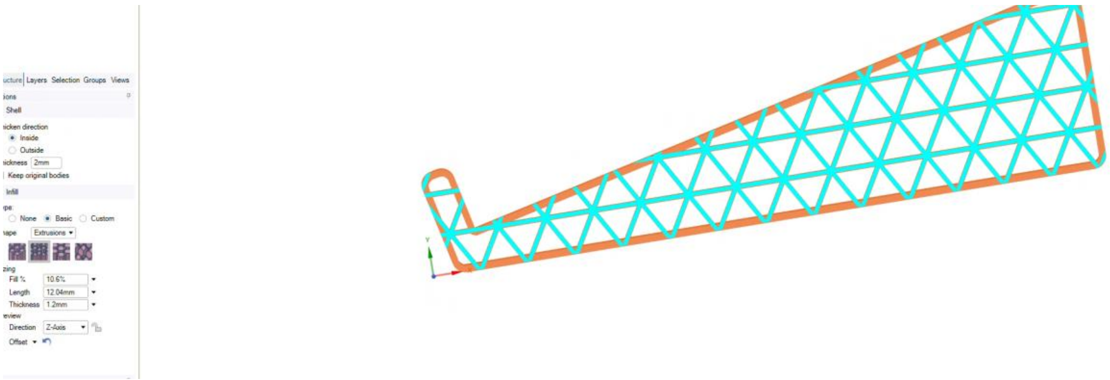


Figure 52 Triangle Extrusion Infill

Infill: 12.04mm length and 1.2mm thickness. 2mm shell thickness.

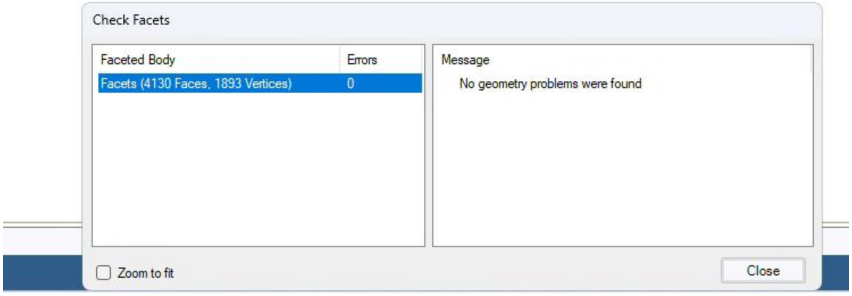


Figure 53 Check Facets

Check facets no issues found

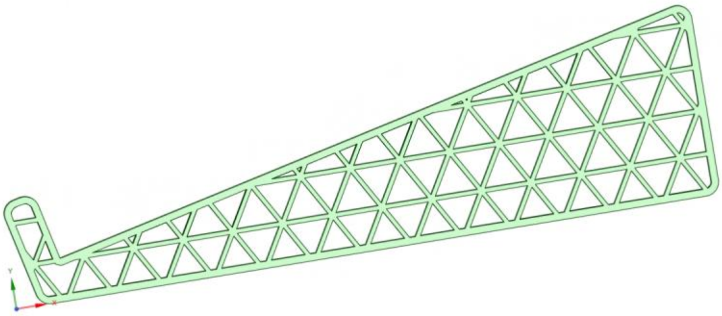


Figure 54 Successfully Converted to Solid after Shell Tool

Converted to solid for FEA.

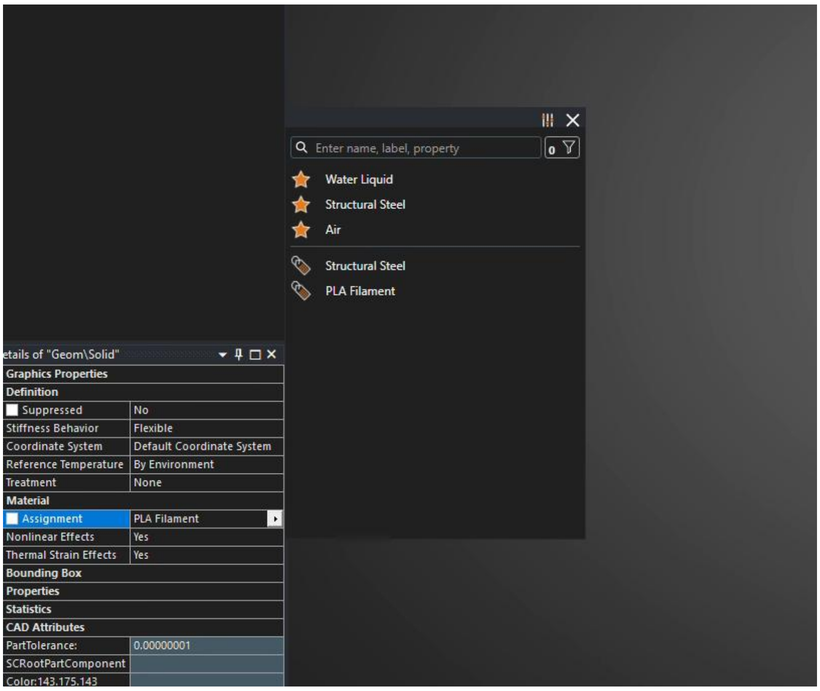


Figure 55 Material Selection in FEA

PLA filament selected

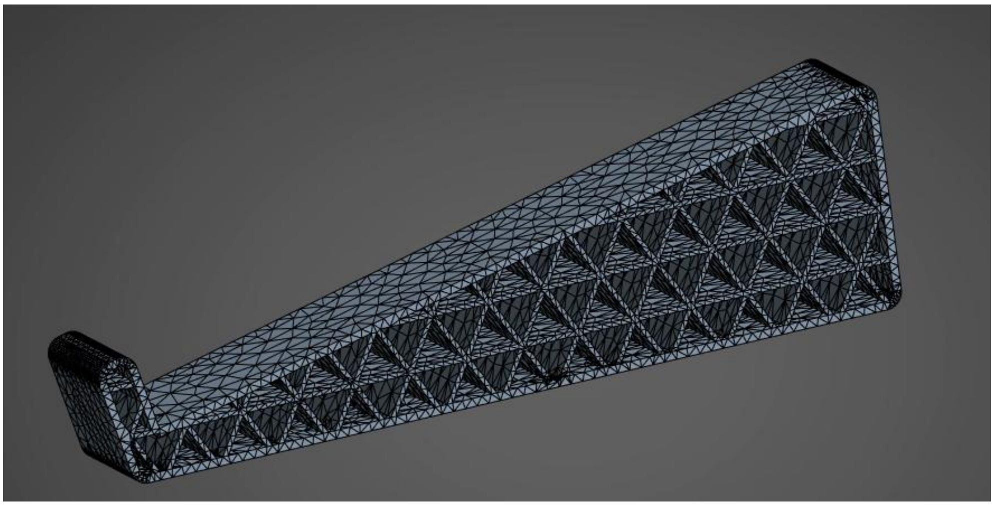


Figure 56 Meshing Successful

Mesh 3mm, adaptive sizing on for accurate round edge meshing.

Figure 57 Force/Fixed Support

Force and fixed support added to static structural.

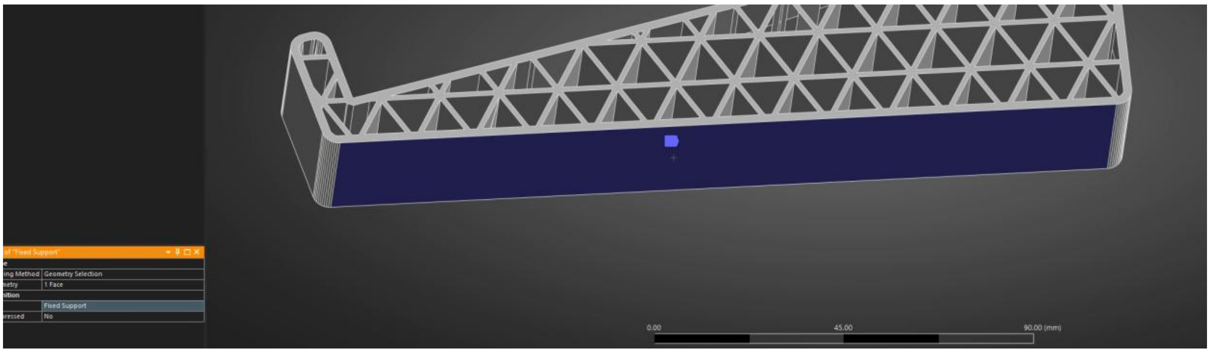


Figure 58 Fixed Support Face

Bottom face selected as fixed support as this will be resting on a table.

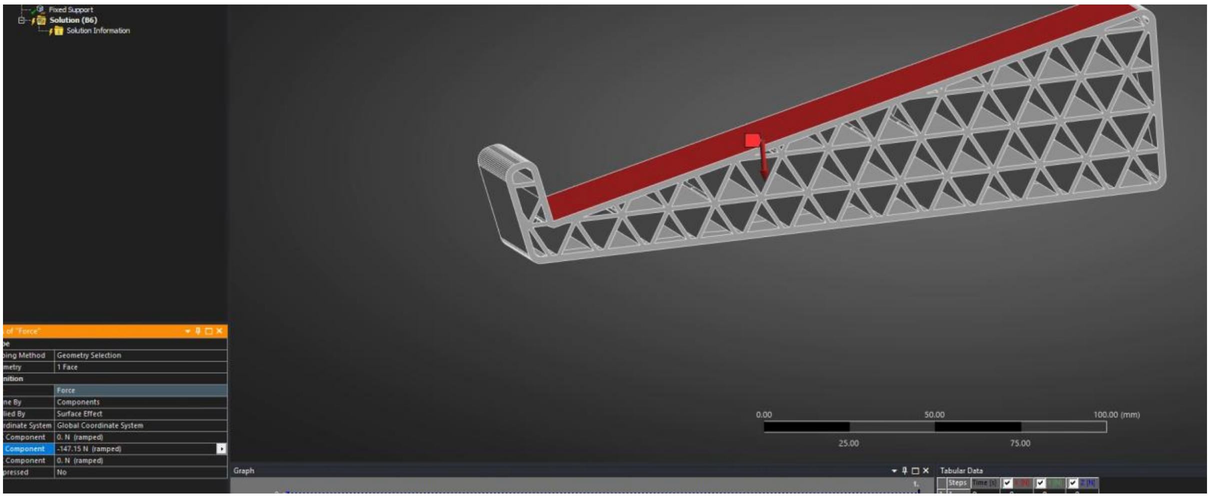


Figure 59 Force Face

Top platform selected for force. 15*9.81=147.15N of applied force. Applied in Y axis as a negative as laptop and user force will be applied downwards.

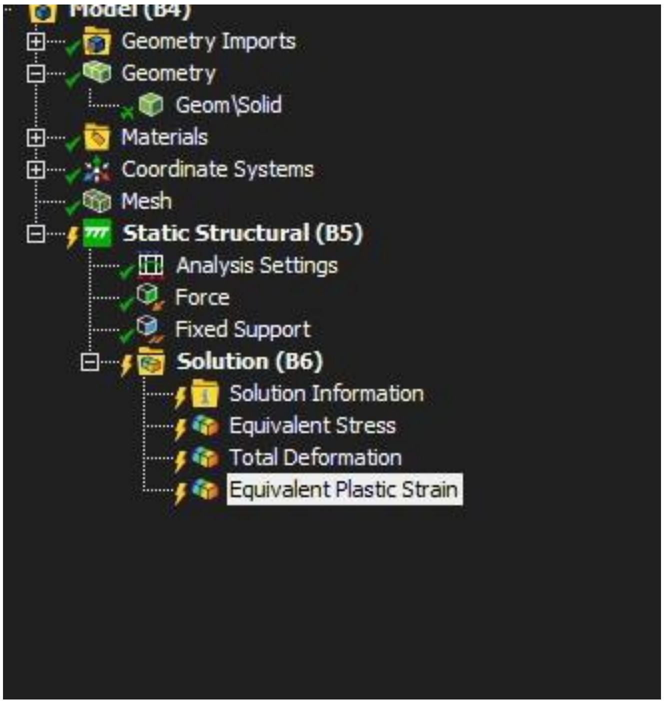


Figure 60 Solution Properties

Equivalent stress, total deformation and plastic strain added in solution. Will show where stress is highest in this geometry, how much it deforms and if any deformation is plastic.

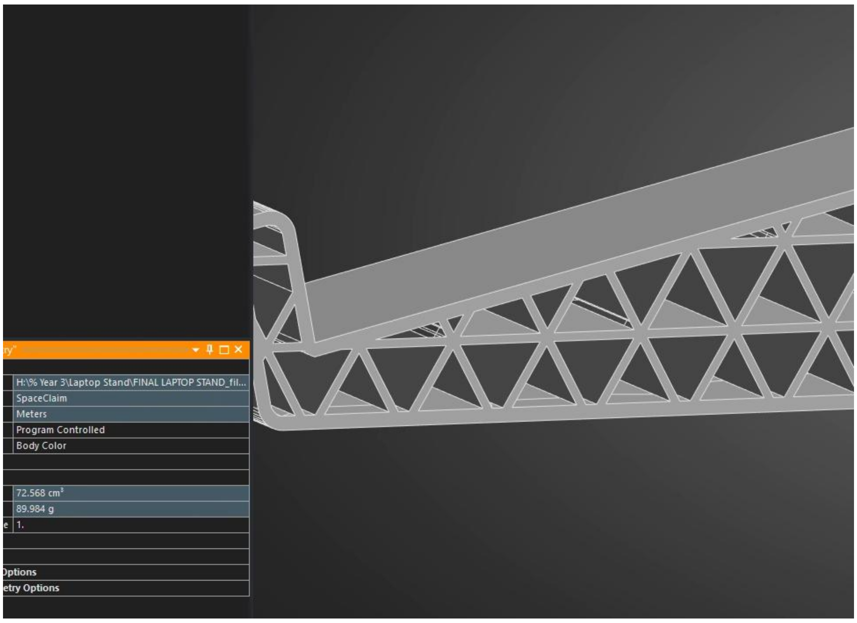


Figure 61 Mass Evidence

Mass of 89.98g.

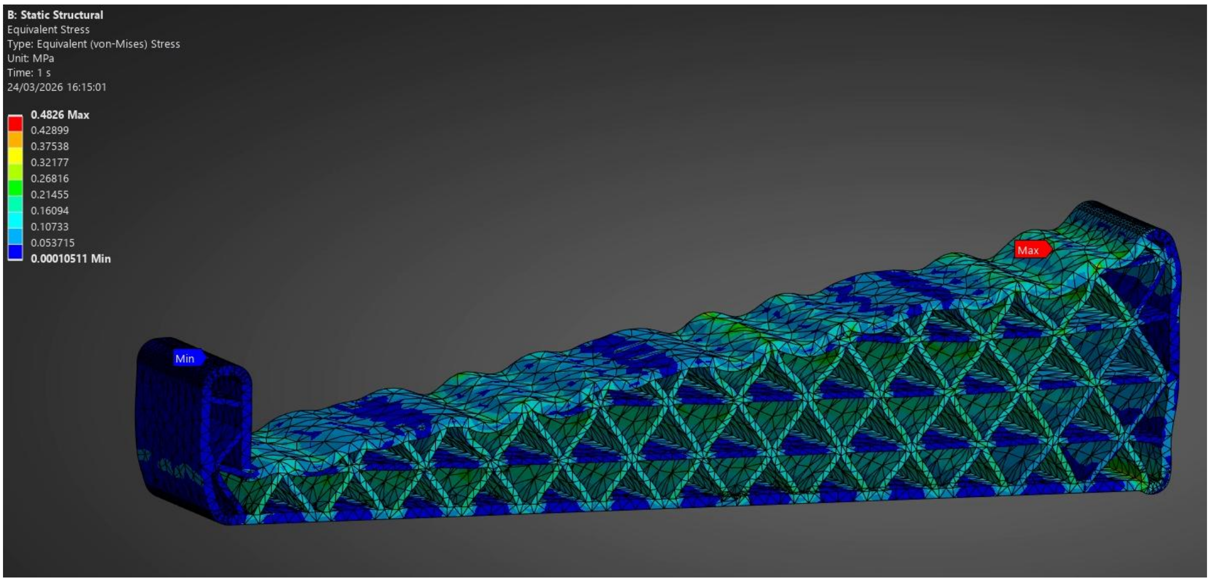


Figure 62 Equivalent Stress Results

Maximum stress = 0.5 MPa. Yield strength = 65MPa. This is a safety factor of 65/0.5=130. This is excessive safety factor, and mass can be reduced to improve cost, 3D print time and sustainability.

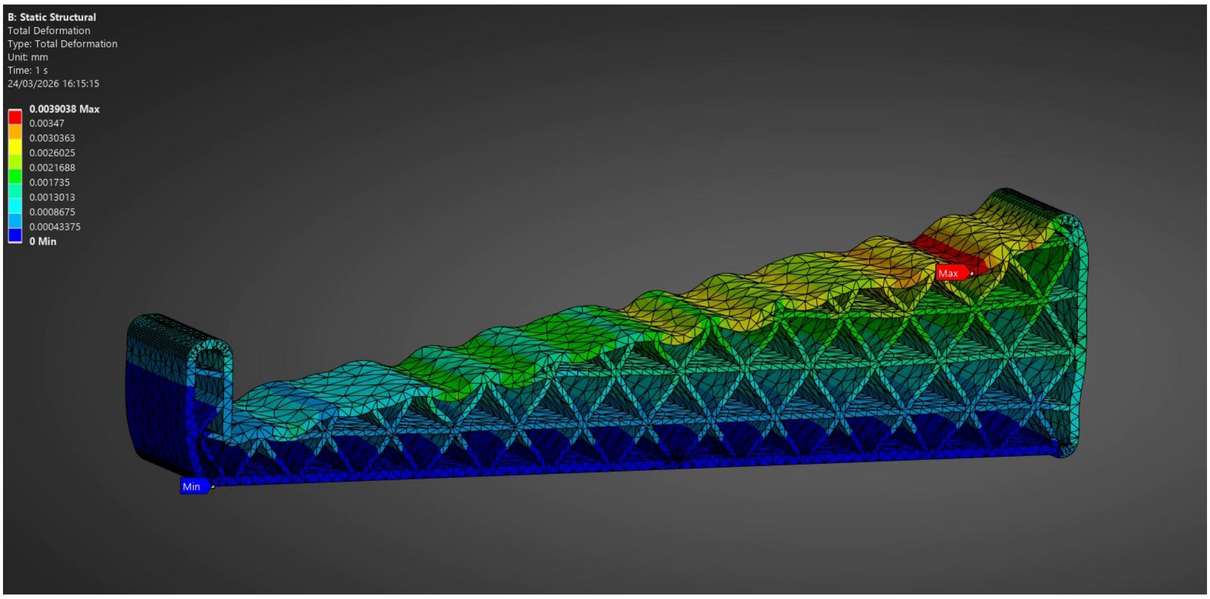


Figure 63 Total Deformation Results

Deformation is minimal at 0.0039mm, this will not cause any “squishiness” when in use, will seem high quality and durable by user.

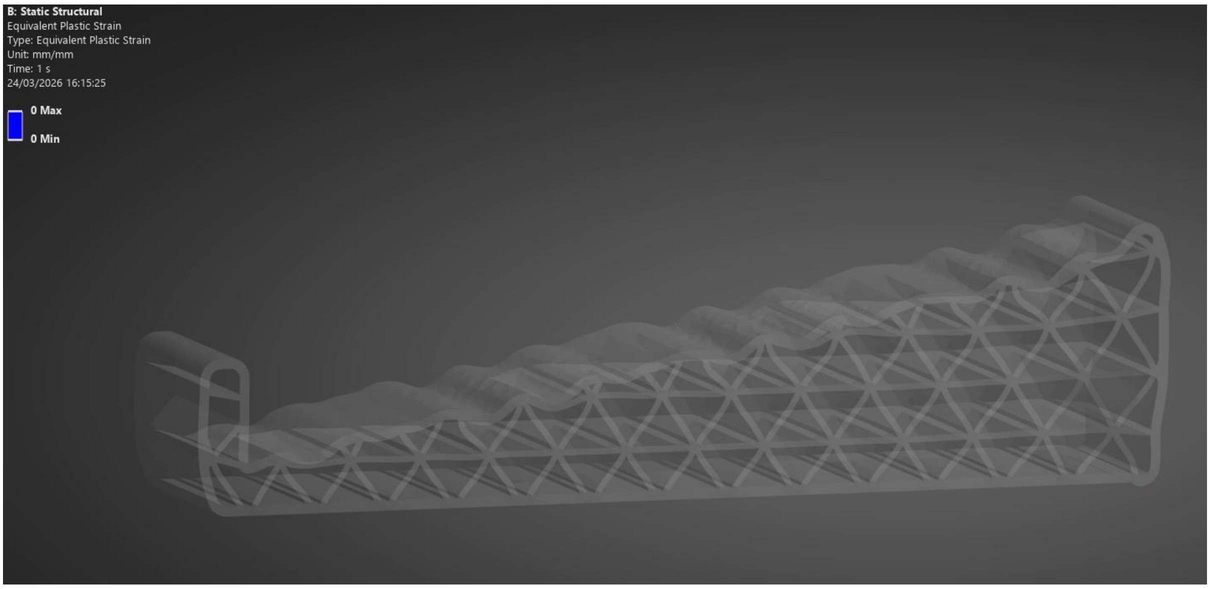


Figure 64 Plastic Deformation Results

No plastic deformation.

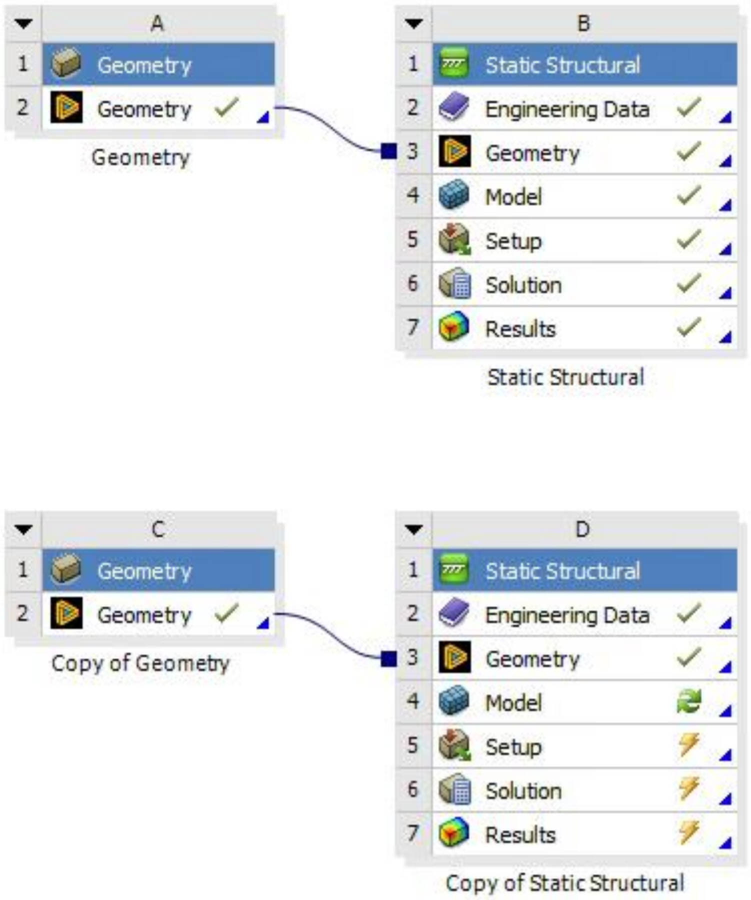


Figure 65 Second Iteration Started in Workbench

Duplicated geometry and static structural for second iteration. Since 15kg is very low even for just one of the two laptop stands that would be used; The target is changed to 45kg with a safety factor of >2, a total of 90kg if both sides are printed and used. This should be possible even with less material than the first iteration.

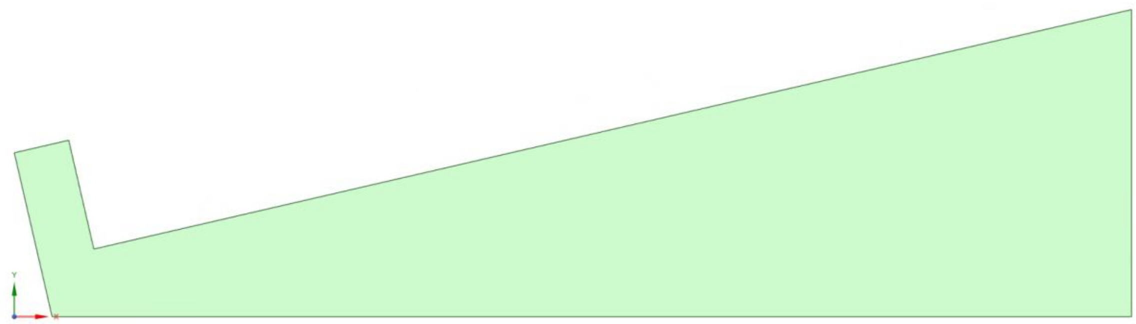


Figure 66 Restart

starting from the beginning this time is fine due to the limited complexity of the first iteration. However, since the geometry was not saved at different steps of the design and only when it was finished, and ANSYS Spaceclaim does not provide saved data after quitting the application. It would be optimal to make saves at different points of the geometry for any possible next iteration.

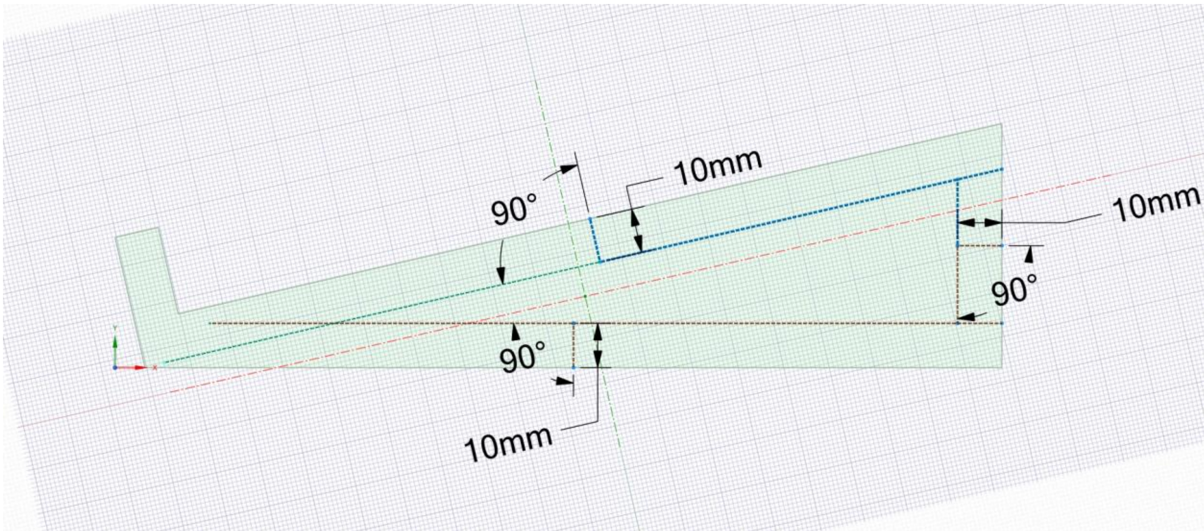


Figure 67 Cutout in Centre to Reduce Mass

construction lines for triangle cut out. 10mm from edge to provide 8mm of infill around the edges since shell thickness will change to 1mm. shell thickness will be reduced since the previous iteration safety factor was excessively high.

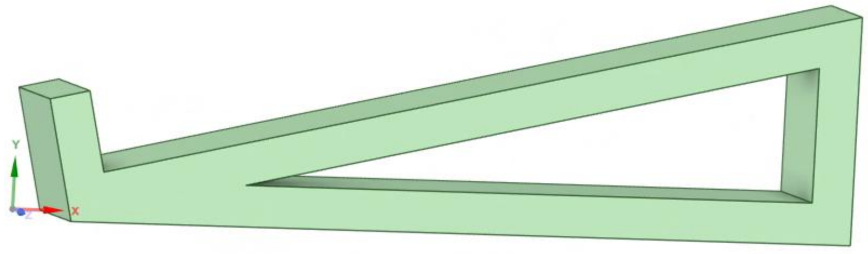


Figure 68 Pull Tool

Pull tool to remove material.

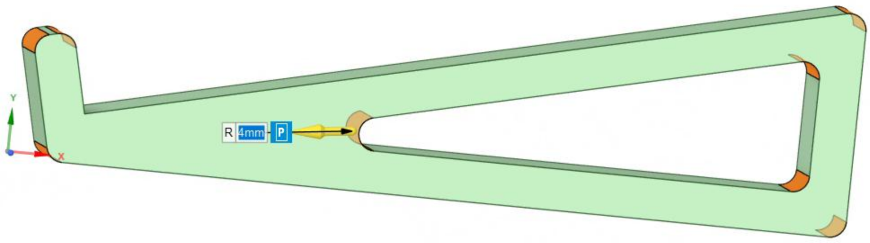


Figure 69 Equal Rounding Everywhere for Simplicity & Easier Computation

pull tool rounding 4mm. aesthetically pleasing and sharp points removed for safety.

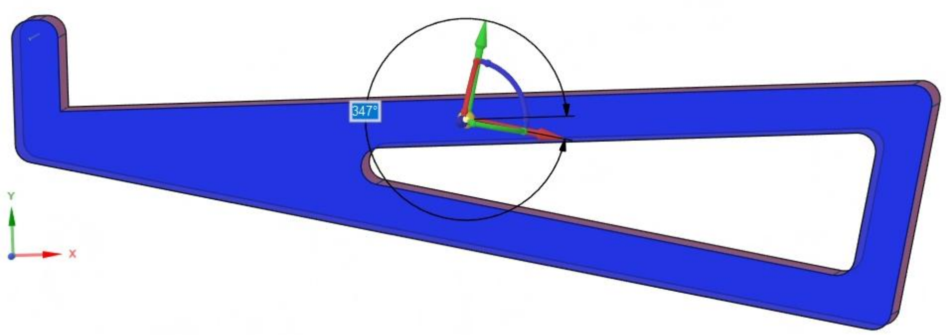


Figure 70 Rotation Using Move Tool

Geometry rotated so that top platform is inline with x axis; the infill triangles will potentially be in a better orientation to provide strength against the force being applied since the surface is angled compared to the ground.

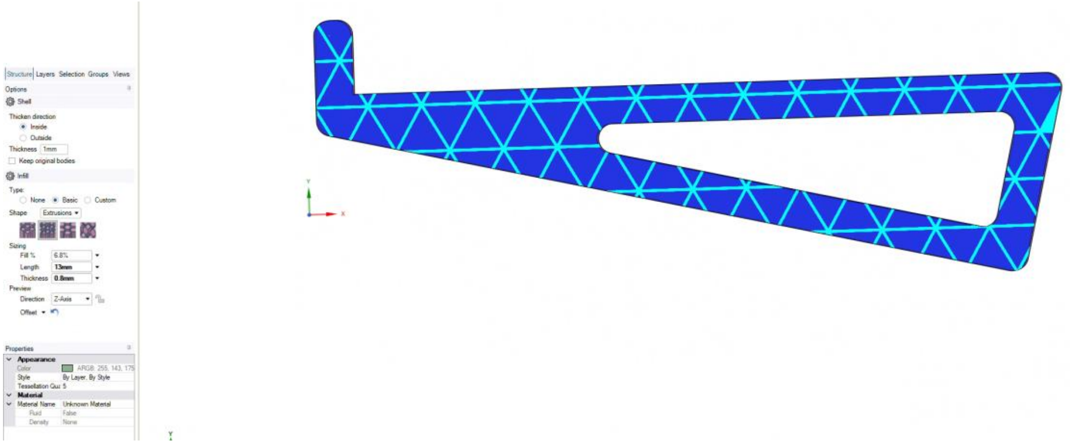


Figure 71 Triangle Infill, Fill % Reduce From Previous Iteration

Fill 6.8%. shell thickness 1mm. Infill length 13mm, thickness 0.8mm.

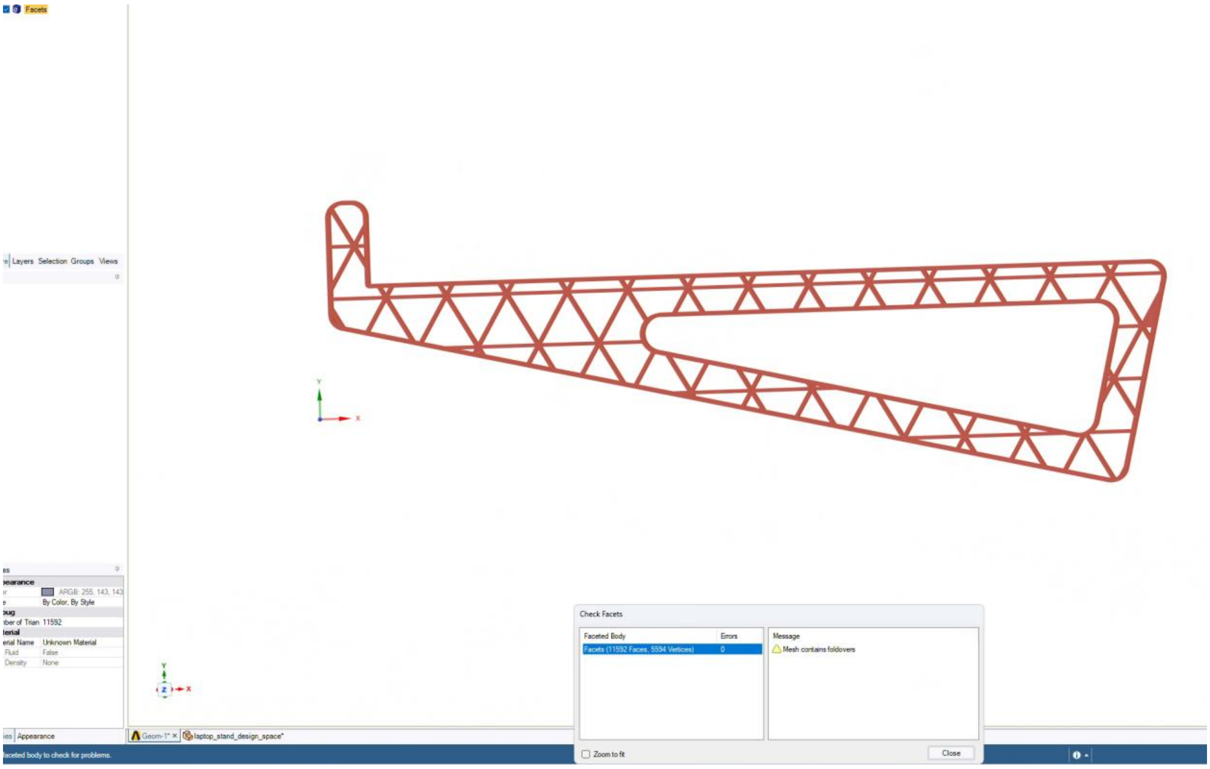


Figure 72 Check Facets, Errors

Errors show up. Mesh contains fold overs.

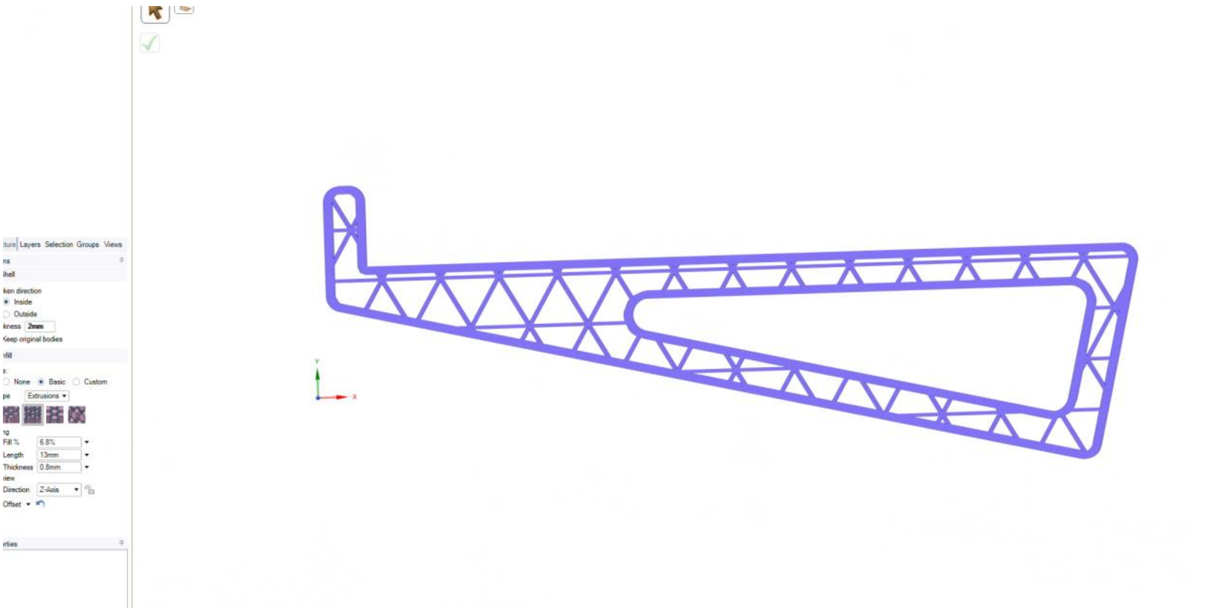


Figure 73 Errors Prevented

Changing shell thickness to 2mm fixes the issue.

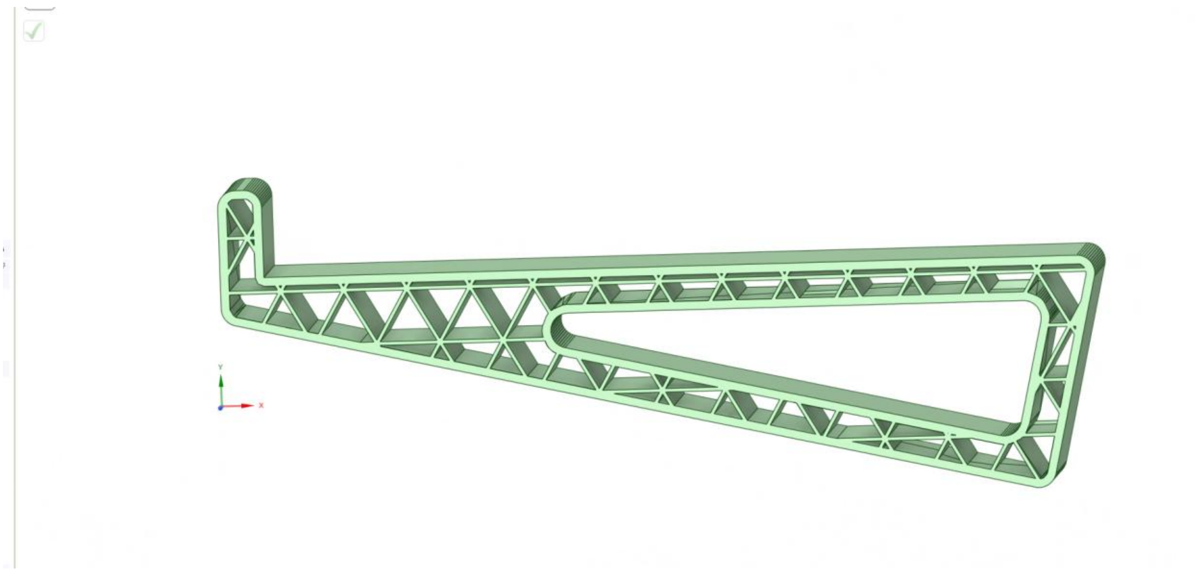


Figure 74 Check Facets, Converted to Solid

Checked for any missing faces. None found, converted to solid and ready for FEA testing.

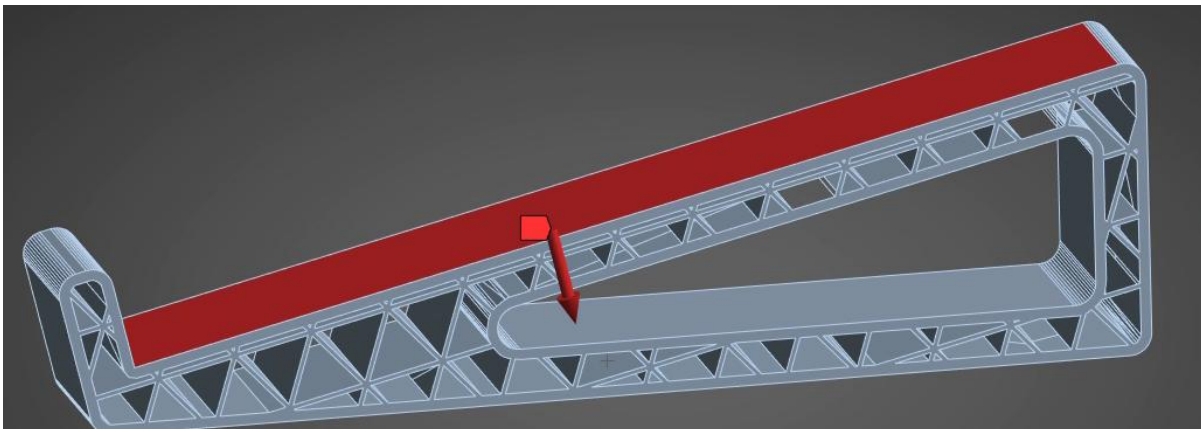


Figure 75 Force Direction Wrong

Mesh settings remain same. PLA Filament selected as material. Only change is force is increased to 45*9.81=441.45N. therefore, -441.45N to be in negative y direction. Problem: orientation was not reverted, so axis is not accurate to real world use.

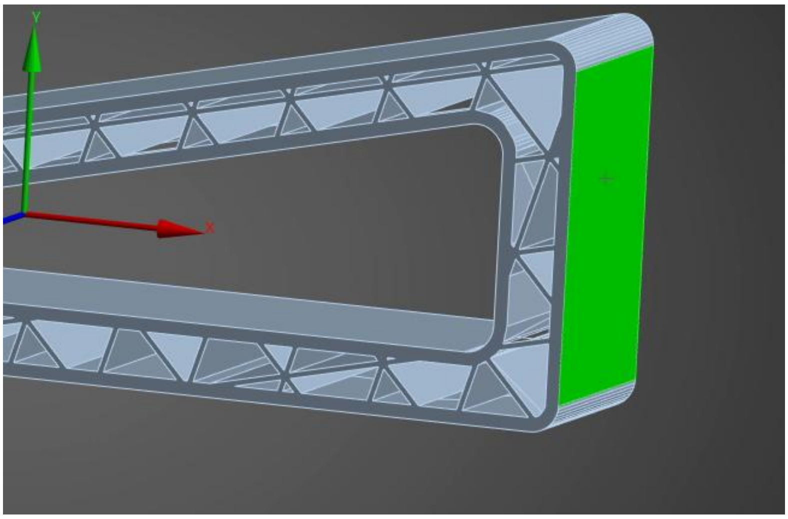


Figure 76 New Coordinate System Applied

inserted new coordinate system. Changed x axis direction to “geometry selection” and selected highlighted side.

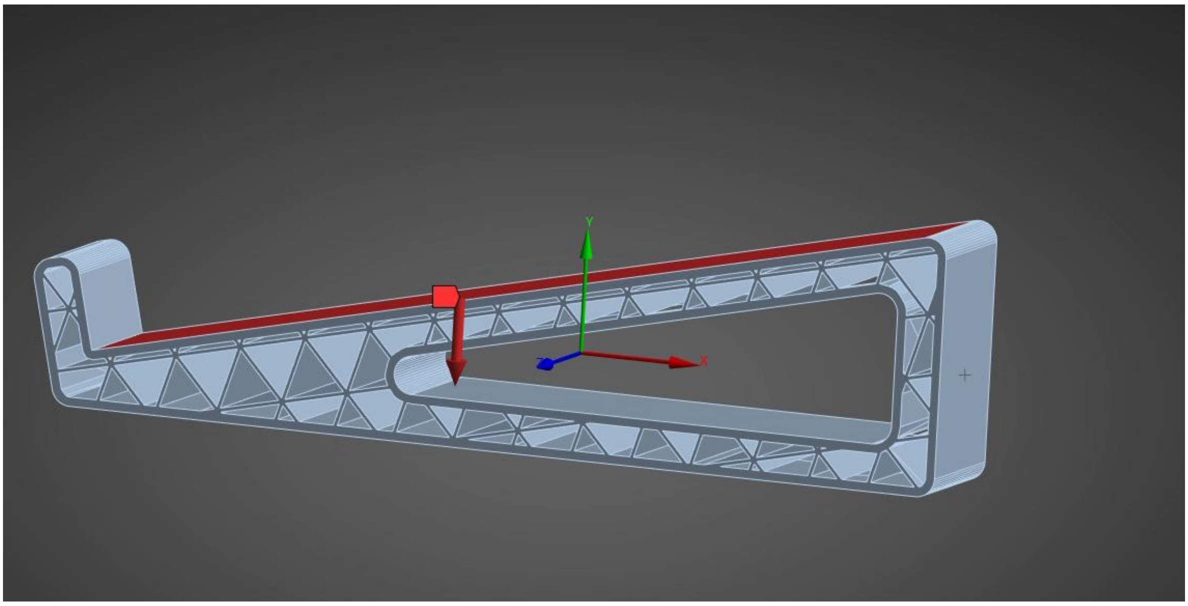


Figure 77 Force Direction Corrected

Selected this new coordinate system in force, direction of force is now correct.

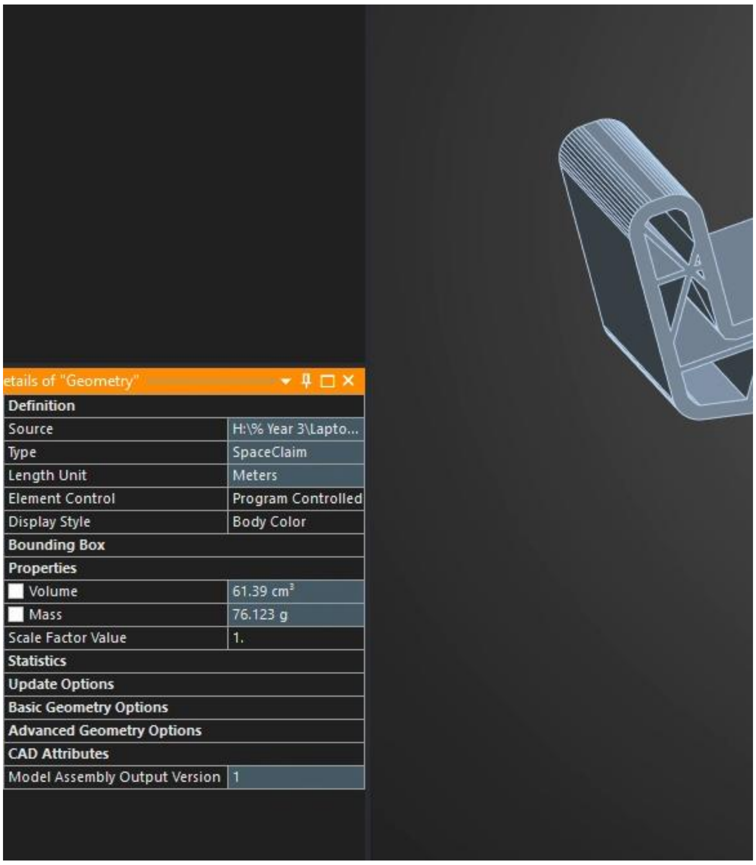


Figure 78 Mass Lowered

Mass at 76.12g

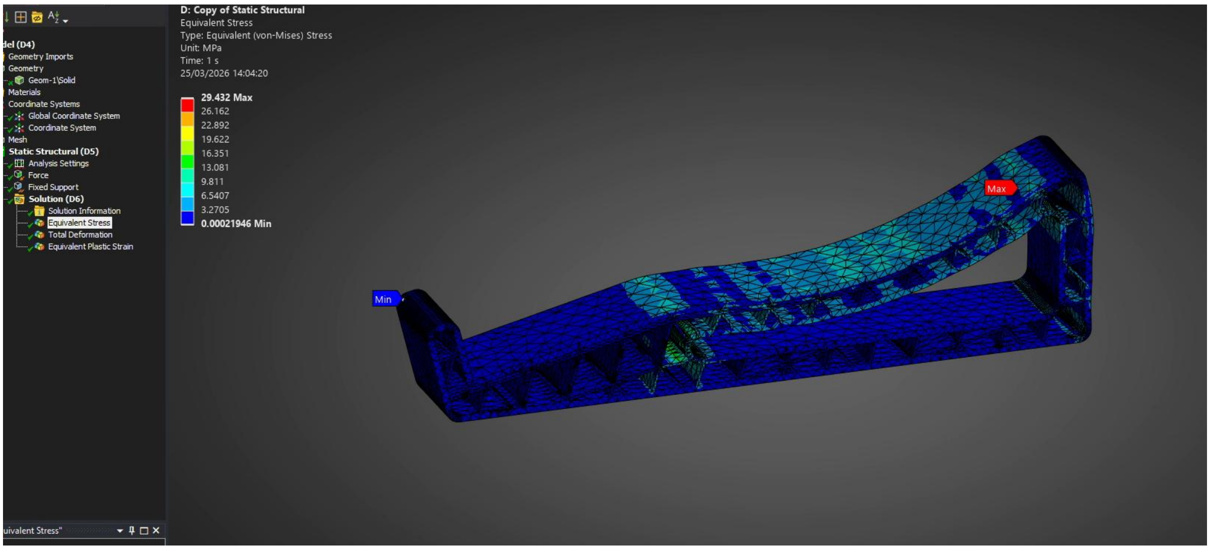


Figure 79 Equivalent Stress Pass

Maximum stress at 29.43 MPa. 65/29.43>2. Therefore, the safety factor of above 2 has been met.

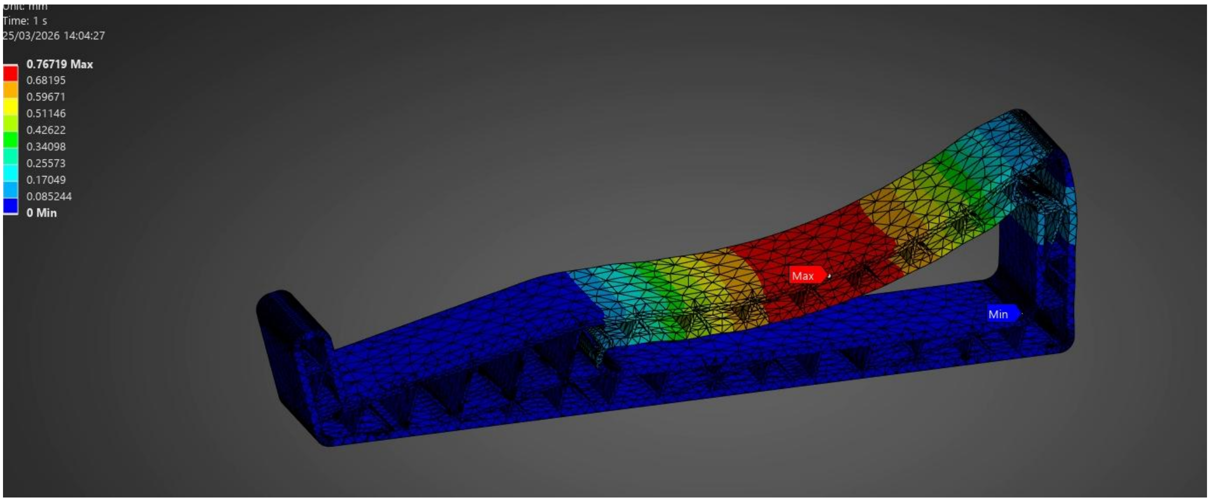


Figure 80 Total Deformation Pass

Theoretical max deformation of 0.767mm. greatly increased but must also factor in the increased force applied.

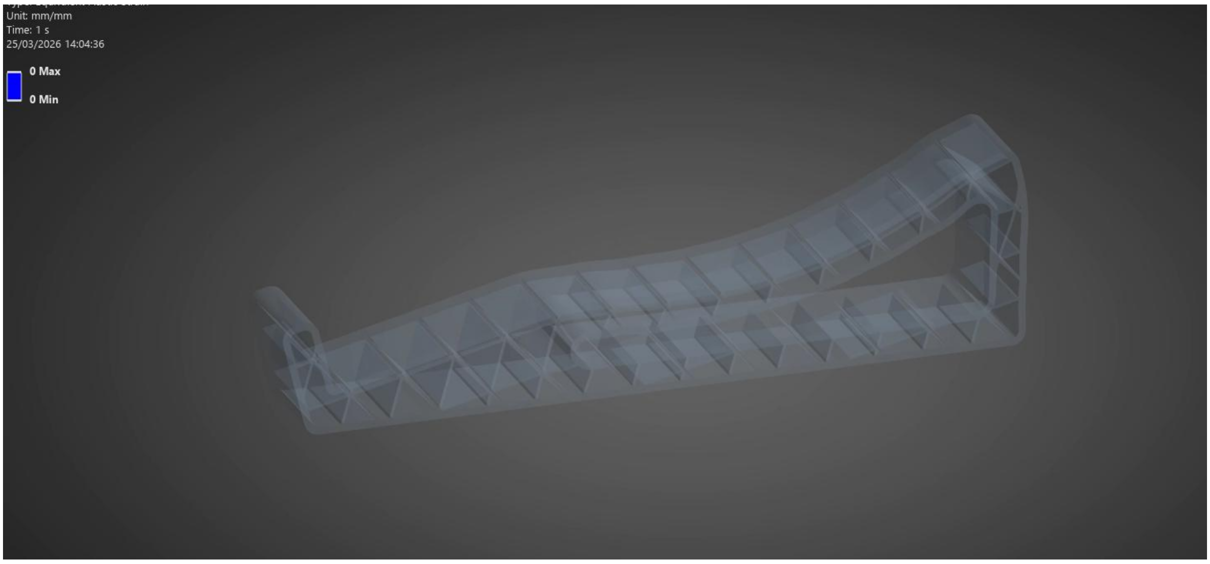


Figure 81 Plastic Deformation Pass

Still no plastic deformation, which means it is theoretically safe against fatigue.

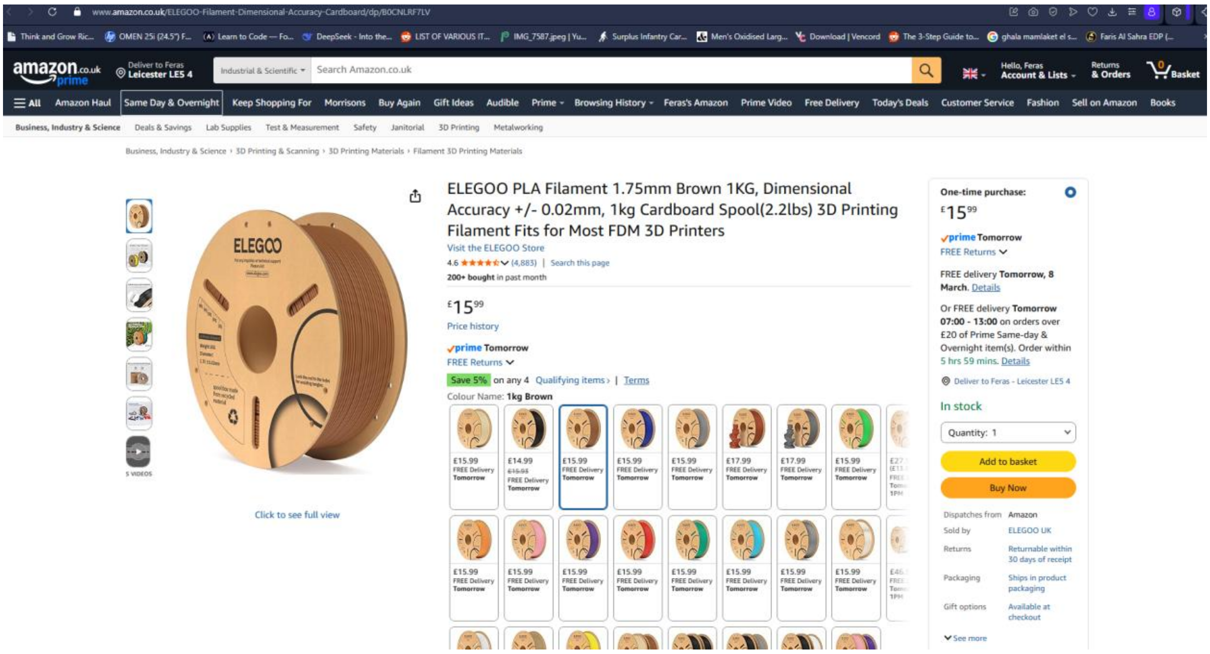


Figure 82 PLA Filament Commercial Price

1kg of PLA costs about £16 on amazon. 0.08998 * 16 = £1.44, 0.07612 * 16 = £1.22. Savings of £0.22 every print. `

ii. 3D Printing preparation: orientation, support structures, estimated filament usage and printing time.

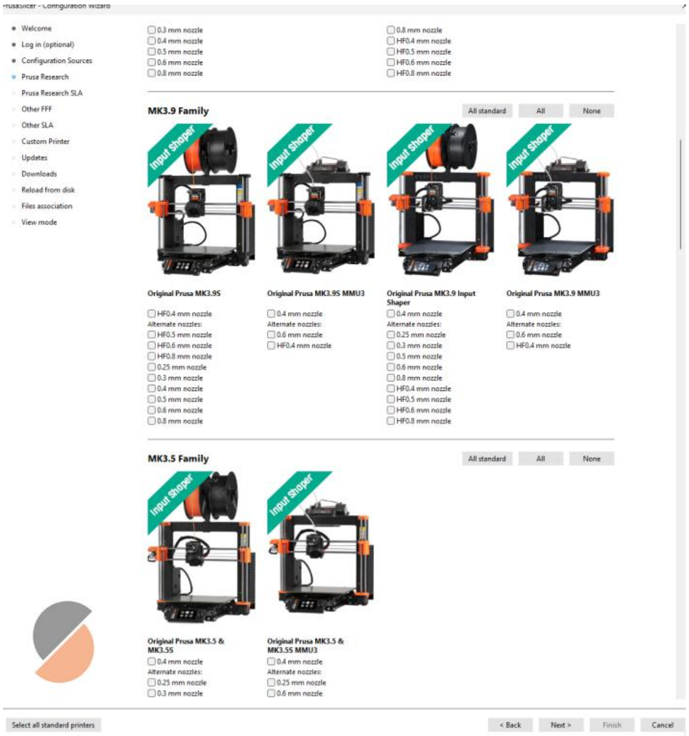


Figure 83 Prusa Slicer Machines Selection

One was not picked, but if the printing was taking place, the printer that would be used would be picked in this menu.

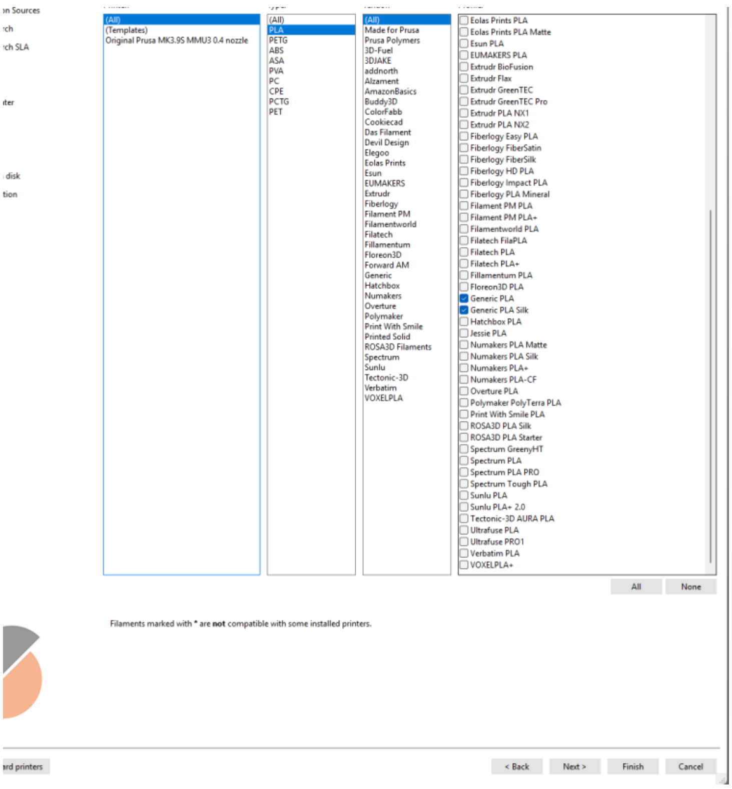


Figure 84 Generic PLA filament Selected

iii. Validity and suitability of design, improvements or refinements that could enhance the design.

Comments

Both sides of laptop stand together can handle 90kg load. This is extremely strong especially since the average laptop is lower than 5kg in weight. Only costs £2.44 in material to produce both sides of the laptop stand. Which is equivalent to 36.9kg/£.

The infill allows for very minimal support structure as it can be printed in its optimal orientation to reduce support requirements in the same orientation as the rest of the design in its optimal orientation. This reduces material use therefore, cost and material waste, improving sustainability.

The mass of the stand is 76g with 45kg strength; equivalent to 1kg of strength per 1.7g. The design is aesthetic. Since it has such high strength, it is likely to last long, a lifetime which makes it extremely sustainable since it will not break and be thrown away. Although PLA Filament is recyclable, it requires specialised industrial equipment to be recycled.

Improvements

The laptop stand requires two separate parts to be in use; this makes it awkward for the user to utilise. The user will have to set the laptop stands up and then carefully place their laptop on top. To prevent this issue, a third piece can be printed to simply connect the separate sides of the laptop stand so that the user can have a smoother experience. This would allow the user to simply place one whole laptop stand on the table rather than preparing two pieces, at the right distance to one another.

Being able to handle 90kg is quite high, a smaller amount of strength may be adequate which could save material, improving sustainability and price.

Section 3 - Topology Optimisation and Testing of Bracket

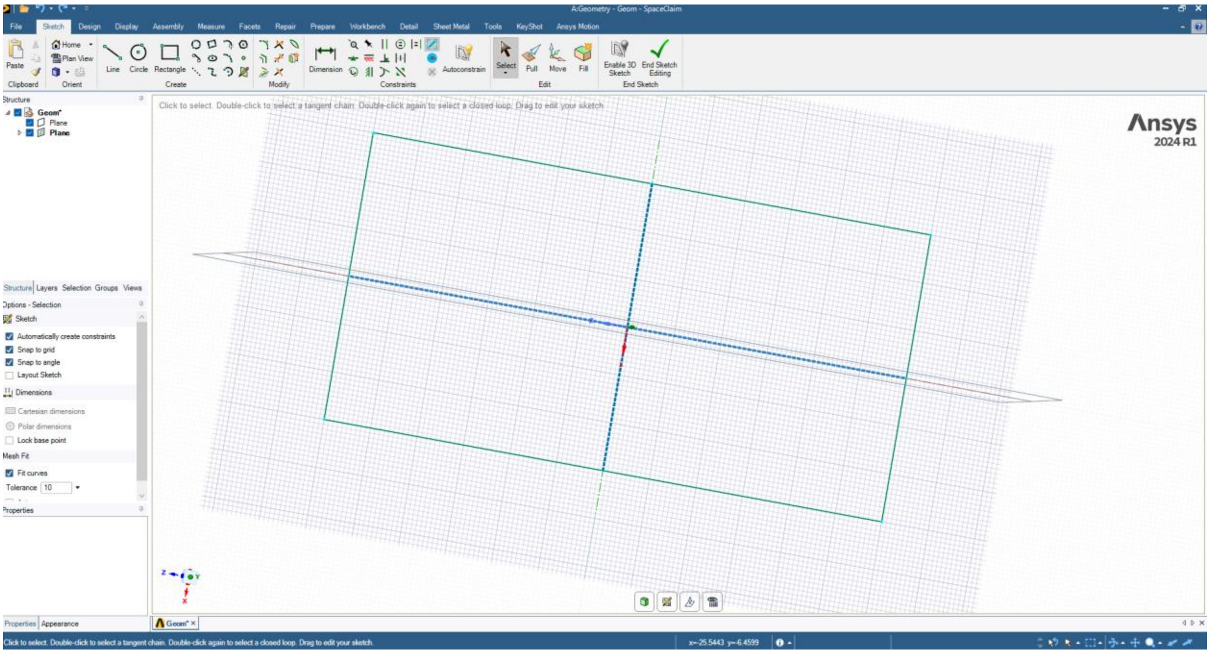


Figure 85 Initial Geometry Plan Drawing

Square drawn centred in axis. 120mm length 60mm.

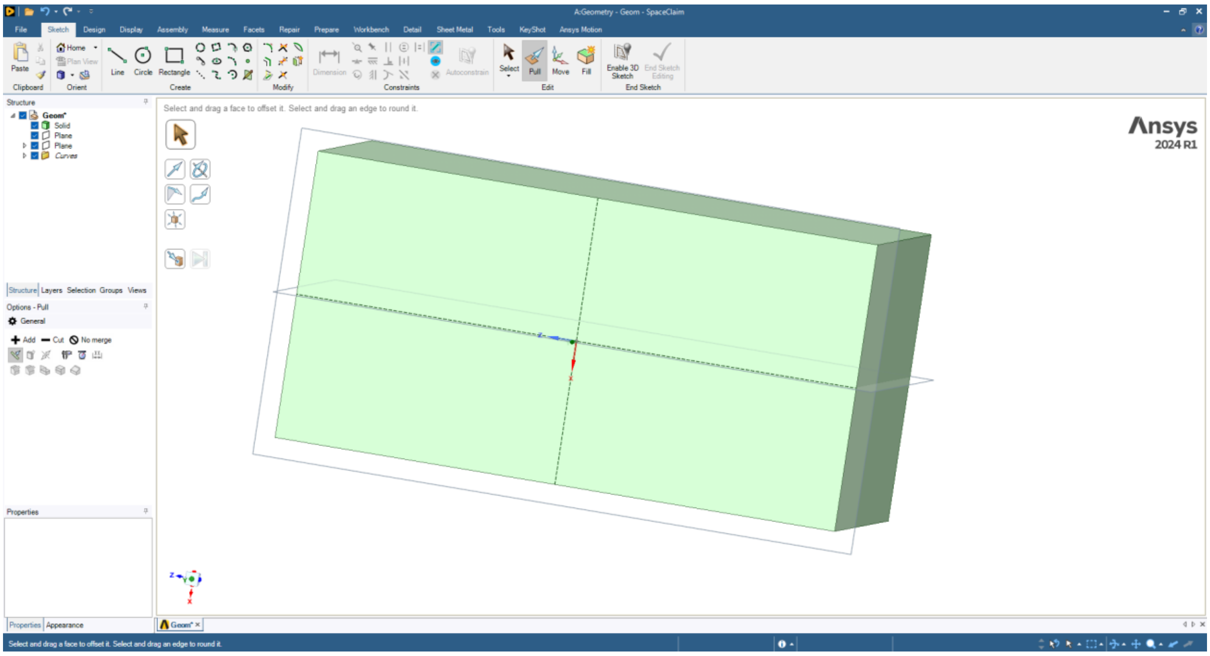


Figure 86 Pull Tool

Pull by 50mm.

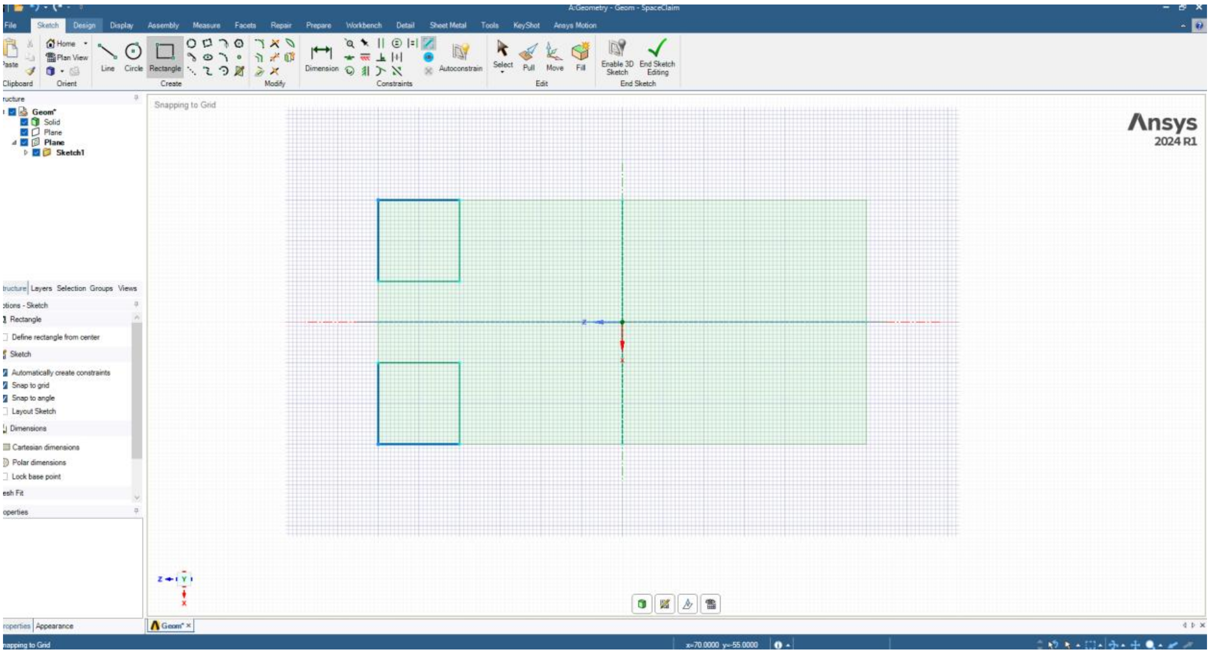


Figure 87 Cutouts Drawn

20mm square tool.

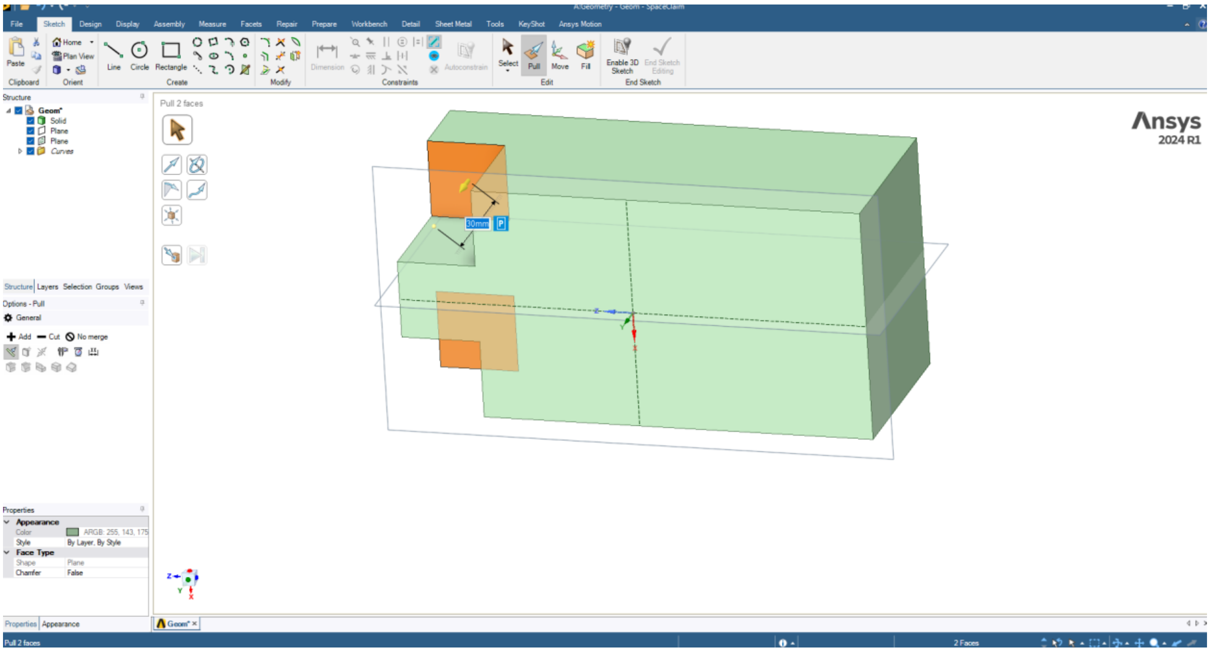


Figure 88 Pull Tool to Remove Geometry

Pull tool 30mm.

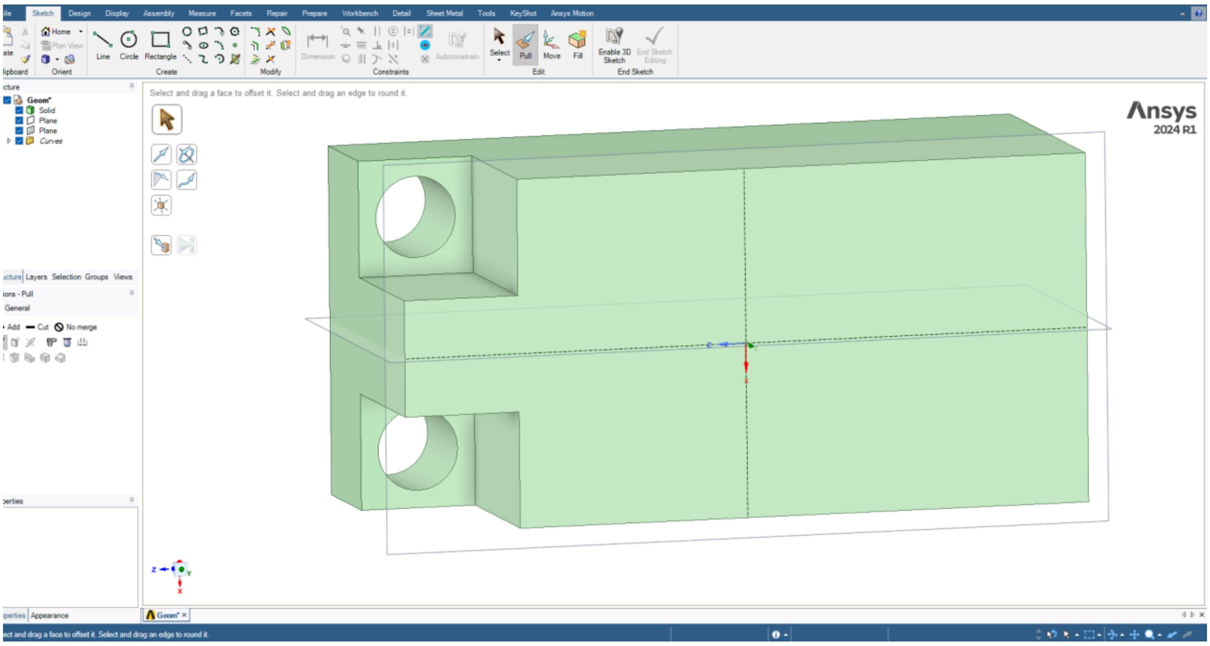


Figure 89 Holes Added

14 mm radius at centre of circle and pull tool.

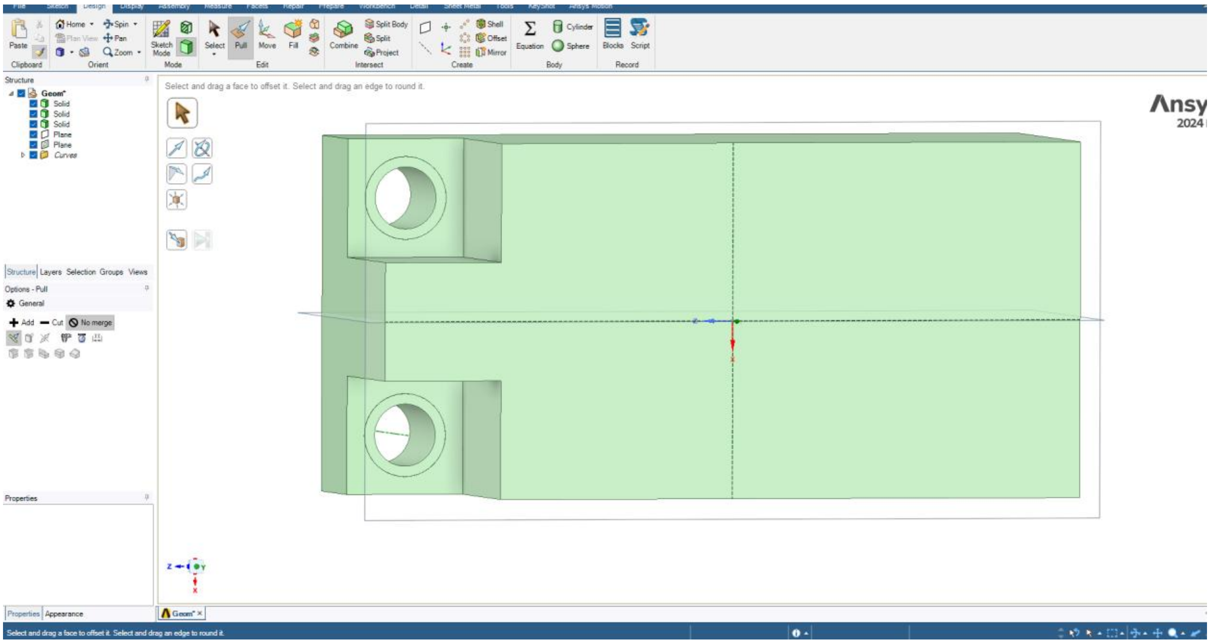


Figure 90 Hole Geometry Added as Separate Part of Geometry for Future Selection

14mm + 10.5mm circle pull tool 20mm. Critical to use “No Merge” while using pull tool to allow for separate body for later selection of holes. 10.5mm is 0.5mm larger than specification; tolerance for m10 screw to fit.

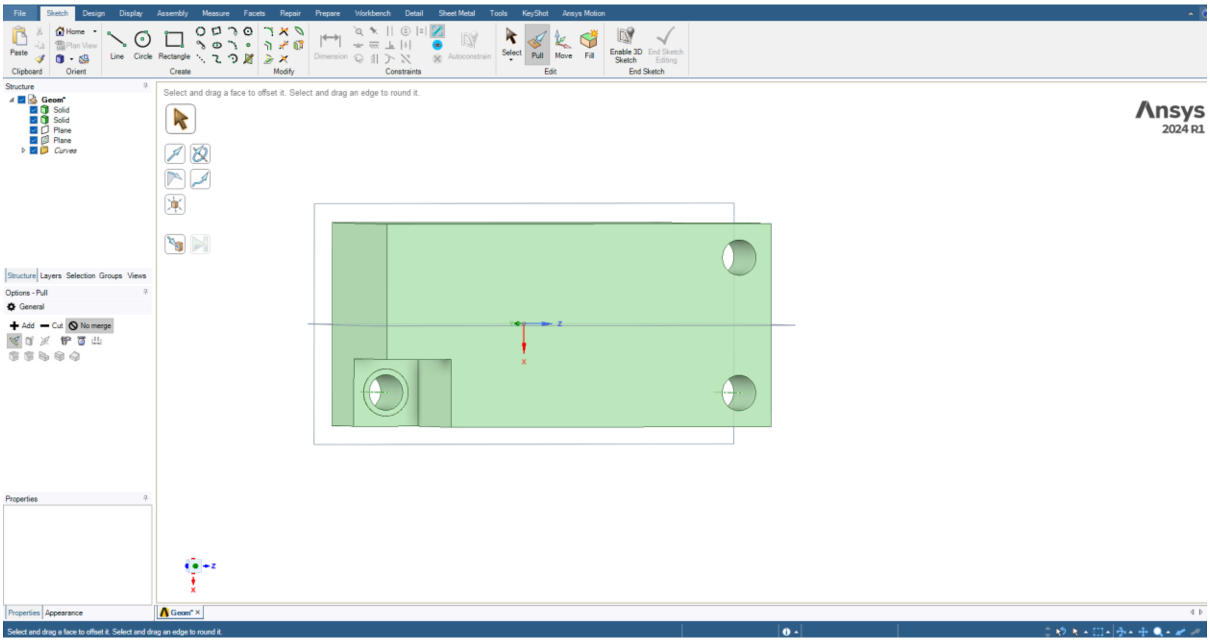


Figure 91 Final Hole

repeated for final hole, “no merge” was forgotten hence, separate hole geometry from previous figure removed.

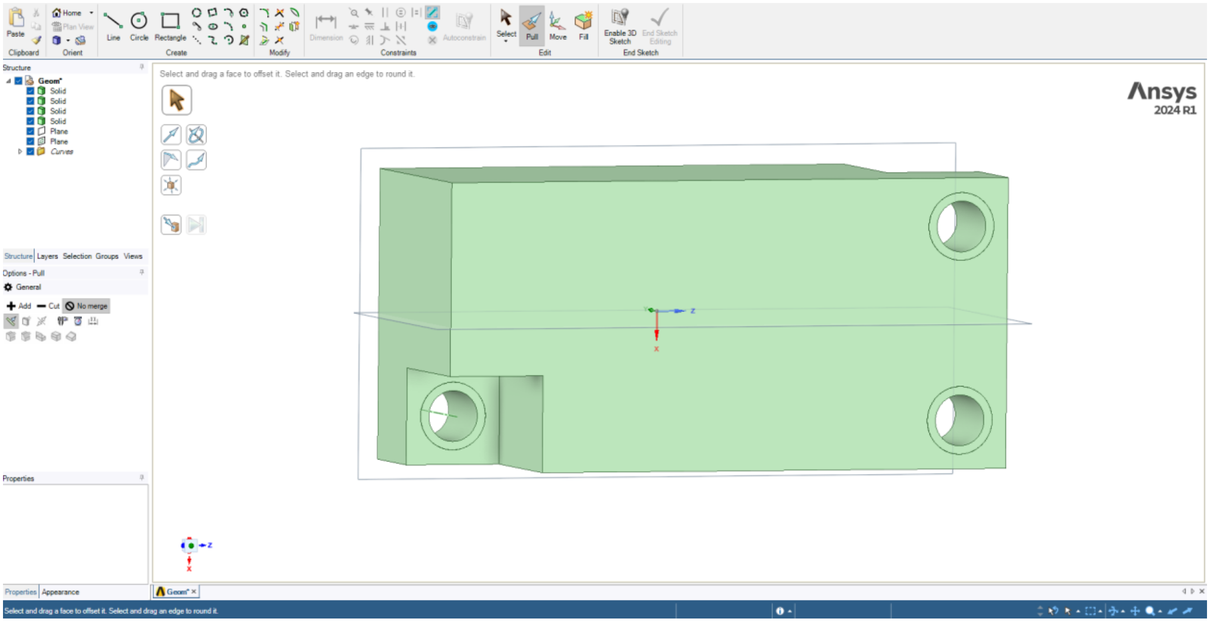


Figure 92 Final Hole Repeated to Maintain Separated Hole Geometry

Failed to use “No Merge”, every step must be pulled with “No Merge” enabled otherwise past separate bodies will be removed; re-did last hole.

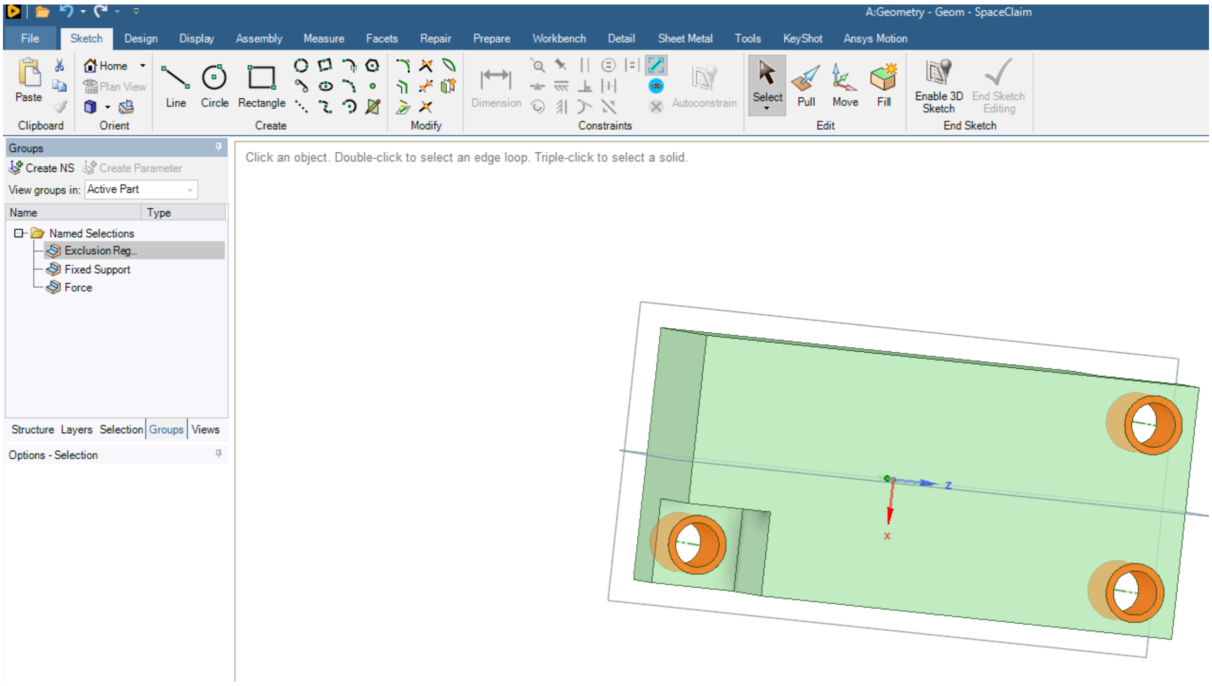


Figure 93 Holes Groups Created

3 named selections. All hole's group for exclusion region when optimising. 2 holes' surface on right for fixed support. Left hole for downwards force.

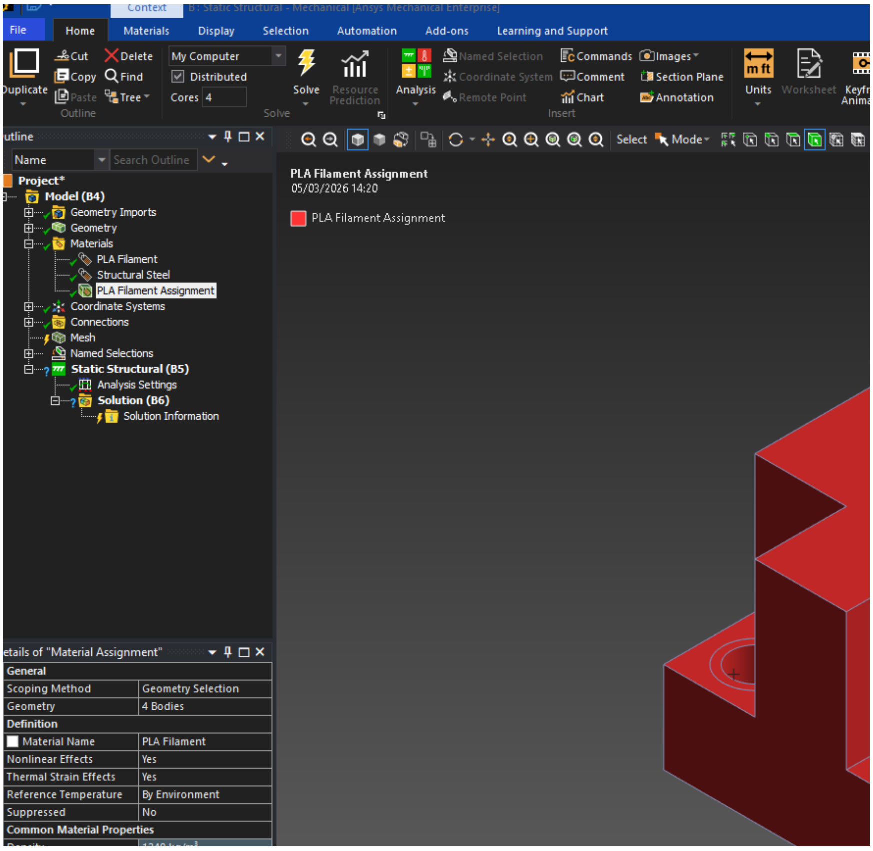


Figure 94 Material Selection

PLA Filament material specifications added. Material assigned to all bodies.

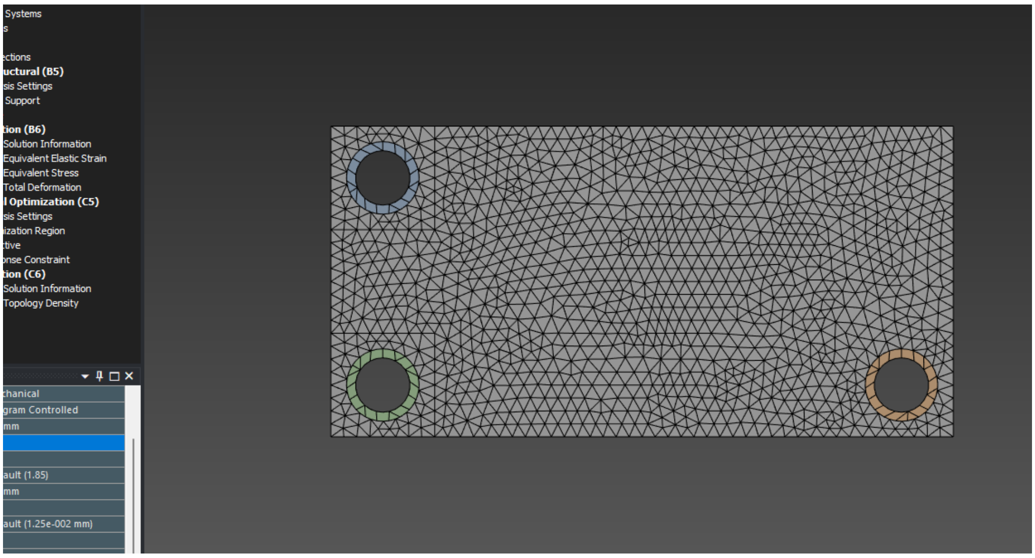


Figure 95 Mesh

Mesh size 2.5mm, max size 2.5mm. adaptive sizing off. To reduce computational power needed and reduce time calculating optimisation.

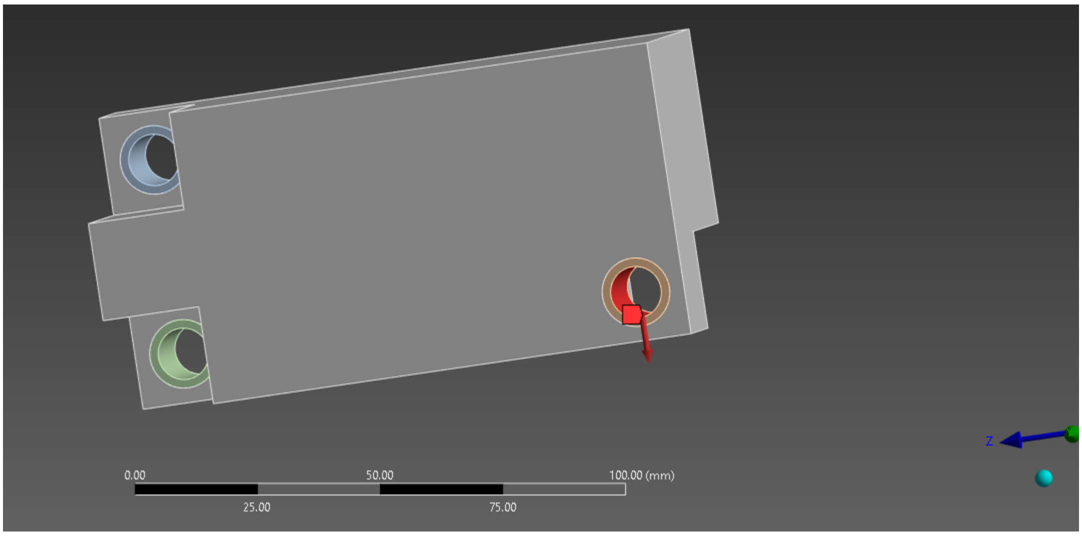


Figure 96 Force

Fixed support & force inserted. Fixed support named group selected. Force named group selected. Force = 30kg * 9.81 = 294.3N. X= positive 294.3N.

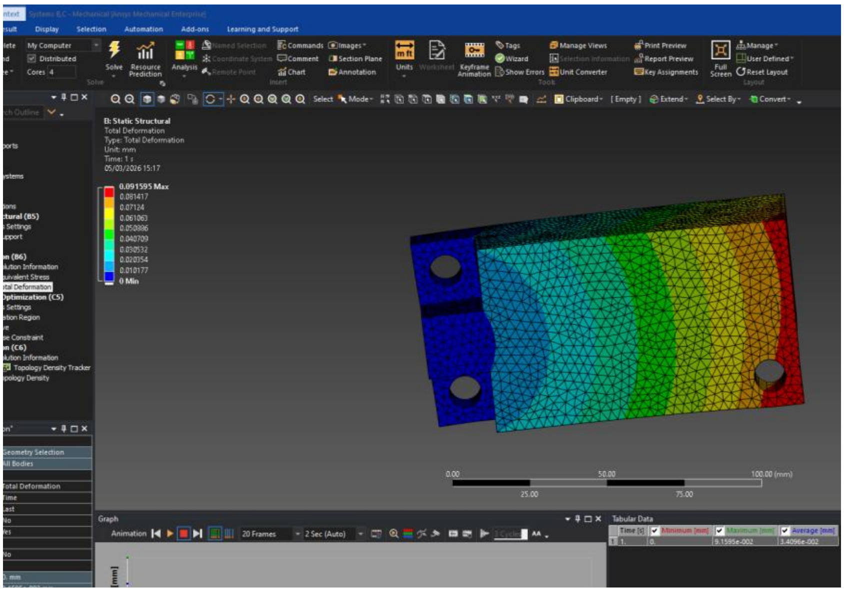


Figure 97 Solution Mode Applied & Solved for Topology Optimisation

Added equivalent stress and total deformation and solved. Deformation is very low: 0.09mm.

Computation took too long. Changed mesh size to 3mm max mesh 3.5mm.

Worksheet

Objective

Right click on the grid to add, modify and delete a row.

Enabled	Response Type	Goal	Criterion	Formulation	Environment Name	Weight	Multiple Sets	Start Step	End Step	Step	Start Mode	End Mode	Mode
☑	Compliance	Minimize	N/A	Program Controlled	Static Structural	N/A	Enabled	1	1	1	N/A	N/A	N/A

Figure 98 Topology Optimisation Settings

Objective of topology optimisation set to compliance. Compliance is a measure of elastic deformation. Minimising compliance means reducing any deformation of the material. However, this may not account for high stress local regions.

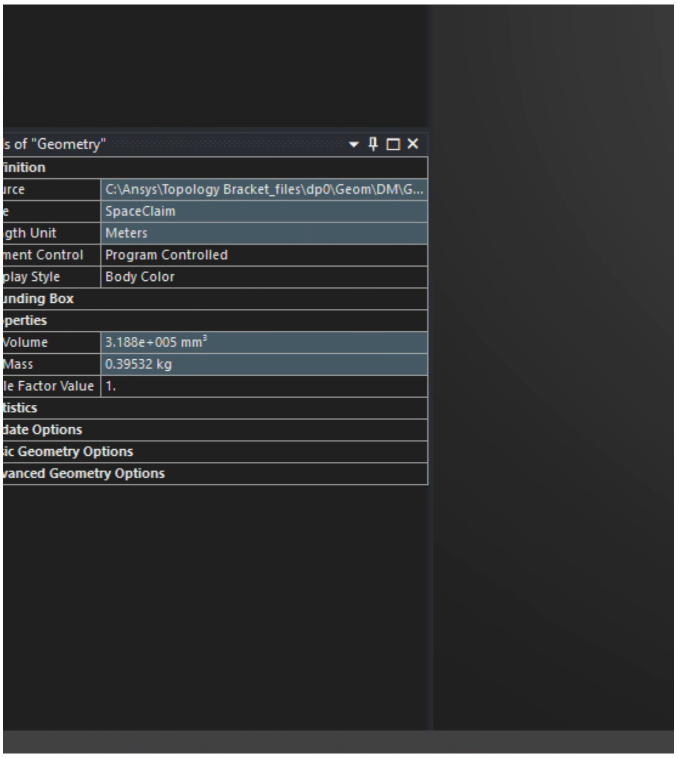


Figure 99 Unoptimized Mass

Current weight before optimisation is 395g.

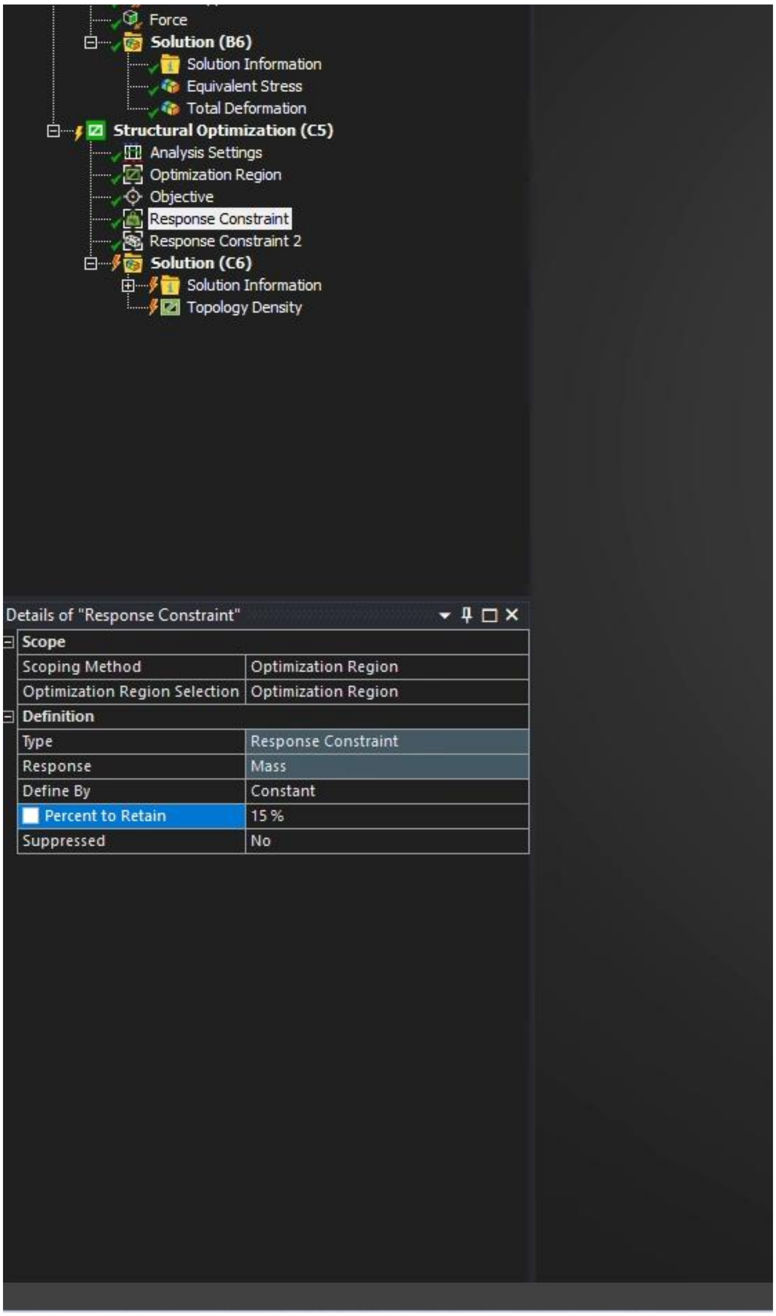


Figure 100 Response Constraint Settings

Added 20% Mass response constraint. Since target is below 60g and current weight is 395g; $\frac{60}{395} = 0.15$. However, for computational speed, 20% will be used and then geometry will be modified after.

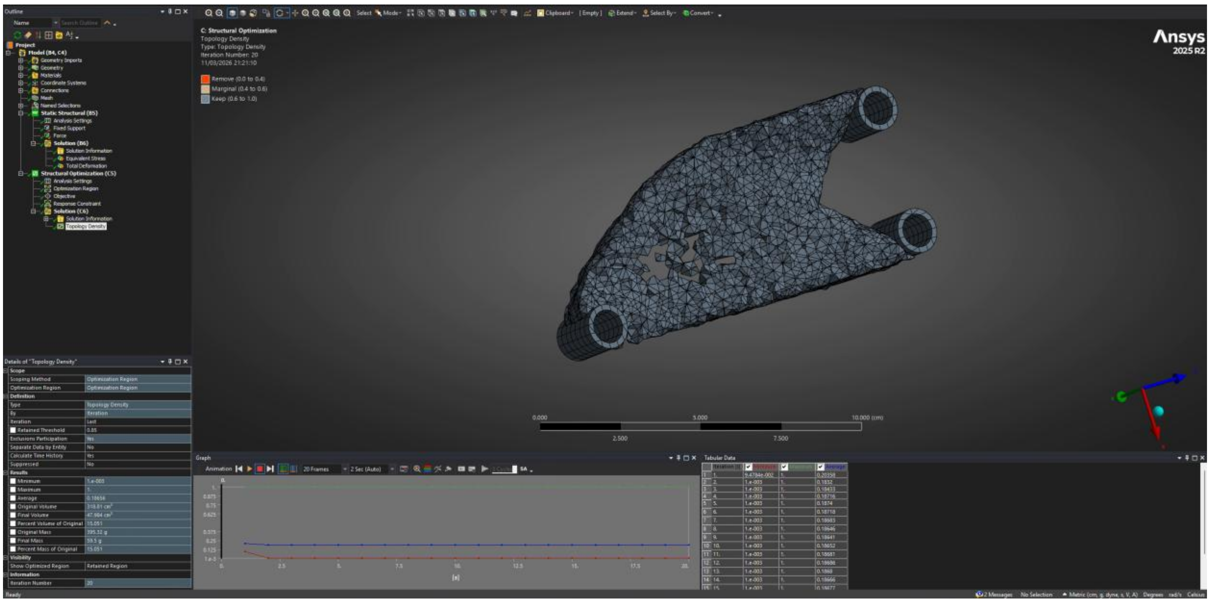
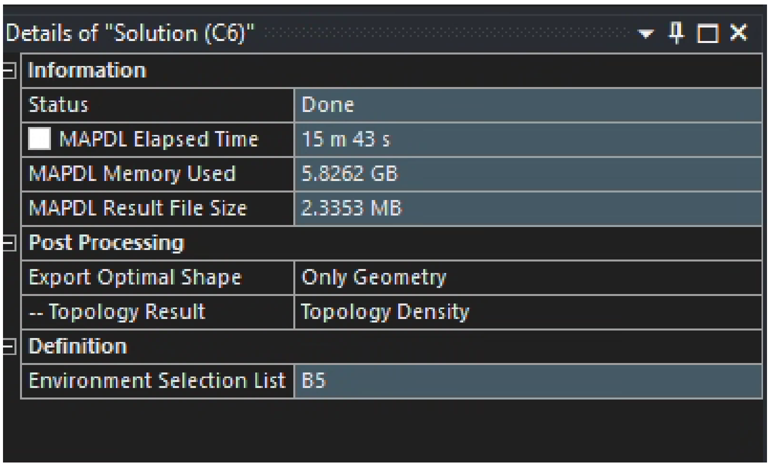


Figure 101 Topology Optimised & Successfully Within Specification

Optimisation completed. Final mass 59.5g which is within specification, achieved using "retained threshold" of 0.85. floating mass can be removed. Areas next to holes not connecting properly, some floating areas.



Details of "Solution (C6)"

Information	
Status	Done
MAPDL Elapsed Time	15 m 43 s
MAPDL Memory Used	5.8262 GB
MAPDL Result File Size	2.3353 MB
Post Processing	
Export Optimal Shape	Only Geometry
-- Topology Result	Topology Density
Definition	
Environment Selection List	B5

Figure 102 Computational Information

15 minutes and 43 seconds elapsed time using 4 cores. Specifications: 14700k, RTX 5070, 48gb DDR4 ram & Samsung SSD. This matters because depending on computer hardware and number of used cores time elapsed can be higher or lower. Design process affects these times and can be modified for improved time and computational efficiency. 15 minutes is reasonable for complex topology optimisation.

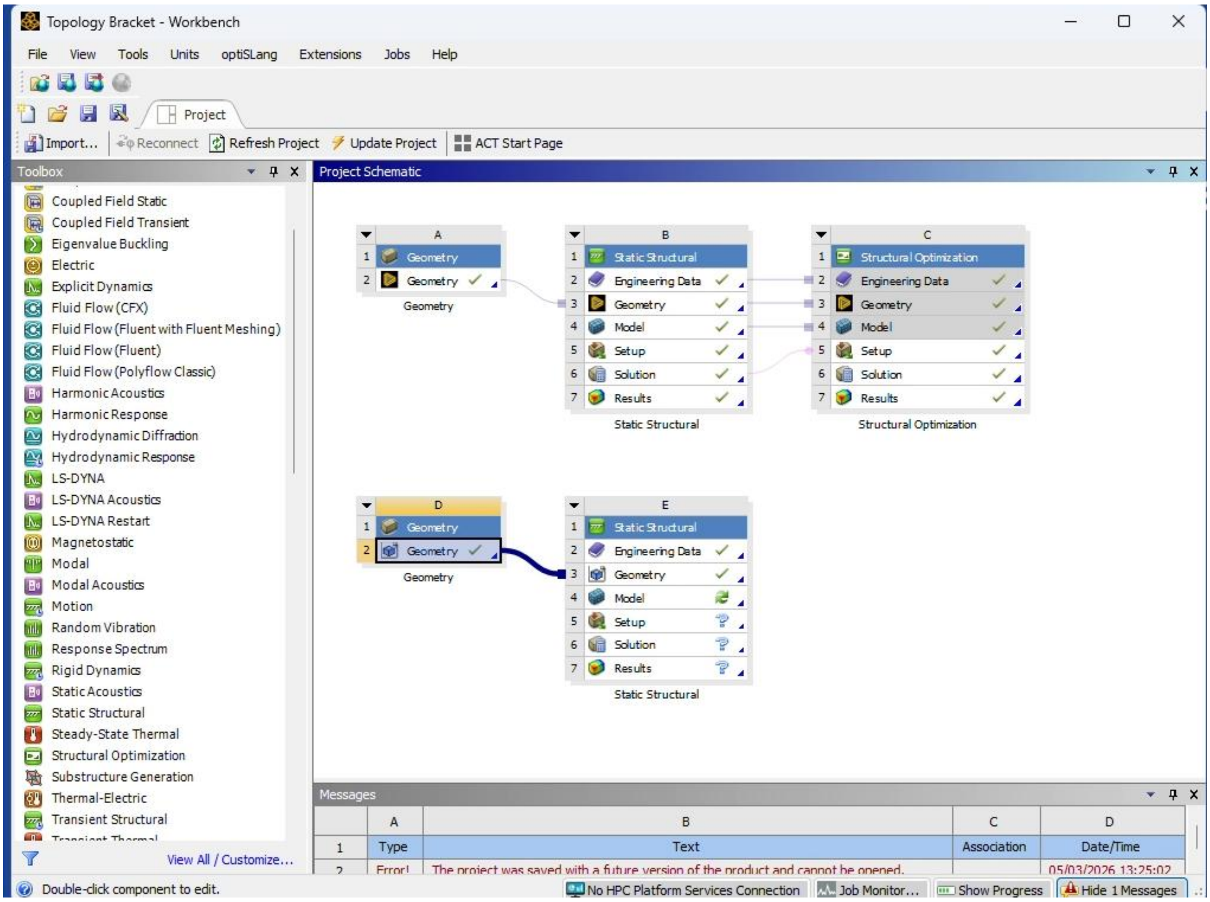


Figure 103 Post Processing Workbench Created

New geometry and structural analysis duplicated for post processing and retesting.

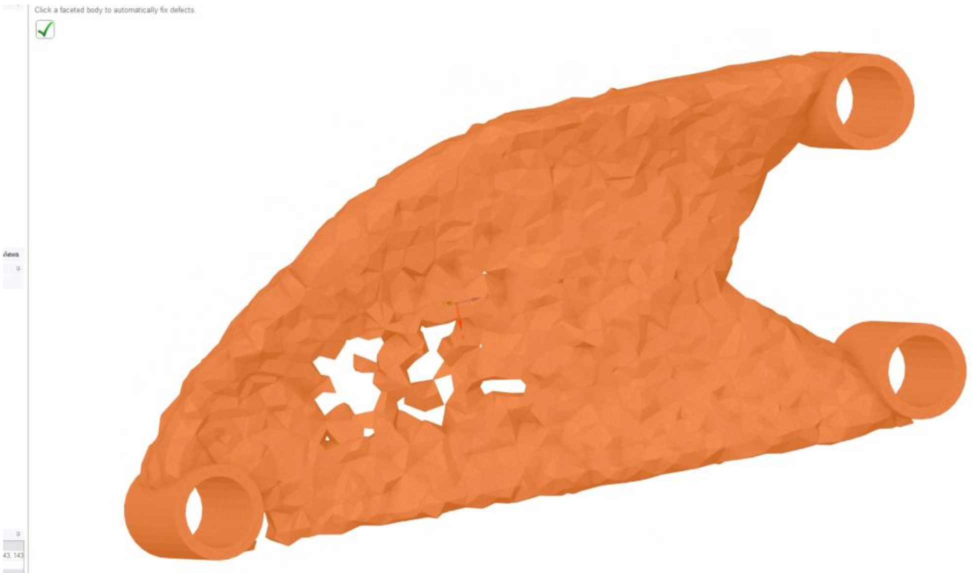


Figure 104 Geometry Exported

Exported as STL file into Spaceclaim. Using check facets there are many errors. Used autofix, only fold overs remained.

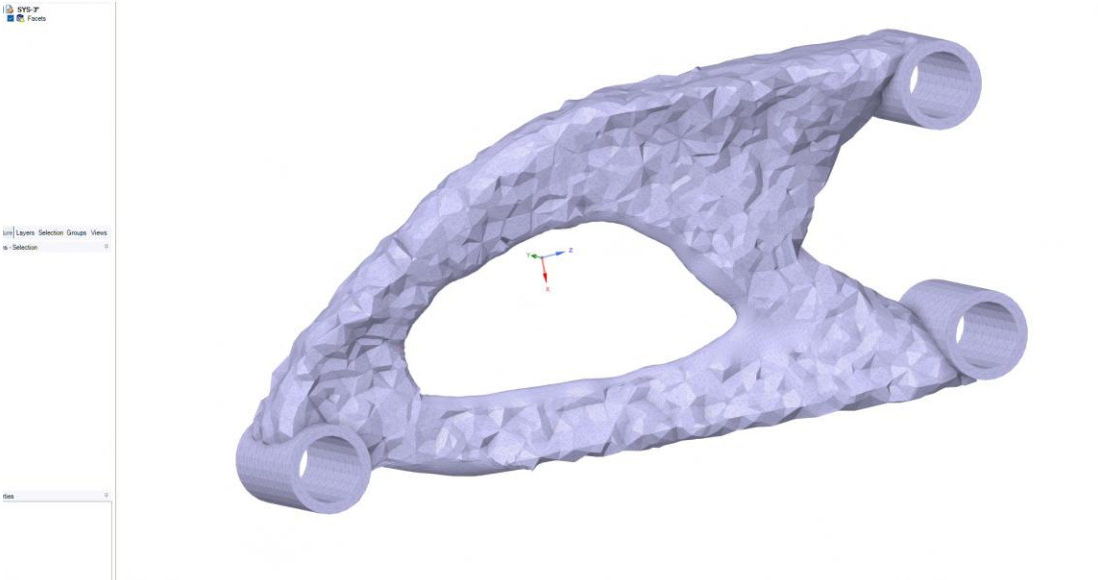


Figure 105 First Large Cutout to Reduce Mass

Selection of geometry removed in places believed to be efficient. To be tested and confirmed or modified later. Used hole tool under facets to close the holes created when cutting out geometry. Left majority of material for upper right screw hole and shaped out a support beam from bottom right hole towards the centre of the upper arch for support; at predicted high stress and high bending region.

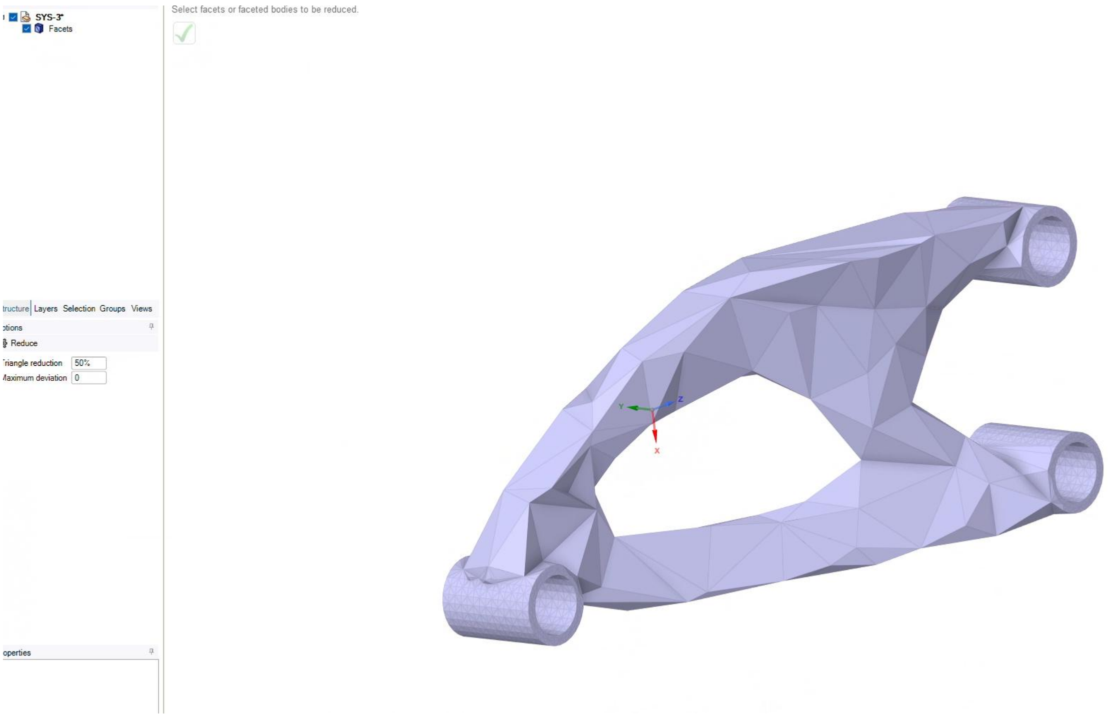


Figure 106 Reduce Tool for Computational Speed

Reduce of 50% used 7 times for easy computation, aesthetics and the stiffness of triangular geometry. Excluded holes when using reduce tool as they are needed for screws to fit in easily.

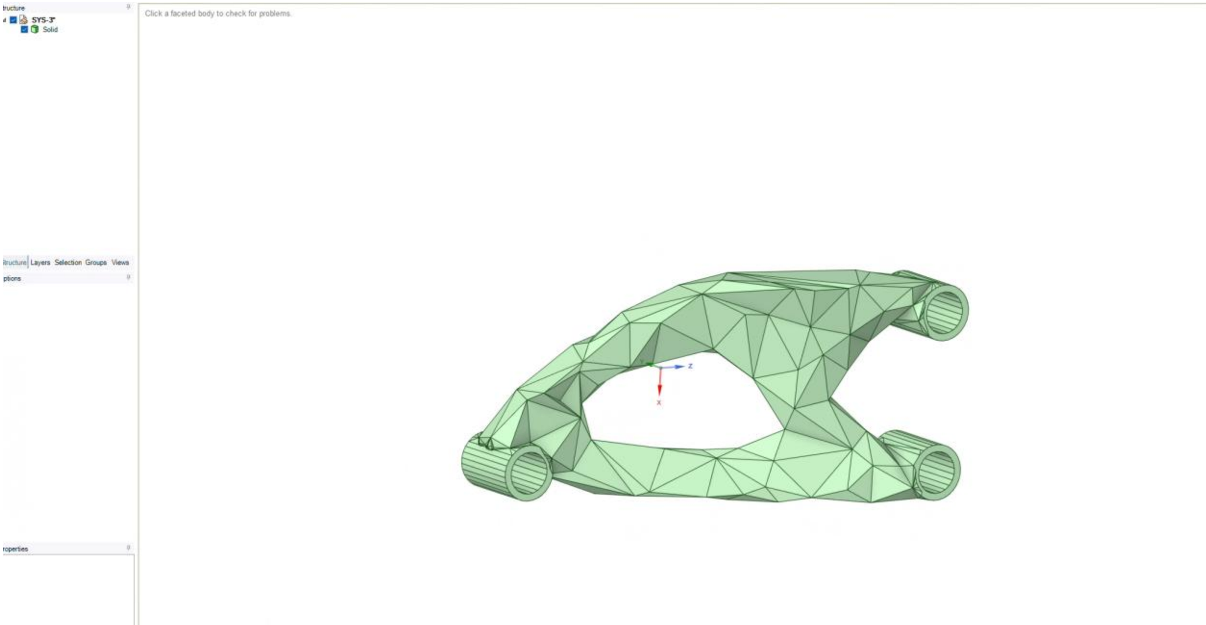


Figure 107 Conversion to Solid Successful

Converted to solid. Added groups for fixed support and force hole.

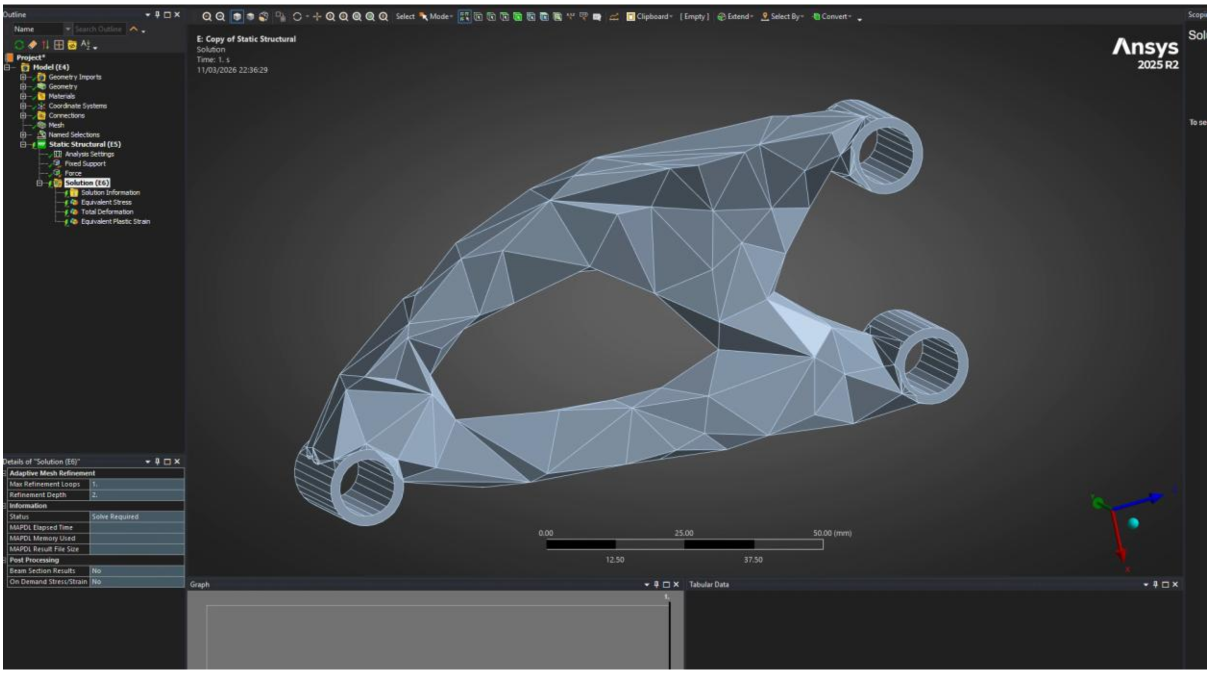


Figure 108 Mesh Settings

Opened in FEA. Mesh 3mm, adaptive sizing on. Fixed support and force holes selected. Material set to PLA filament.

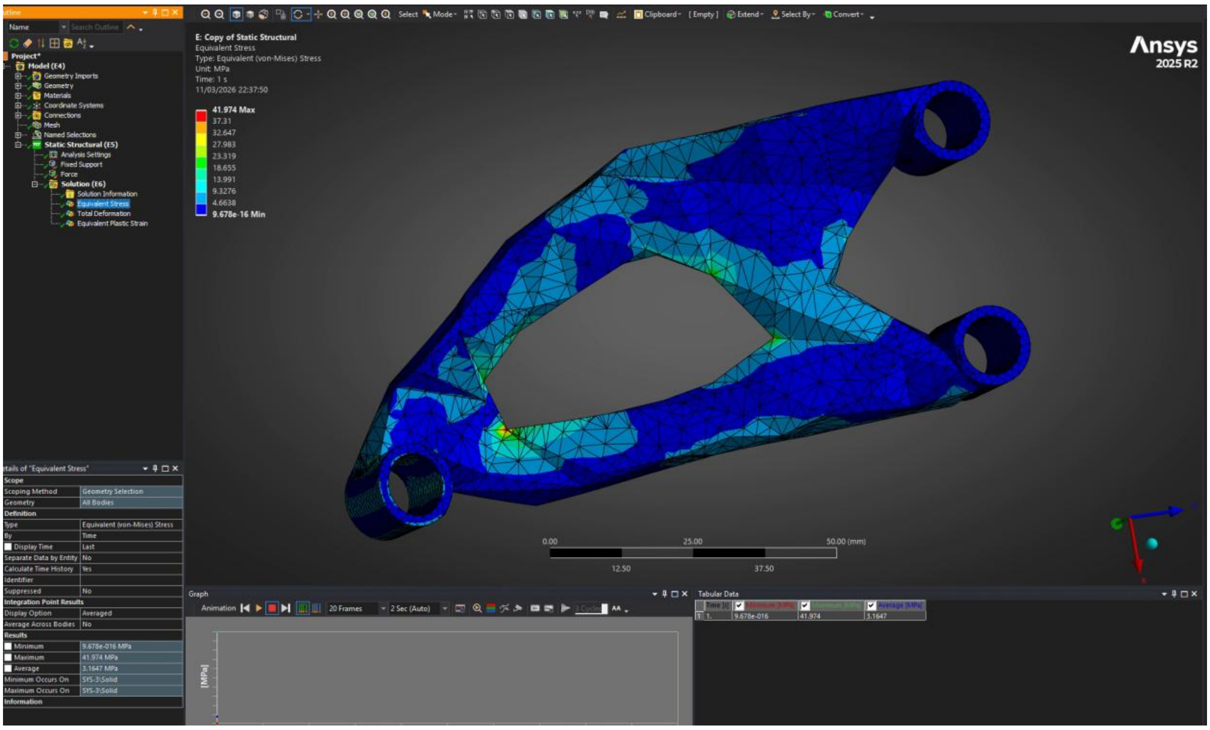


Figure 109 Max Stress Passed But Could Be Reduced

Maximum equivalent stress 42MPa. <65MPa which means it shouldn't fail under the load. Maximum stresses are at sharp corners. Specifically, the corner nearest the hole where the force is applied. the triangle shapes are useful for stiffness, but sharp edges can fail due to high stress concentration; rounded corners are a better alternative for handling high stress.

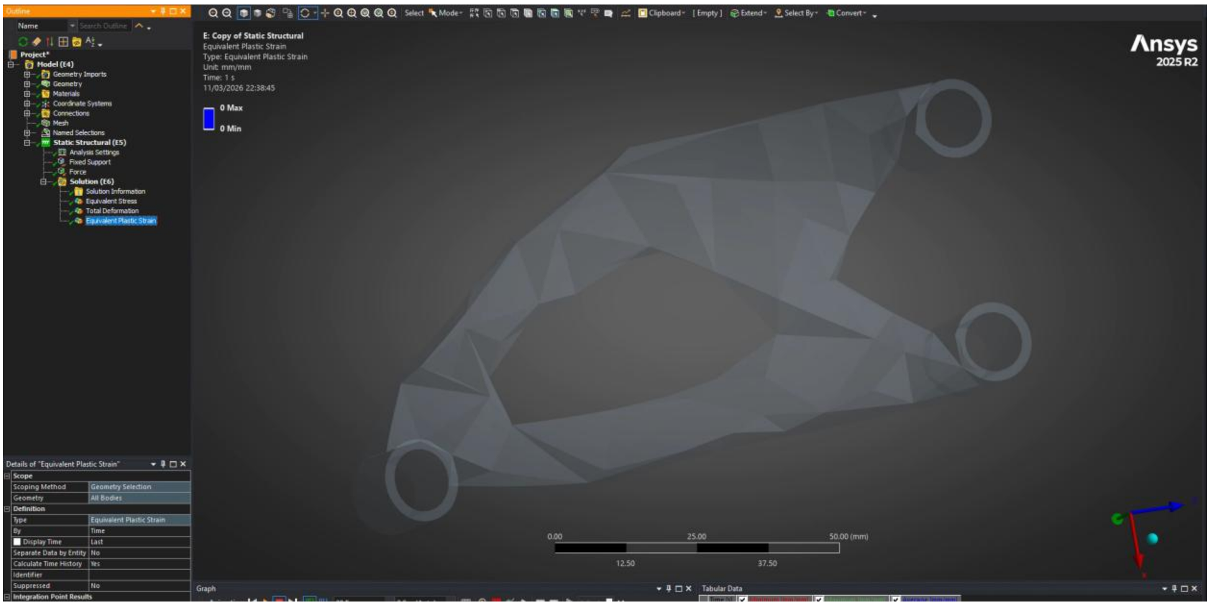


Figure 110 No Plastic Strain

No plastic strain, which means it won't fail from plastic deformation.

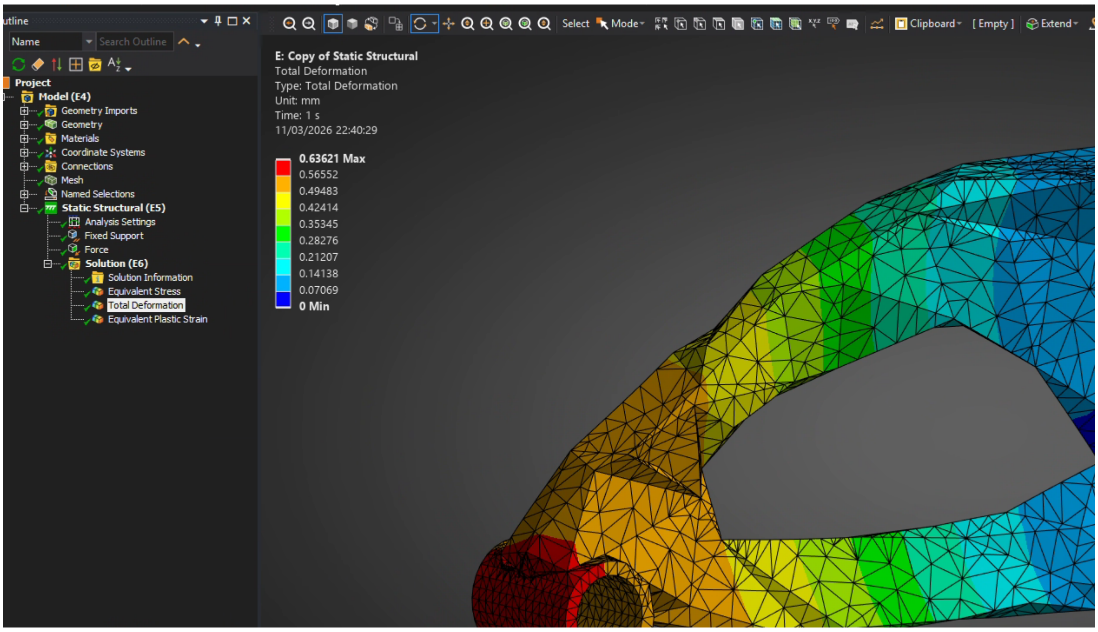


Figure 111 Total Deformation

Highest elastic deformation at force hole. 0.64mm of elastic deformation. This will not be a problem.

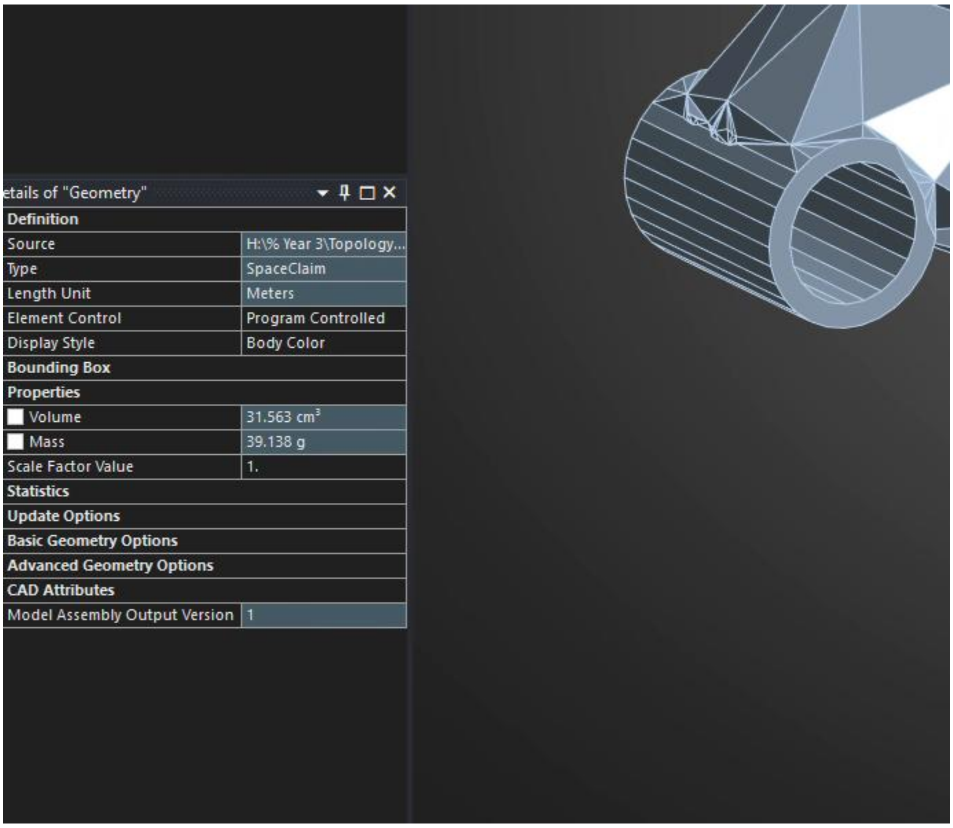


Figure 112 Mass

Mass reduced to 39.14g. this is below the 60g threshold.

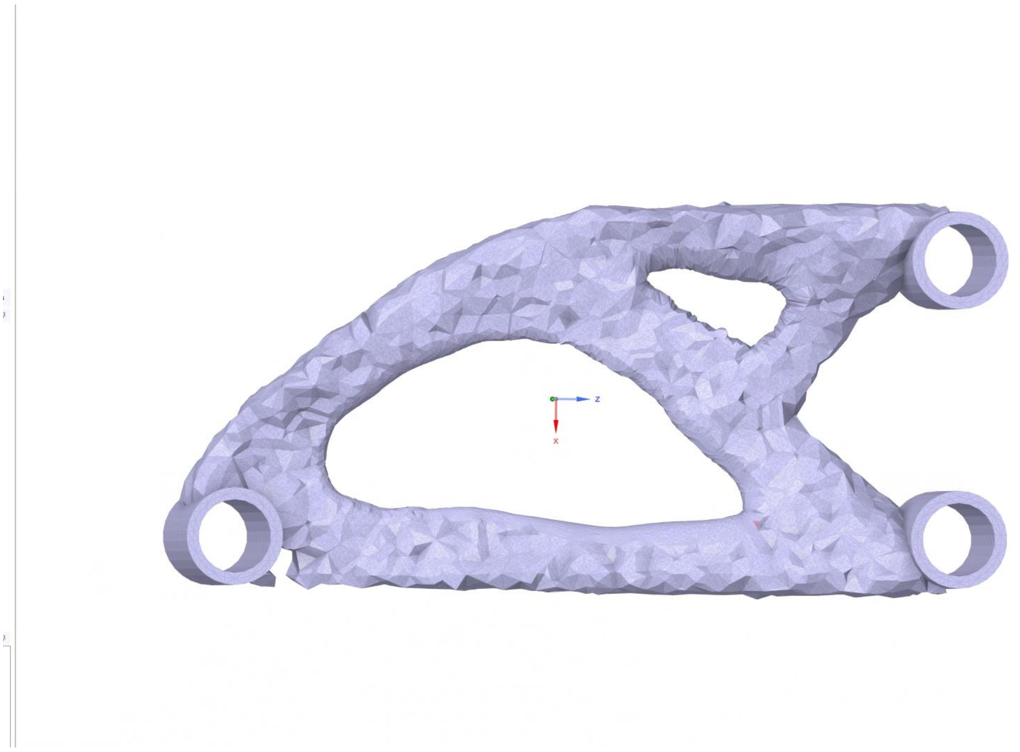


Figure 113 Second Iteration Additional Hole Created

Second iteration design started. Removed more material in low stress regions such as in the top right corner according to analysis.

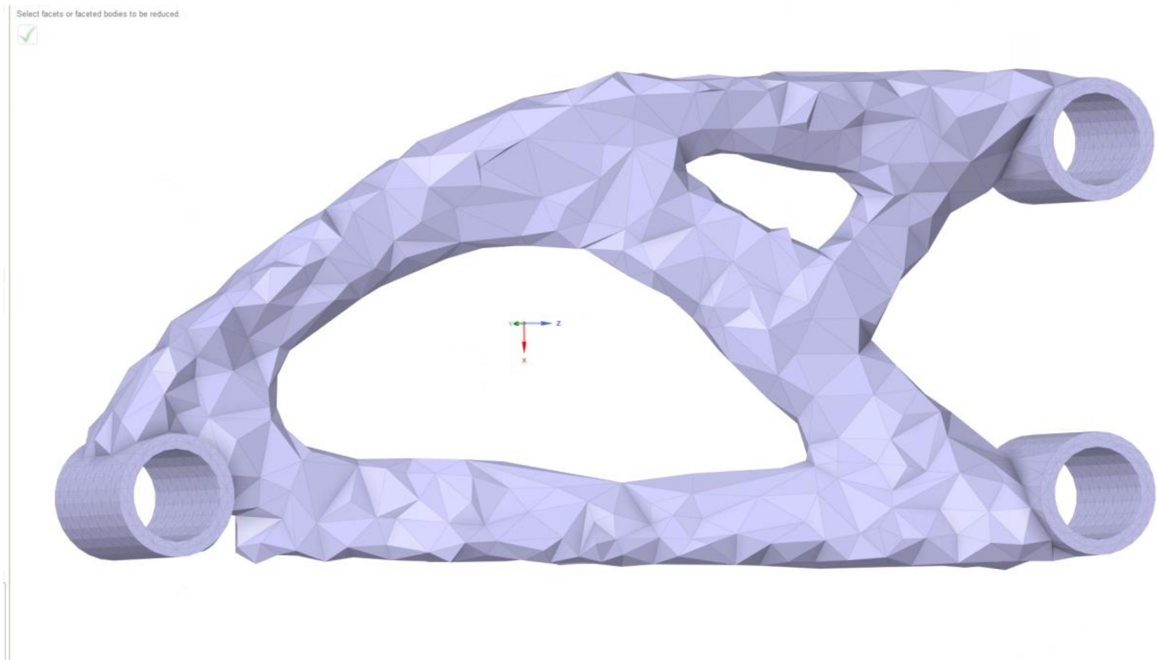


Figure 114 Reduce

Reduce 60% 5 times for computational efficiency and aesthetics.

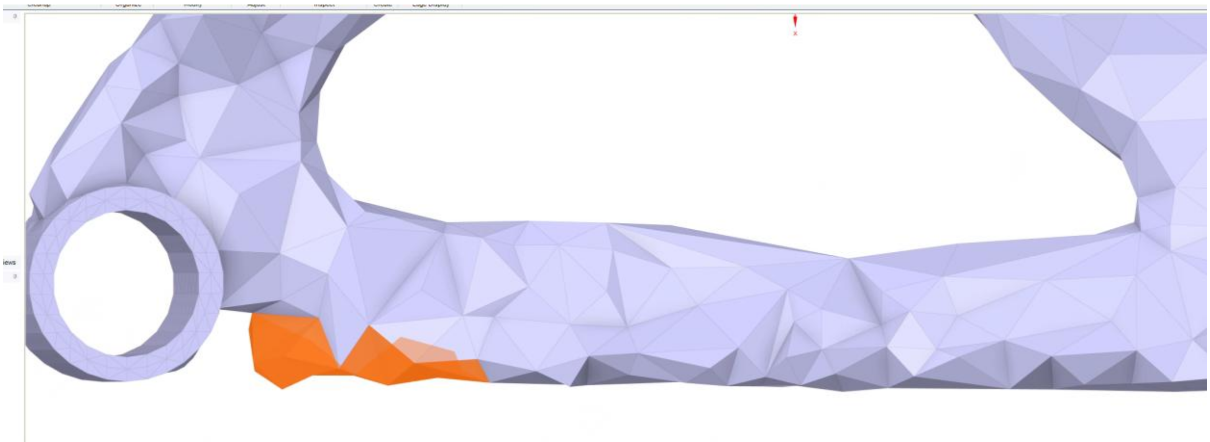


Figure 115 Excess Removed

Unnecessary section removed and hole tool to repair.

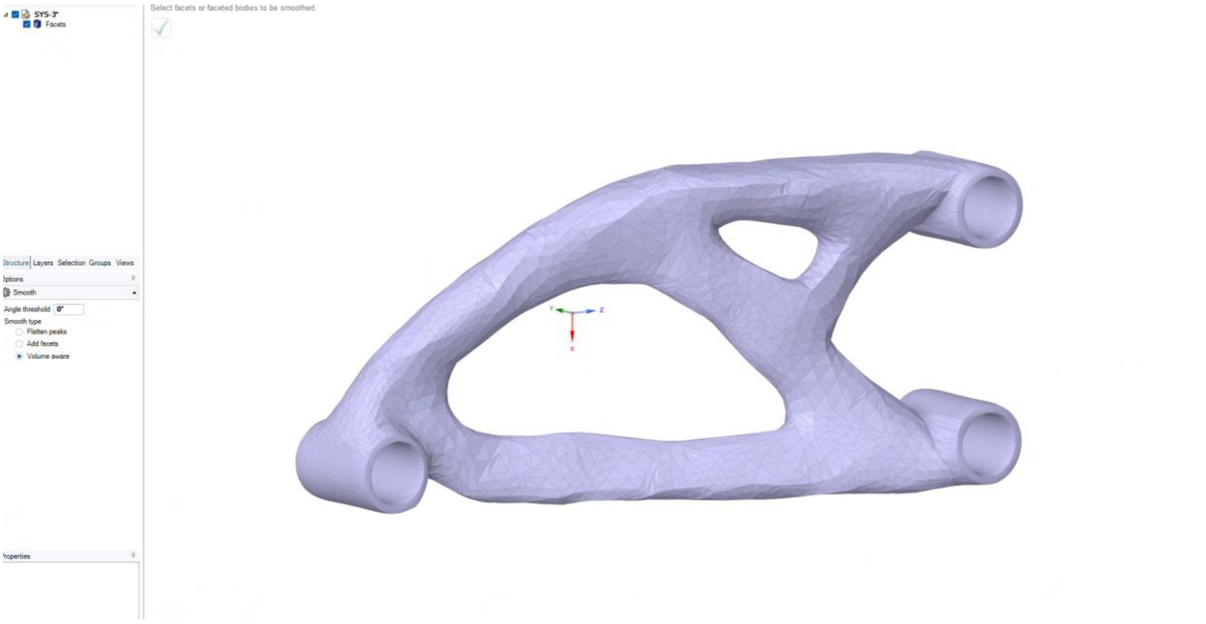


Figure 116 Smoothing/Rounding for Optimised Stress Distribution

Round shapes reduce localised stress the most effectively. So smooth tool used with an angle threshold of 0 degrees and volume aware selection applied.

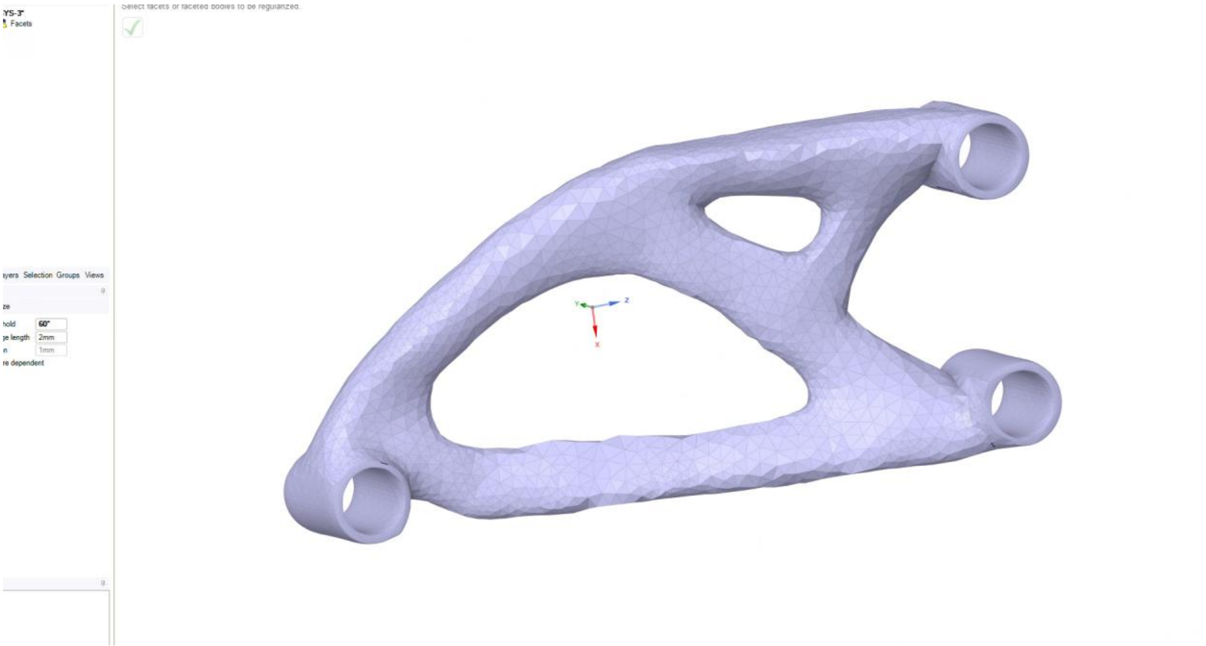


Figure 117 Regularise

Regularise applied to reduce computational stress. Applied on geometry except the holes. Edge length 2mm angle threshold 60 degrees.

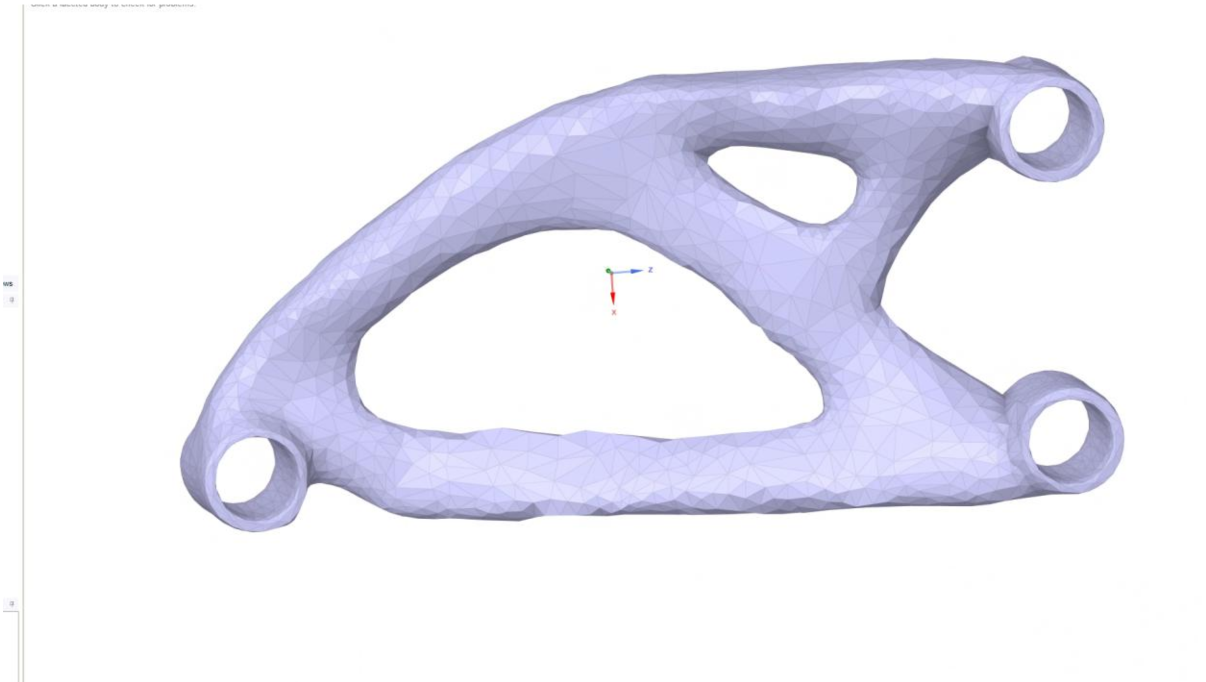


Figure 118 Reduce

Reduce 50% used 3 times to remove errors when using “check facets” where auto fix did not work. And converted to solid for testing.

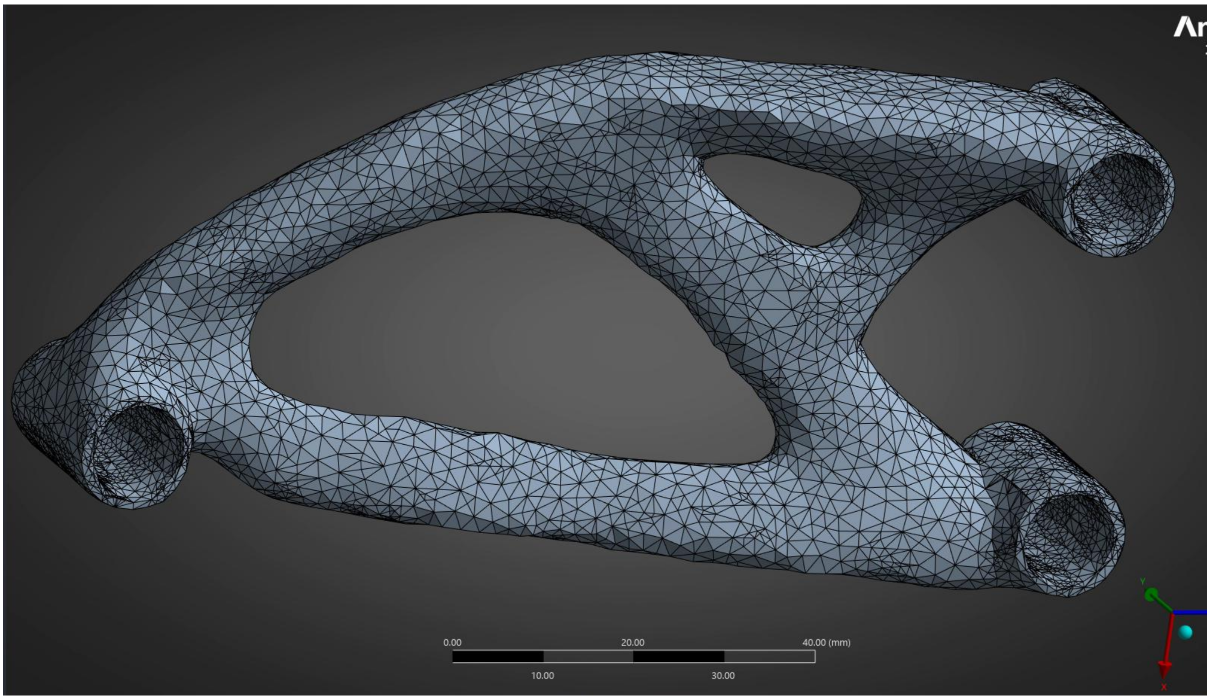


Figure 119 Adaptive Sizing Potential Excess Computation Stress

Adaptive sizing causing high density mesh at holes. This may cause computational errors and extend simulation time unnecessarily.

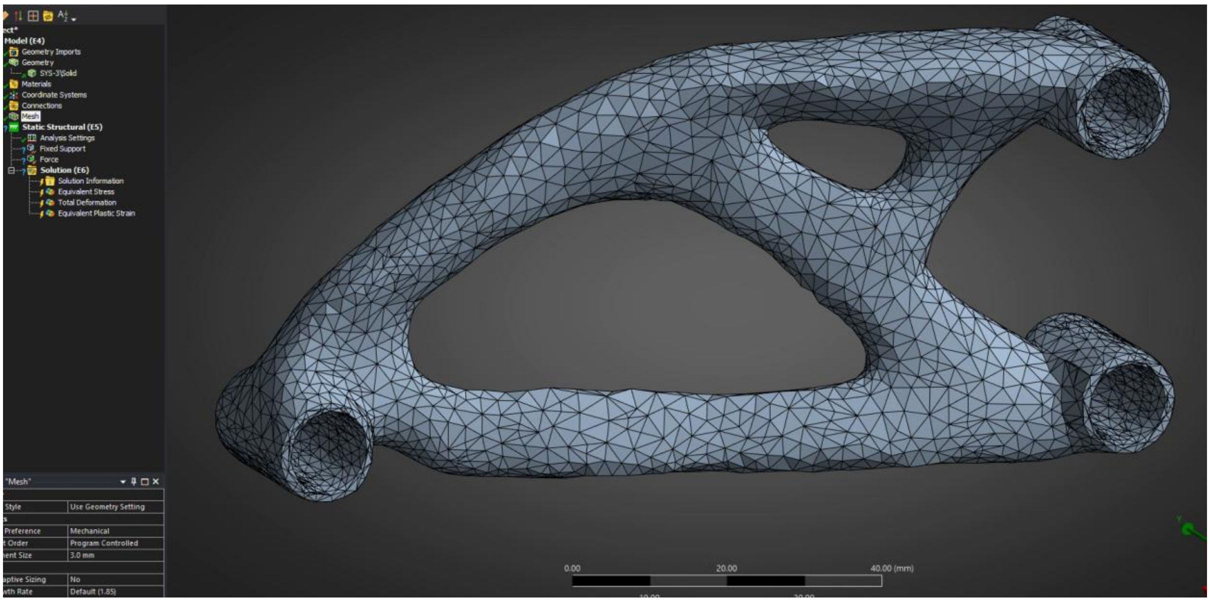


Figure 120 Finalised Mesh

Adaptive sizing removed and max size set to 3.5mm. During the design process before this point complexity was added to the holes reducing the efficiency of computation, complexity still there. Must go back and modify for better computation.

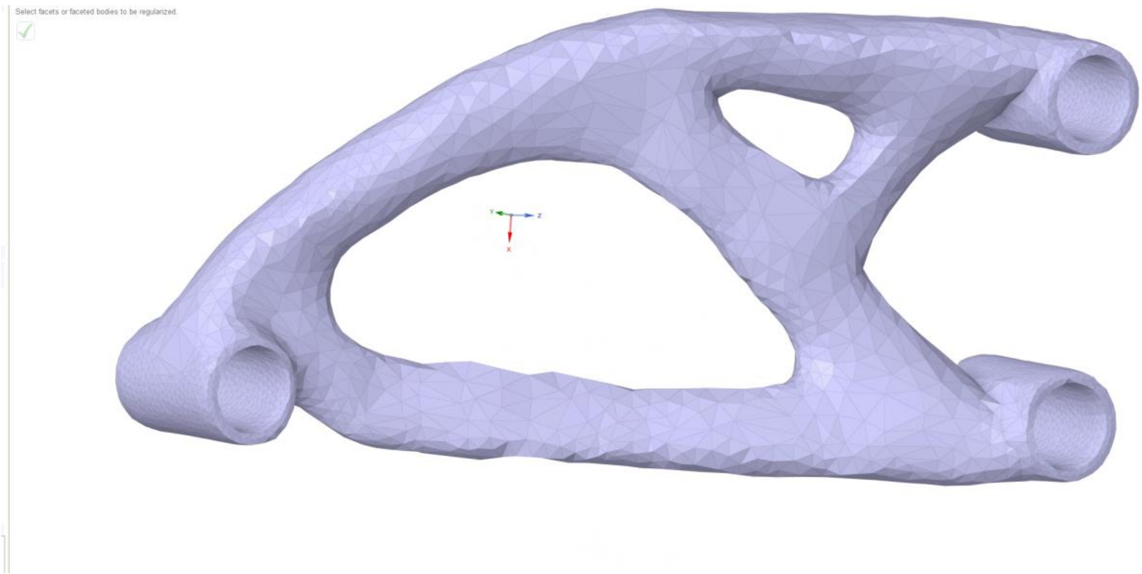


Figure 121 Reduce for Further Computation Ease

Geometry modified in Spaceclaim once again; reduce applied exclusively to holes at edge length 1mm and 60 degrees angle threshold. Geometry complexity greatly reduced.

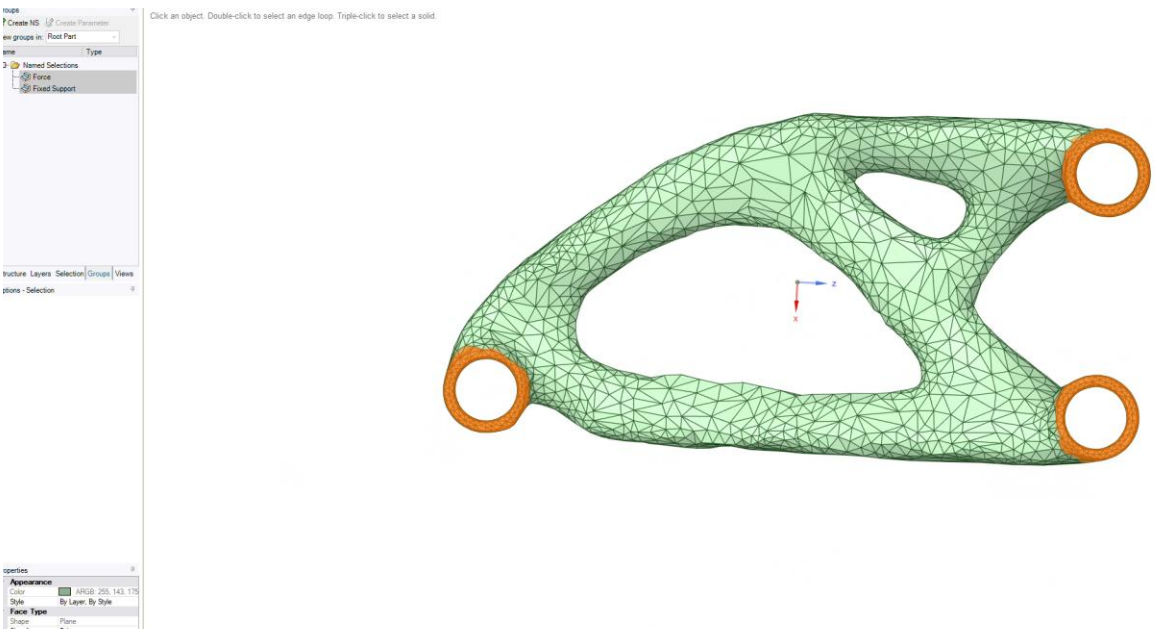


Figure 122 Groups Created for FEA

Groups created for force and fixed support once again. Lasso tool and through selection.

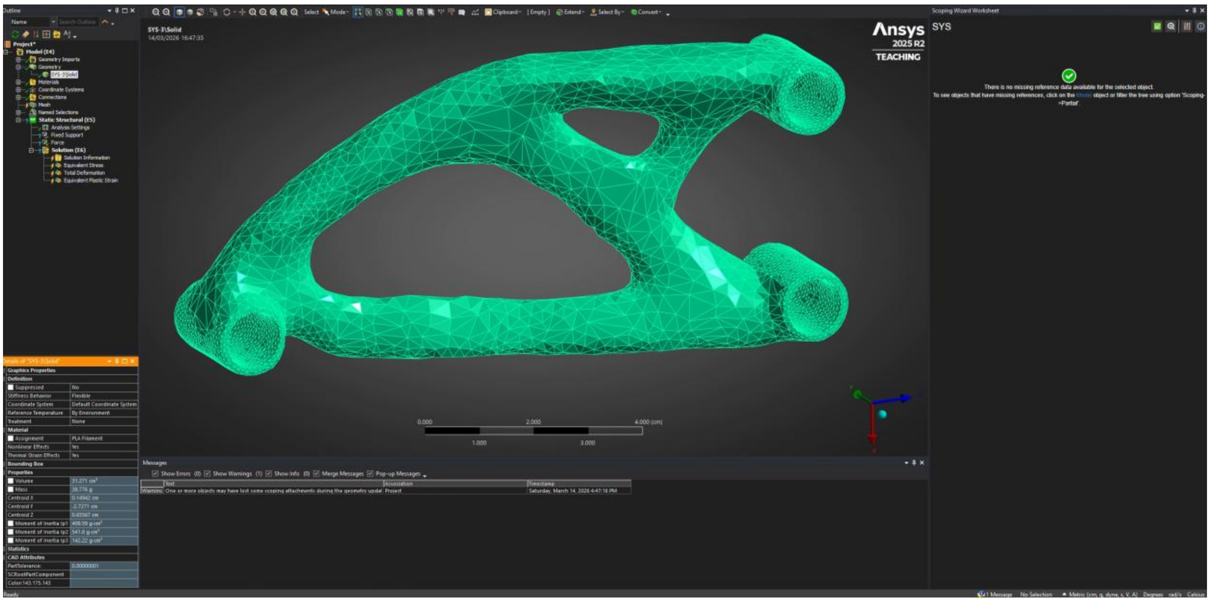


Figure 123 Material Selection & Mass

Material set to PLA filament. Weight reduced even further from first iteration to 38.8g, which is a 35.3% reduction from target weight of 60g.

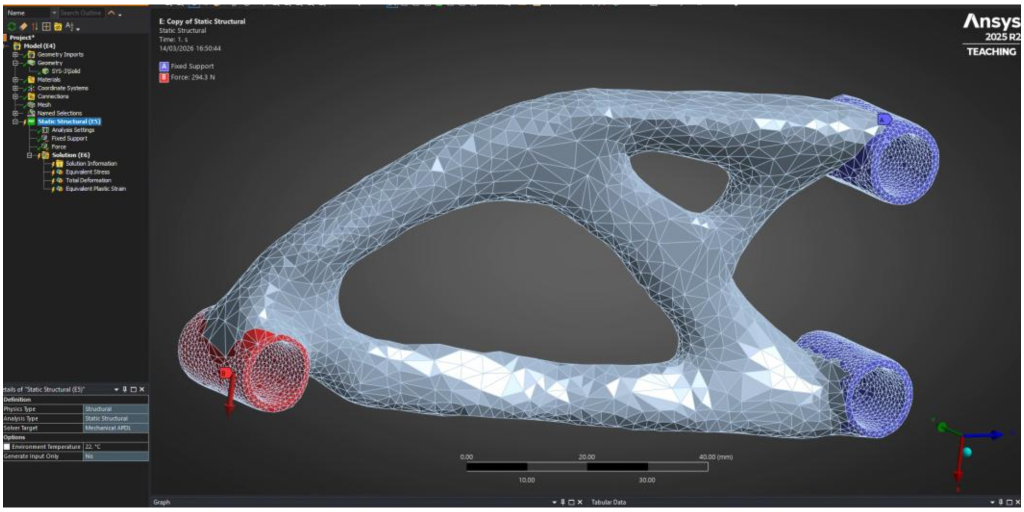


Figure 124 Mesh & Force/Fixed Supports

Mesh 3mm adaptive sizing off, 3.5mm maximum. Force and fixed support applied.

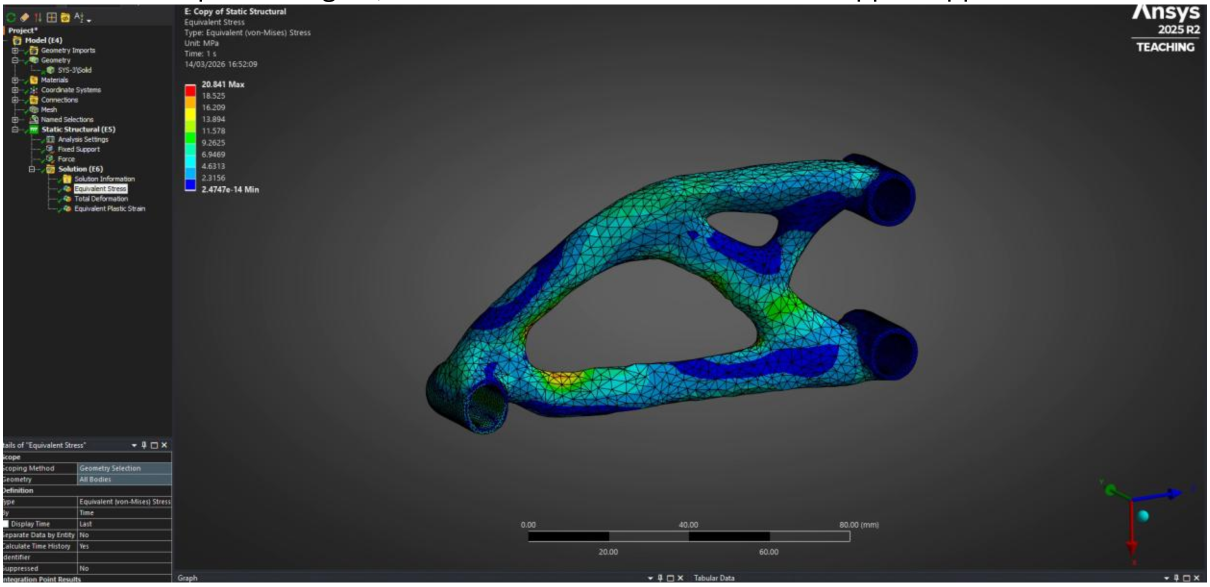


Figure 125 Stress Greatly Reduced

Maximum stress 20.8MPa. This is approximately a 50% reduction in previous equivalent stress while keeping the weight relatively the same, specifically 0.3g lower. This marks significant improvement and extremely stress levels compared to the rated ultimate strength of PLA filament (65MPa).

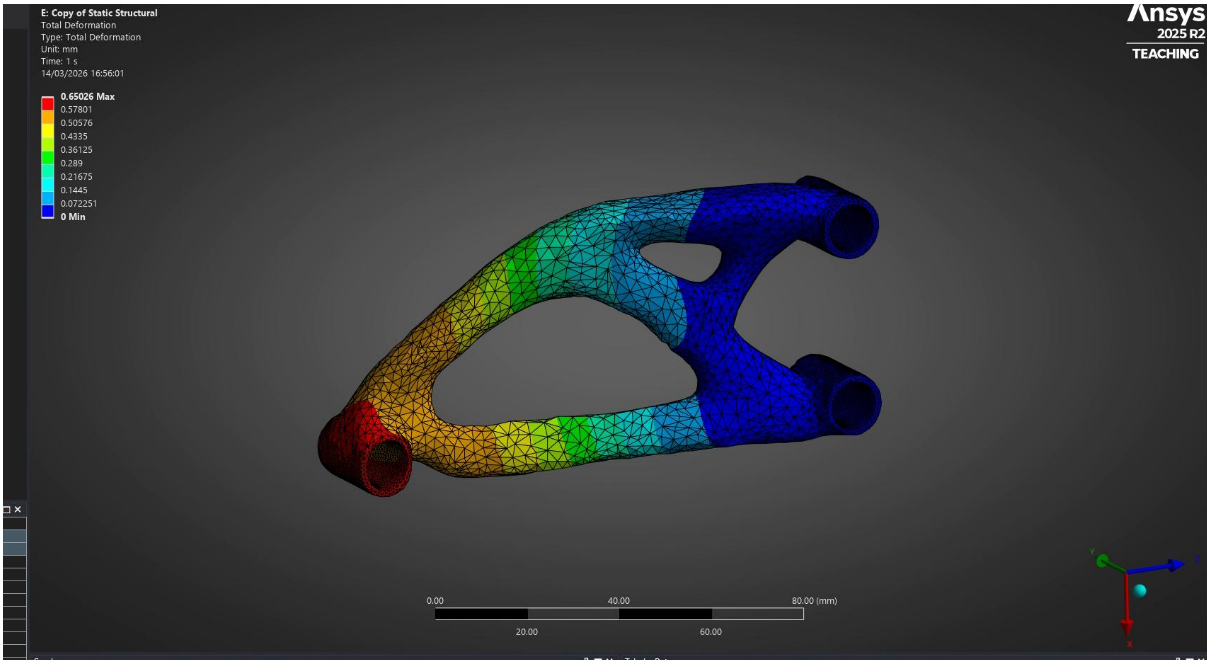


Figure 126 Deformation Maintained

0.65mm of deformation is approximately the same as past iteration of 0.64. This means the design did not improve resistance to bending in the maximum area. Area could be improved by increasing stiffness of geometry towards the force hole and removing some further away from it.

This final iteration holds a design score of $\frac{1}{0.0388 \times 0.65} = 39.65$

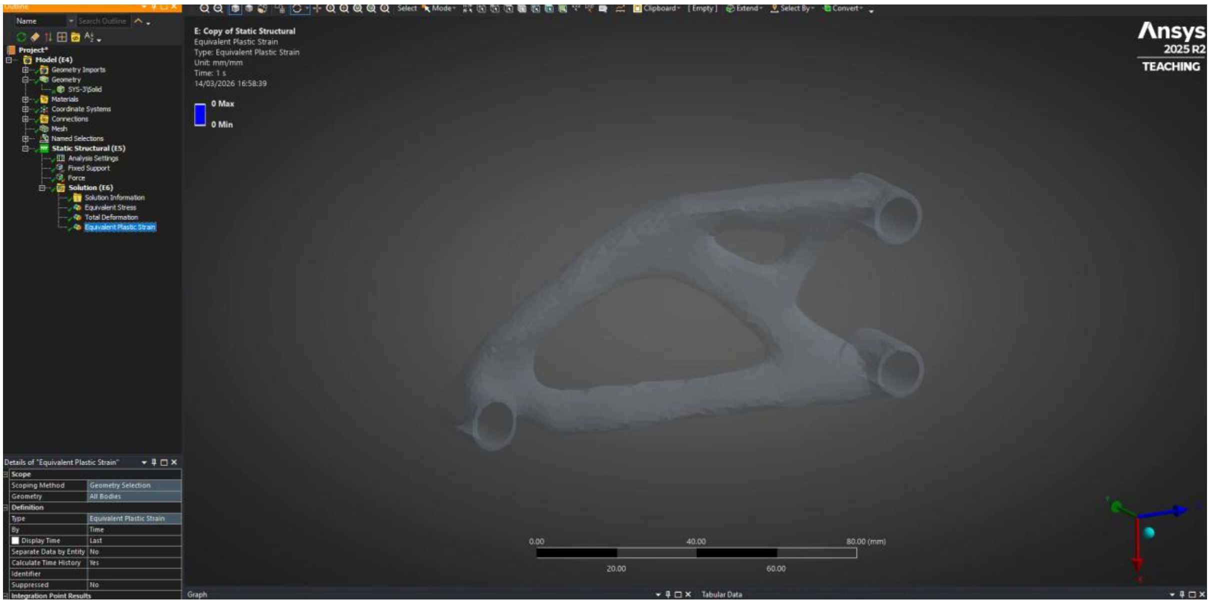


Figure 127 No Plastic Deformation

No plastic deformation of course since stress and deformation is relatively low.

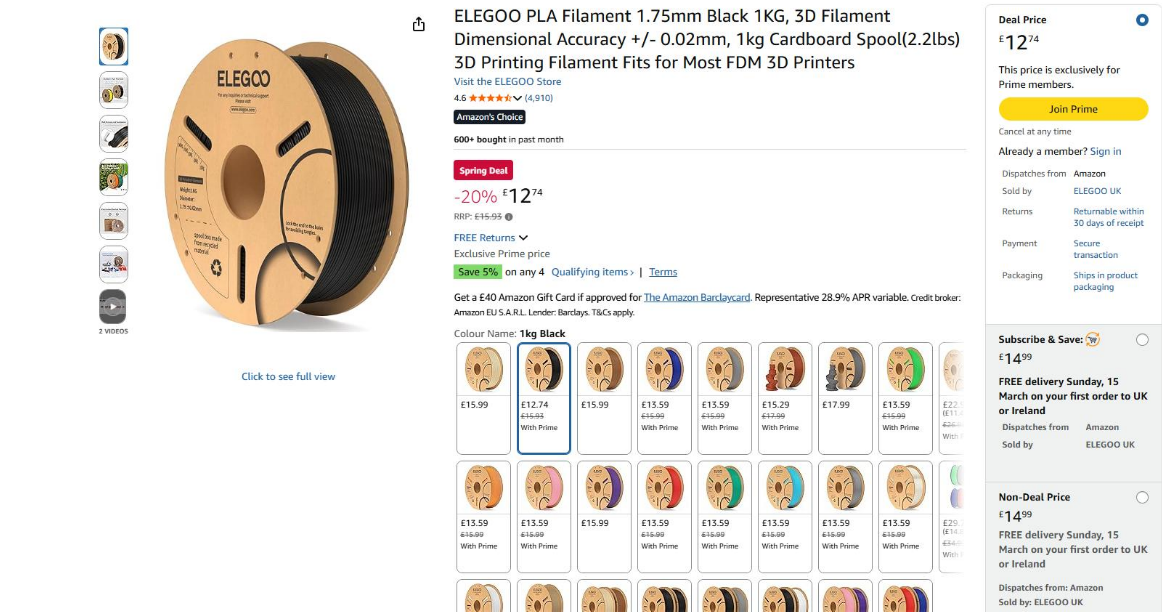


Figure 128 Commercial PLA Filament Price

High quality PLA Filament with a tolerance of 0.02mm costs around £13 for 1kg. This means this support costs $0.0388 \times 13 = £0.504$ or 51p. 51p and not 50p because rounding down money excludes the extra cost from multiple prints.

Section 4 – 3D Printing

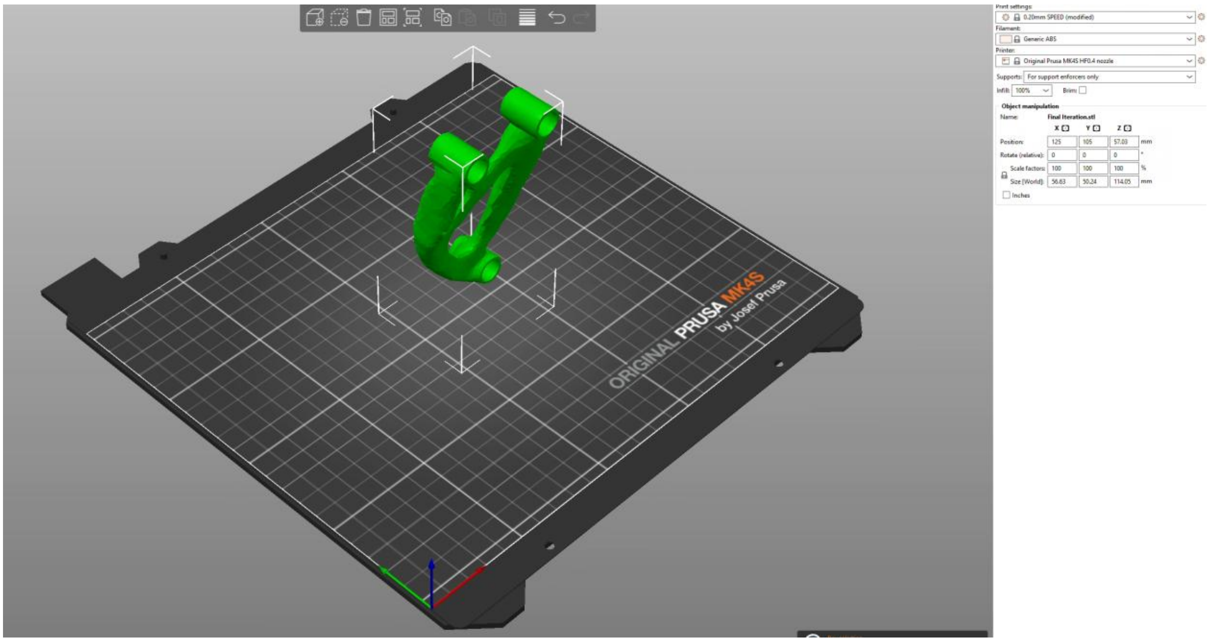


Figure 129 Imported to Prusa Slicer

Prusa slicer opened. Infill set to 100% and STL geometry file imported.

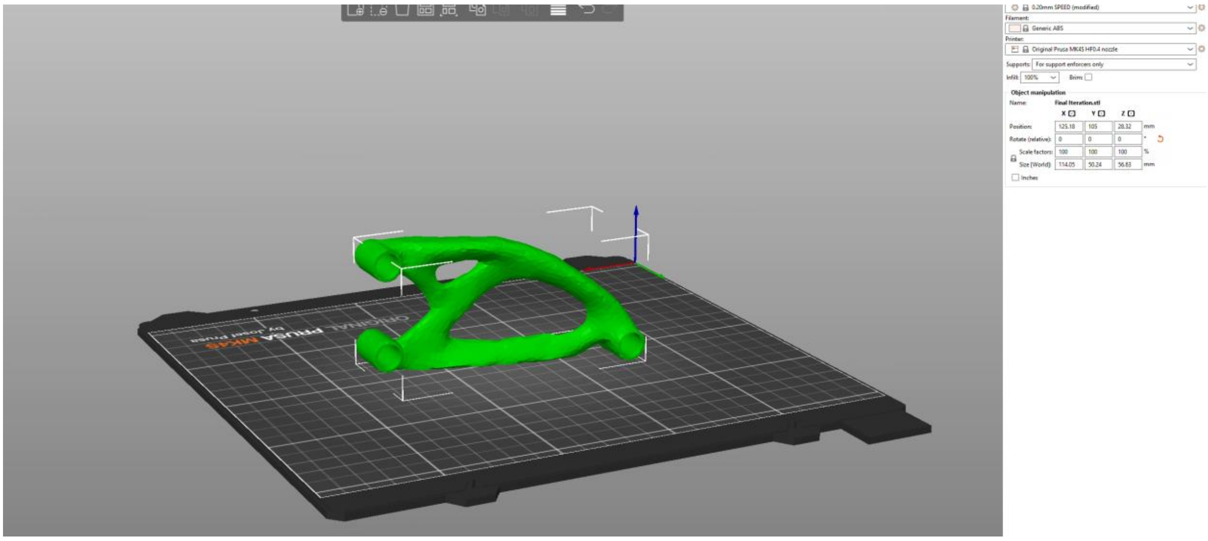


Figure 130 Rotation 1

Rotate by 90 degrees, y axis.

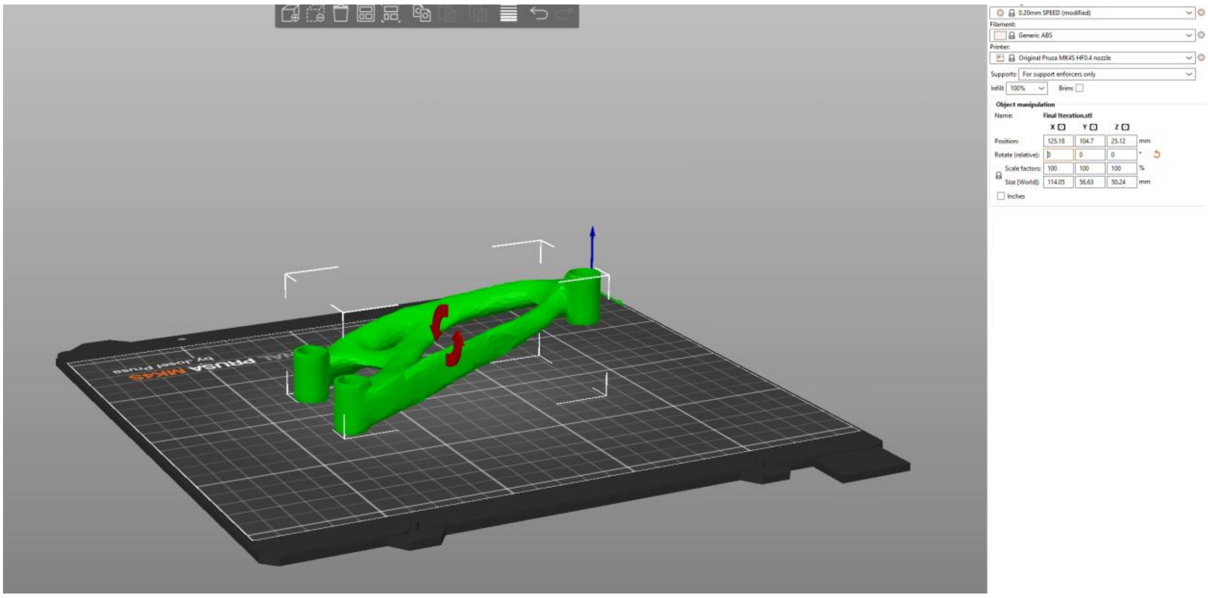


Figure 131 Rotation 2

Rotate 90 degrees, x axis.

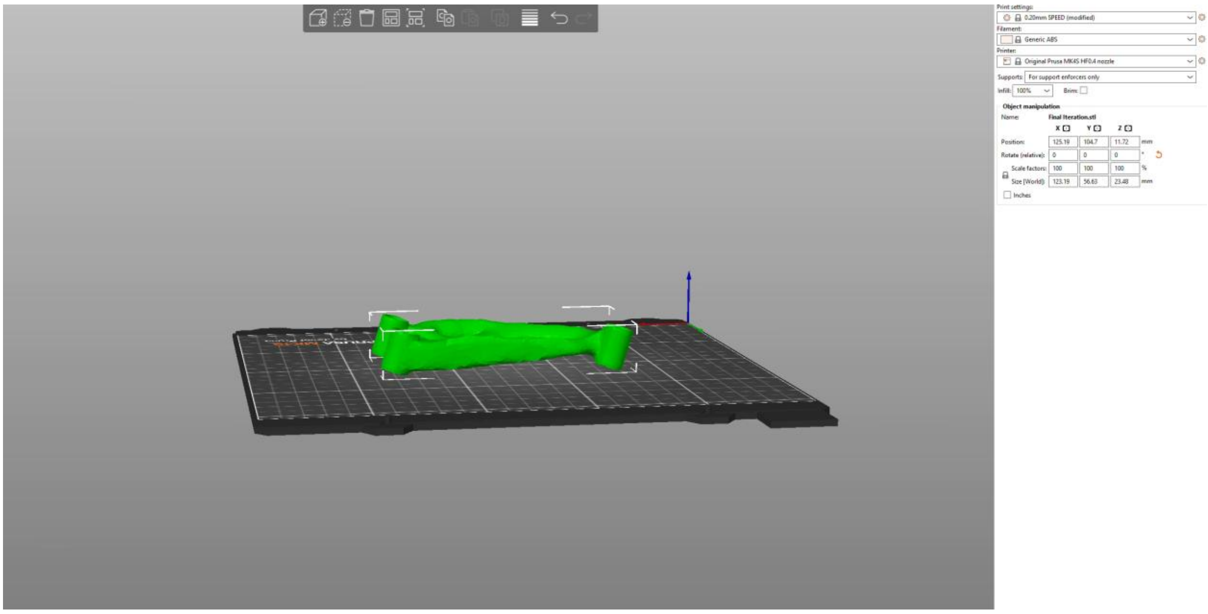


Figure 132 Rotation 3

Rotated by -14 once, then -1 degrees until all holes are touching printing floor. A total of -17 degrees in the y axis. This orientation where the holes are facing up would require less support material because of the large gaps in the centre of the geometry.

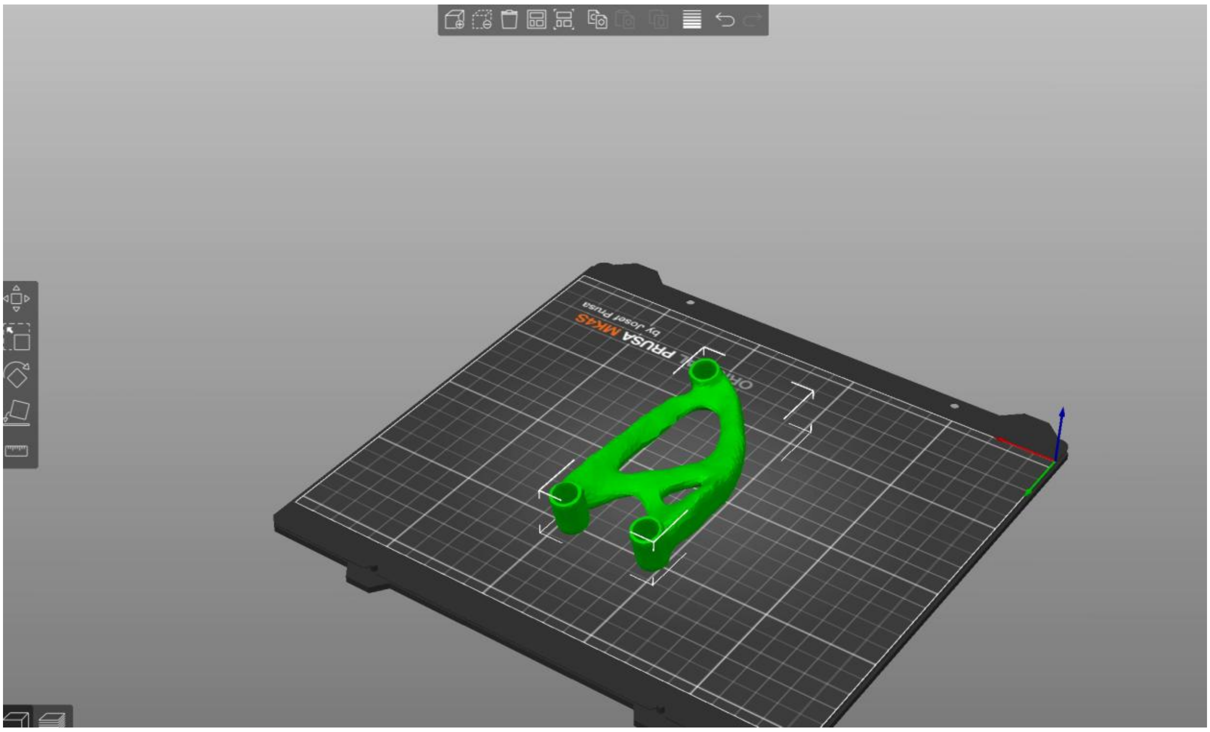


Figure 133 Rotation 4 – Face Tool

Upon discovery of place on face tool, the face desired can be selected in which the program will orientate it perfectly in line to touch the selected face with the platform. This is much easier and time efficient.

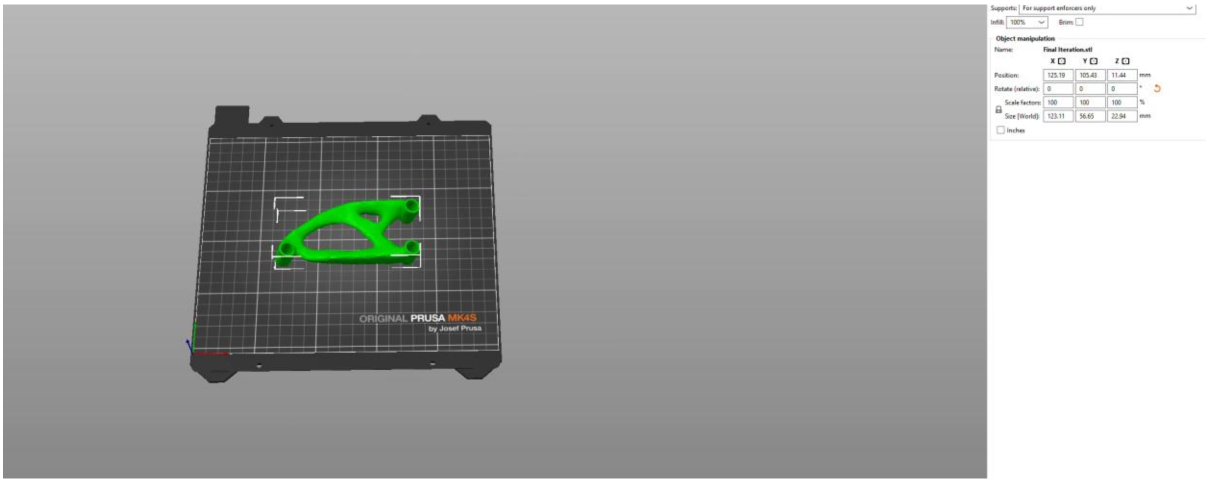


Figure 134 Rotation 5 – Perfectly Sideways

After which, geometry can be orientated to be facing viewer as printed from the front. Rotated -90 degrees, z axis.

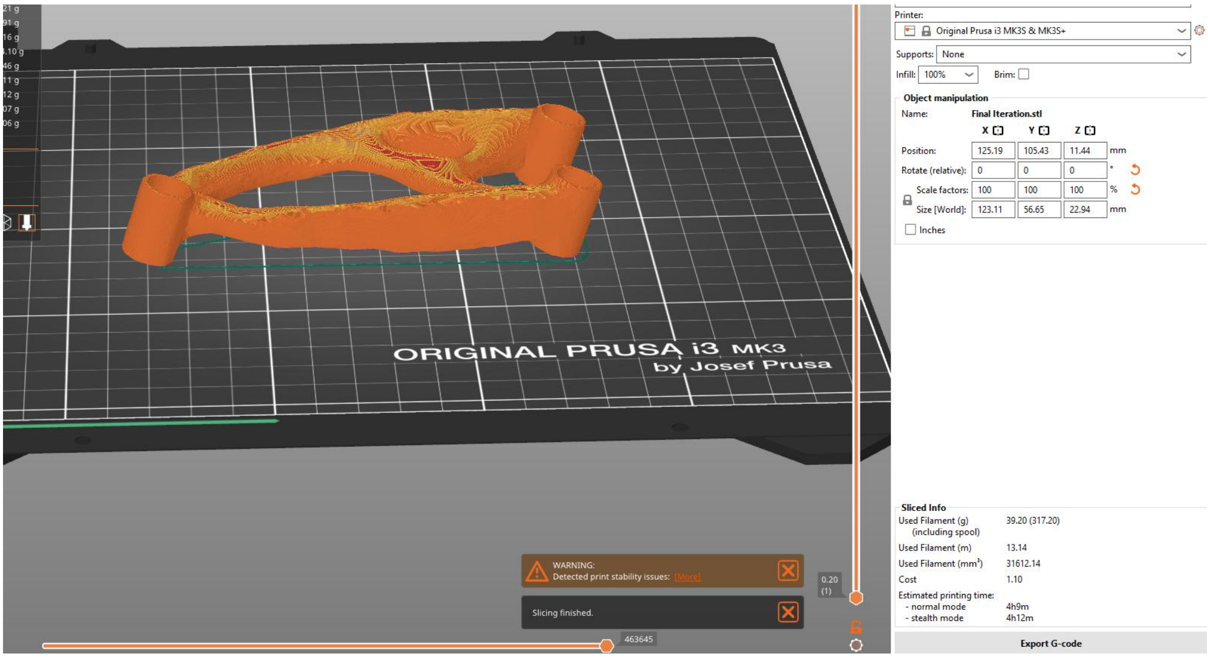


Figure 135 Mass Without Supports

Used filament in grams on Prusa slicer only at 32.73g which is much lower than on ANSYS.

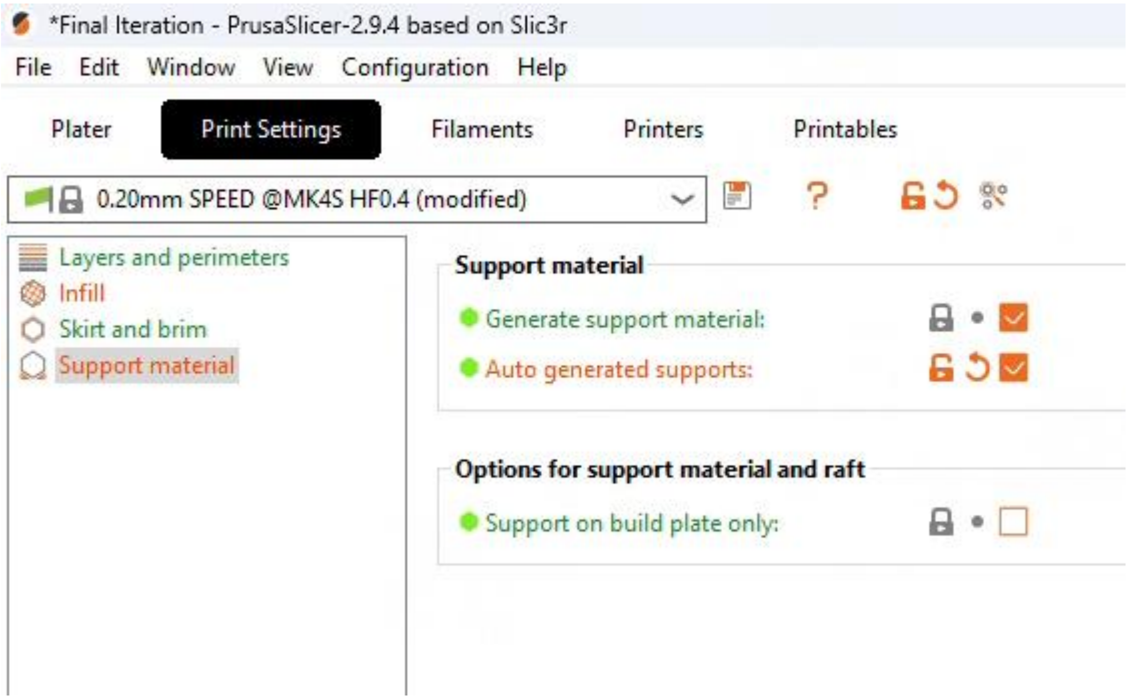


Figure 136 Support Settings

Auto generate supports enabled.

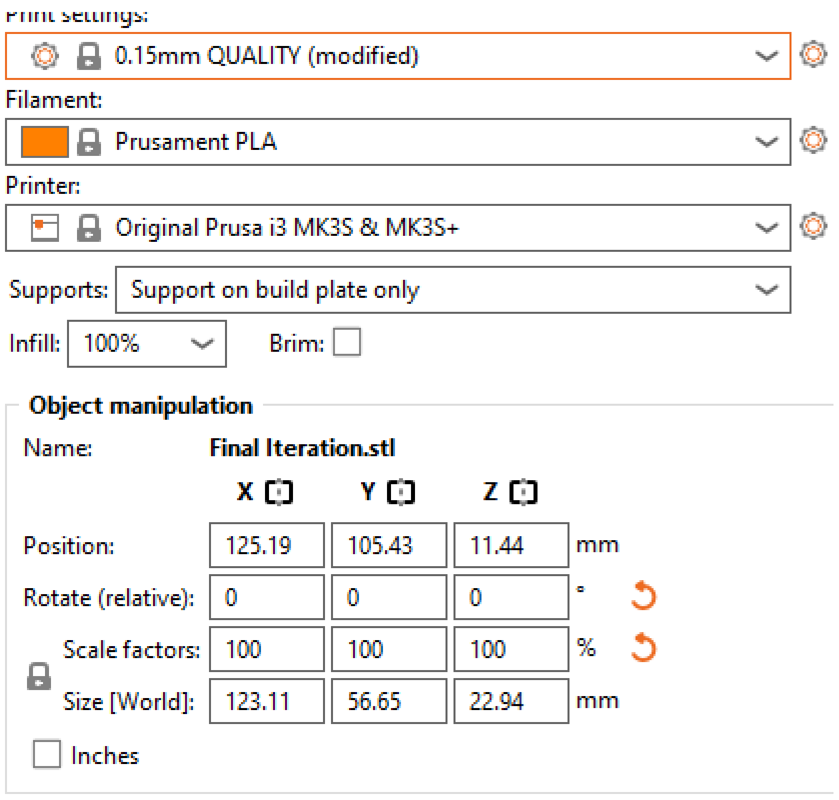


Figure 137 Support, Material & Infill%

Filament set to Prusament PLA, supports set to support on build plate only and 100% infill.

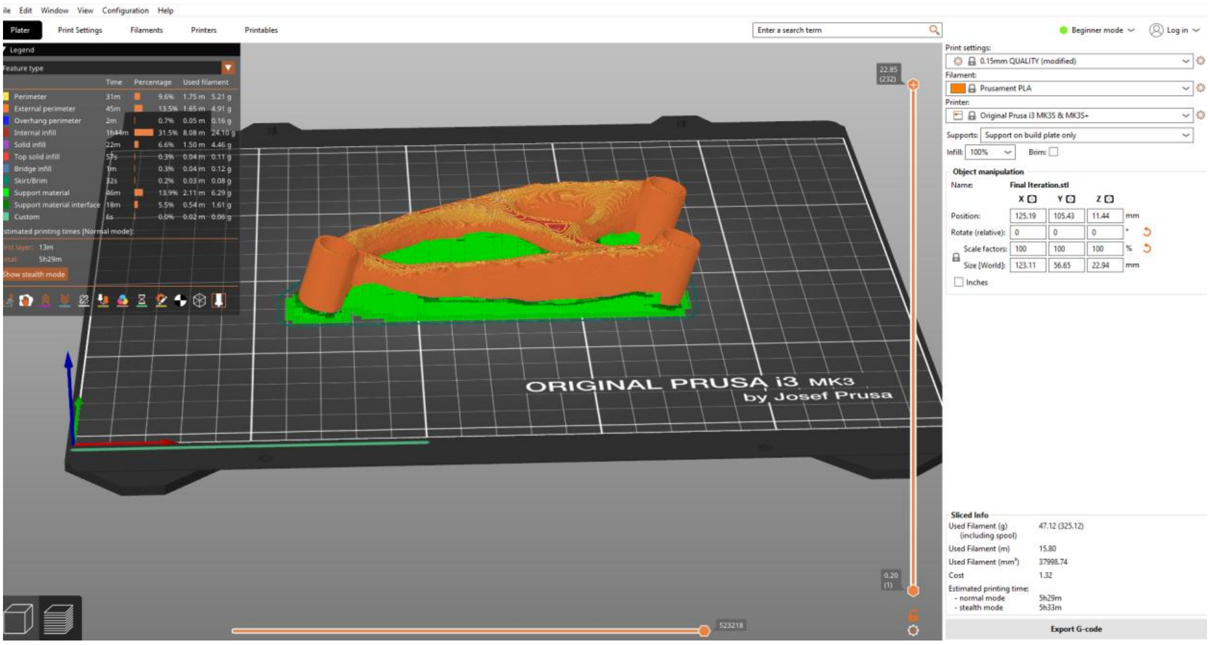


Figure 138 Mass Including Support

Supports set to everywhere and sliced. Relatively low mass in supports required for complexity of geometry. At 37.41g total mass this means $37.41g - 32.73g = 4.68g$ is the mass of support required. G-code exported to USB for insertion and printing.

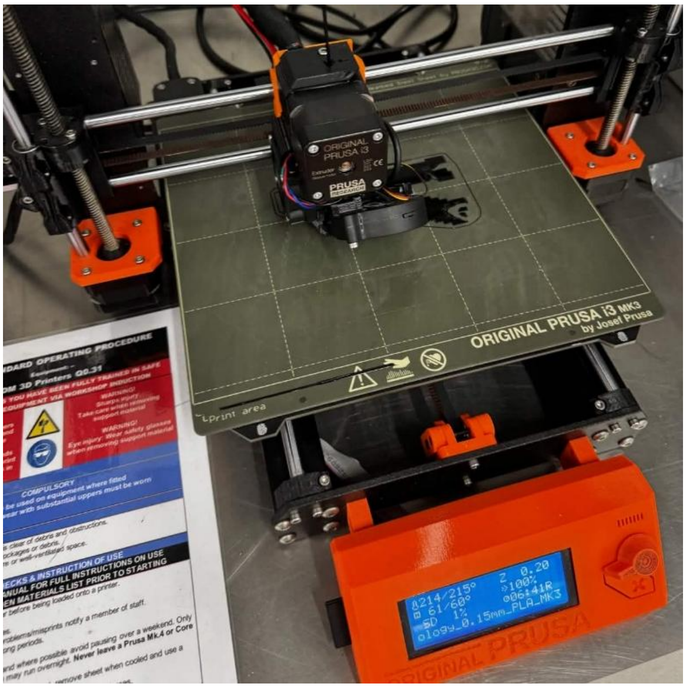


Figure 139 During Printing

Figure 140 After Printed, with Supports Still Attached

Very little supports, efficient printing time, efficient use of material. Reduces cost and less environmental impact. More sustainable.

Figure 141 After Supports Have Been Removed

Holes tested by attempting to fit the testing screw, passed.

Section 5 – Comparison of Theoretical/Experimental Values & Discussion

OrgDefine	Bracket weight (	Bracket weight (	x0 = deflection at preload 5kg (mm)	x1 = deflection at full load 25 +5kg (mm)	bracket deflection corresponding to 25kg load d = (x1-x0)	design score = 1/(d*m)	hole clearan	Comments
2674080	56.4	0.0564	0.6168	1.9448	1.328	13.35	ok	1st
2729950	44.2	0.0442	3.3491	5.1612	1.8121	12.49	ok	2nd
2792586	38.8	0.0388	3.3856	5.4601	2.0745	12.42	ok	3rd
2722743	58.2	0.0582	0.4504	1.9159	1.4655	11.72	ok	
2810934	51.7	0.0517	2.3037	3.9615	1.6578	11.67	ok	
2795067	42.4	0.0424	3.6513	5.6898	2.0385	11.57	ok	
2731284	36.4	0.0364	3.5701	6.0332	2.4631	11.15	ok	
2714192	47.7	0.0477	2.6433	4.5461	1.9028	11.02	filing needed	One cylindrical hole is not aligned
2784060	42.2	0.0422	3.1275	5.2963	2.1688	10.93	ok	
2721953	50.7	0.0507	0.6057	2.4295	1.8238	10.81	filing needed	
2788263	47.2	0.0472	1.2461	3.2142	1.9681	10.76	filing needed	
2856412	56.1	0.0561	0.8221	2.5524	1.7303	10.3	ok	
2832381	48.7	0.0487	2.8662	4.8662	2	10.27	ok	
2752640	50.8	0.0508	4.3351	6.2559	1.9208	10.25	ok	
2799782	59.6	0.0596	1.3886	3.0266	1.638	10.24	filing needed	
2939580	51	0.051	1.7787	3.704	1.9253	10.18	ok	
2789059	52.9	0.0529	3.7161	5.5858	1.8697	10.11	ok	Scaling issue
2792456	47.2	0.0472	0.632	2.774	2.142	9.89	ok	Bracket is mirrored
2788668	36.2	0.0362	2.5285	5.3248	2.7963	9.88	filing needed	
2712591	55	0.055	1.9215	3.7765	1.855	9.8	ok	
2766328	54.2	0.0542	4.1416	6.0527	1.9111	9.65	ok	
2797578	41.3	0.0413	4.2432	6.7934	2.5502	9.49	ok	
2832388	52.9	0.0529	3.9682	6.0229	2.0547	9.2	ok	Scaling issue
2801376	55.9	0.0559	3.1205	5.0841	1.9636	9.11	ok	
2717904	48.7	0.0487	2.1177	4.5044	2.3867	8.6	ok	
2860504	53	0.053	2.4147	4.6422	2.2275	8.47	ok	
2824444	49.1	0.0491	3.6204	6.0288	2.4084	8.46	ok	
2792395	52.2	0.0522	3.1504	5.5139	2.3635	8.11	ok	
2797185	55.8	0.0558	1.4985	3.7435	2.245	7.98	filing needed	
2715909	57.2	0.0572	2.9211	5.1342	2.2131	7.9	filing needed	
2799226	50.9	0.0509	3.4267	5.9205	2.4938	7.88	ok	
2814069	51.8	0.0518	1.1091	3.6395	2.5304	7.63	filing needed	
2850510	69.6	0.0696	1.9411	3.8311	1.89	7.6	filing needed	Exceeded mass limit
2796544	71.1	0.0711	2.7922	4.6551	1.8629	7.55	ok	Exceeded mass limit
2857629	61.5	0.0615	2.7432	5.0058	2.2626	7.19	ok	Exceeded mass limit
2794828	46.2	0.0462	1.6599	4.7354	3.0755	7.04	ok	Scaling issue; cylindrical wholes orientation iss
2765248	49.3	0.0493	7.7917	11.4516	3.6599	5.54	ok	Scaling issue
2727526	53.1	0.0531	1.9497	6.1121	4.1624	4.52	ok	
2715219	60.1	0.0601	1.8422	7.3643	5.5221	3.01	filing needed	Not topology optimised. exceeded mass limit
2798756	53.2	0.0532	2.6973	N/A	N/A	0	filing needed	Failed the test. Rough boundaries
2768657								
2727526								
2760024								
2805866								
2760024								
1722192								
2791707								
2658776								
2731738								

Figure 142 Design Score/Results

My design achieved 7the place in terms of design score. Design score is calculated using $\frac{1}{m \times d}$, where m is mass and d is deflection. My design was the second highest scoring design below 40g where there were only 3.

Experimental design score of 11.15, theoretical FEA design score of 39.7. this is nearly a multiplication of three in difference. This is likely due to the absence of “slack” in 3D CAD software like ANSYS FEA. Where in reality, materials are not perfect; the hint of this is during the 5kg load faze where over half of the total 6mm deflection at 3.5mm deflection occurs. If slack were non-existent, majority of the deflection would occur towards the higher forces, yet at 1/6th the maximum force caused more than half the deflection. Also, the preload was not accounted for in the FEA analysis, which means the peak deflection at 30kg load would not be accounted for.

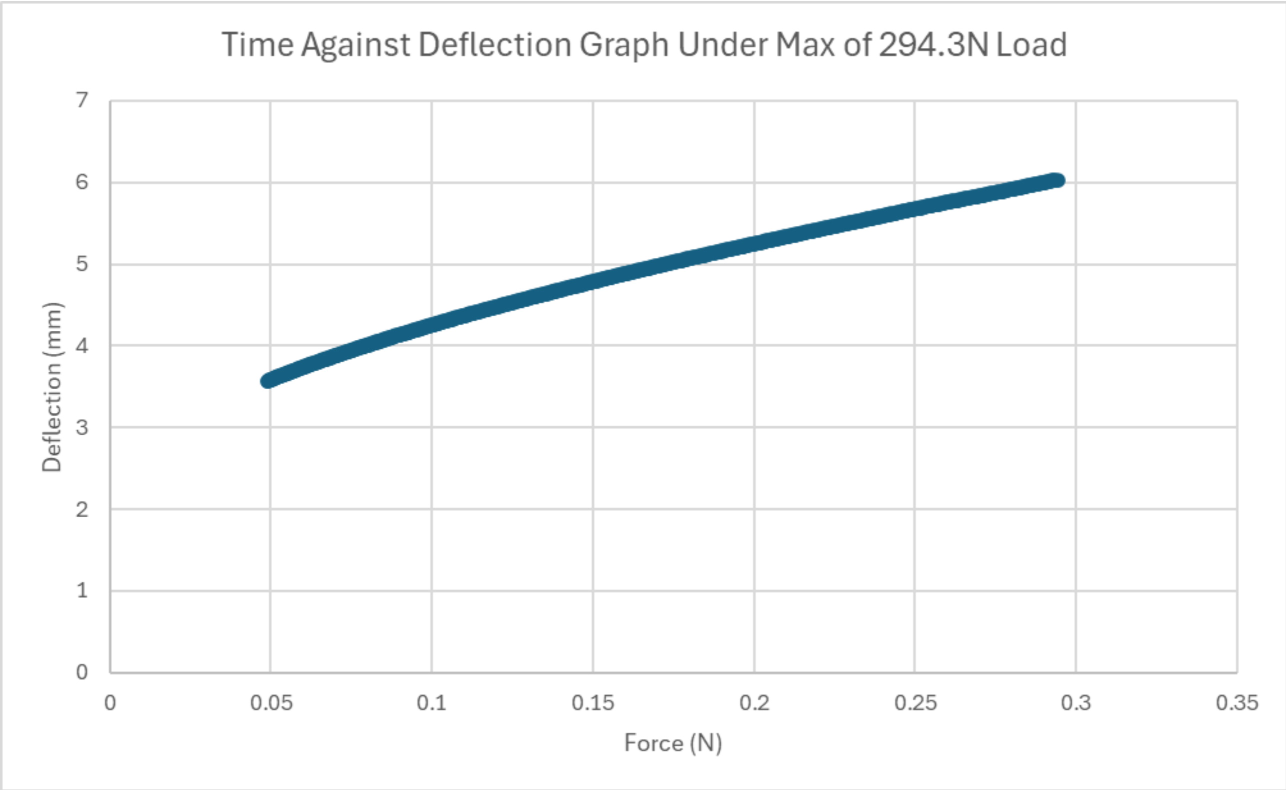


Figure 143 Experimental Time Against Deflection Graph